MIL-HDBK-157

15 OCTOBER 1979

MILITARY HANDBOOK

TRANSPORTABILITY CRITERIA

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DEPARTMENT OF DEFENSE
Washington, DC 20301

Transportability Criteria

MIL-HDBK-157

1. This standardization handbook was developed by the Department of Defense with the assistance of several industry associations, in accordance with established procedure.

2. This publication was approved on 17 July 1979 for printing and inclusion in the military standardization handbook series.

3. This document provides basic transportability criteria for all modes. It will provide valuable information and guidance to personnel concerned with the design, development, and procurement of military materiel. The handbook is not intended to be referenced in purchase specifications except for informational purposes, nor shall it supersede any specification requirements.

4. Every effort has been made to reflect current criteria for the various transportation modes. This handbook will be periodically reviewed by its proponent to insure its completeness and currency. Beneficial comments (recommendations, additions, deletions) and any pertinent data that may be used to improve this document should be addressed to: Director, Military Traffic Management Command Transportation Engineering Agency, PO Box 6276, Newport News, Virginia 23606, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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FOREWORD

1. This handbook provides transportability criteria and test information for guidance in the design, development, and procurement of military materiel within the framework of the DOD Engineering for Transportability program, and in compliance with the requirements of DOD Directive 3224.1 and AR 70-44/OPNAVINST 4600.22B/AFR 80-18/MCO 4610.14C/DLAR 4500.25. Emphasis is placed upon the maximum dimensions and weights that worldwide transportation systems can accept for transportability of materiel or equipment by all modes. Constraints applicable to each transportation mode are outlined, and a discussion is included for intermodal cargo containers and pallets. The natural and induced environmental considerations that affect the reliability of military materiel when moving through the transportation systems are provided. Finally, guidance on test plan development is provided. This guidance will assist in evaluating the transportability characteristics of military materiel and the effect those characteristics will have on functional performance of the materiel.

2. Some of the material used is copyrighted; permission for its use, given by those associations, industries, and individuals who responded generously to all requests for information, is gratefully acknowledged. In particular, appreciation is expressed to the American Association of State Highway and Transportation Officials (AASHTO) for table II and to the International Road Federation (IFR) for figure 23 and tables III through VI.

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1. SCOPE

1.1 General. In consonance with the DOD Engineering for Transportability Program (paragraph 4.1) and MIL-STD-1366A, this handbook covers fundamental transportability criteria for the development of items of materiel prepared for shipment, and for the evaluation of the transportation capabilities of commercial nondevelopmental items. Included are general considerations on dimension and weight limitations, restraint, shock and vibration, and test plan development so that the materiel-developing and procurement agencies can meet the transportability requirements of the various transport modes.

1.2 Application. The handbook is intended as a reference document for use by the research, development, test and evaluation community (RDTE); transportation engineers and specialists; contractors; and contracting officers and their representatives. It is intended to provide general guidance on transportability technology to be used in the design, development, and procurement of all military materiel. Solution of specific transportability problems should consider costs, schedule, environment, and the other constraints of the specific item and its application, as delineated in specific operational requirements (SOR), required operational capabilities (ROC) documents, or other requirements documents that are beyond the scope of this handbook. Specific problems should be coordinated with the agency that is responsible for the items or functions with which the problem can be classified.

1.3 Metric equivalents. Where appropriate, metric equivalents are used as a secondary standard, in parentheses, and conform to the Standard for Metric Practice of the American Society for Testing and Materials (ASTM) E380-76. Conversion tables are in the appendix.

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2. REFERENCED DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on the date of invitation for bids or request for proposals, form a part of this handbook to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	- Preservation - Packaging, Methods of
MIL-A-8421	- Air Transportability Requirements and Tests
MIL-H-9884	- Honeycomb Materiel, Cushioning Paper
MIL-P-27443	- Pallet, Cargo, Aircraft, Type HCU-6/E, HCU-12/E, and HCU-10/C
MIL-N-27444	- Net, Cargo Tiedown, Pallets, HCU-7/E, HCU-15/C, HCU-11/C, and HCU-16/C

STANDARDS

MILITARY

MIL-STD-147	- Palletized Unit Loads on 40" x 48" Pallets
MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment
MIL-STD-210	- Climatic Extremes for Military Equipment
MIL-STD-648	- Design Criteria for Specialized Shipping Containers
MIL-STD-669	- Airdrop Loading, Environment, and Related Requirements for Military Materiel
MIL-STD-810	- Environment Test Methods
MIL-STD-814	- Requirements for Tiedown, Suspension, and Extraction Provisions on Military Materiel for Airdrop
MIL-STD-1319	- Item Characteristics Affecting Transportability and Packaging and Handling Equipment Design

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MIL-STD-1366	- Packaging, Handling, Storage, and Transportation System Dimensional and Weight Constraints, Definition of
MIL-STD-1660	- Design Criteria for Ammunition Unit Loads
MIL-STD-1670	- Environmental Criteria and Guidelines for Air-Launched Weapons

INTERNATIONAL

AIR/STD 44/21	- American, British, Canadian, Australian Air Forces Air Standardization Coordination Committee Criteria for Air Transportability and Air Drop of Equipment
STANAG 2175, NATO	- Flatcars used for Military Equipment
STANAG 2828, NATO	- Military Pallets, Packaging, and Containers
STANAG 2832, NATO	- Restrictions for the Transport of Military Equipment by Rail on European Railways
STANAG 3548, NATO	- Tiedown Fittings on Air Transportable and Air Droppable Cargo

MANUALS

MILITARY

FM 10-500	- Airdrop of Supplies and Equipment: General Information for Rigging Airdrop Platforms
FM 57-1/AFM 2-51	- US Army/US Air Force Doctrine for Airborne Operations
FM 101-20	- US Army Aviation Planning Manual
TM 55-1520	- Army Helicopters, Operator's Manual Series
NAVAIR 01-85SAB-509	- C1-A Aircraft Cargo Loading Manual
NAVAIR 01-85WBB-1-1	- C2-A Aircraft Cargo Loading (Supplemental NATOPS Manual)
TO 1C-5A-9	- Loading Instructions, USAF Series C-5 Aircraft
TO 1C-130A-9	- Loading Instructions, USAF Series C-130A, RC-130A, C-130B, C-130D, C-130E, and C-130H Aircraft

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TO 1C-141A-9 - Loading Instructions, USAF Series C-141A Aircraft

TO 36M-1-141 - Operating Instructions, 463L Materials Handling Equipment System

HANDBOOKS

MILITARY

AFSC DH1-11 - USAF Systems Command Design Handbook - Air Transportability

MIL-HDBK-304 - Package Cushioning Design

MIL-HDBK-318 - Cargo Aircraft Compartment Dimensional Data

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or, as directed, by the contracting officer.)

2.2 Other publications. The following documents form a part of this handbook to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

GOVERNMENTAL

FEDERAL STANDARDS

FED-STD-101 - Preservation, Packaging and Packing-Material Test Procedures

CODE OF FEDERAL REGULATIONS (CFR's)

CFR Title 10 - Energy
CFR Title 39 - Postal Service
CFR Title 49 - Transportation

(Requests for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

NOTE: Regulations in CFR 49 are also published in the R. M. Graziano Water Carrier Tariff and the R. M. Graziano tariff entitled Hazardous Materials Regulations of the Department of Transportation by

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Air, Rail, Highway, Water, and Military Explosives by Water including Specifications for Shipping Containers, respectively.

(Requests for copies should be addressed to R. M. Graziano, Agent, 1920 L St, NW, Washington, DC 20036.)

JOINT MILITARY

DOD Directive 3224.1 - Engineering for Transportability.

AR 70-44/OPNAVINST 4600.22b/AFR 80-18/MCO 4610.14C/DLAR 4500.25 - Research and Development, DOD Engineering for Transportability.

AFR 71-4/TM 38-250/NAVSUP Pub 505/MCOP 4030.19/DSAM 4145.3 - Packaging and Materials Handling--Preparation of Hazardous Materiels for Military Air Shipment.

AR 55-162/OPNAVINST 4600.11B/AFR 75-24/MCO 4643.5A/DSAR 4540.18 - Permits for Oversize, Overweight, or Other Special Military Movements on Public Highways in the Contiguous States and the District of Columbia of the United States.

(Requests for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

MILITARY AIRLIFT COMMAND REGULATIONS

MACR 55-130 - Safety Isle Requirements.

(Requests for copies should be addressed to Headquarters, US Air Force Military Airlift Command, Scott AFB, IL 62225.)

ARMY

AR 70-38 - Research, Development, Test, and Evaluation of Materiel for Extreme Climatic Conditions.

AR 70-47 - Research and Development, Engineering for Transportability.

(Requests for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

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MILITARY TRAFFIC MANAGEMENT COMMAND

MTMC Pamphlet 55-3 - Barge-Ship Systems: Description, Guidance, and Transporting of Military Cargo.

(Requests for copies should be addressed to the Director, Military Traffic Management Command Transportation Engineering Agency, 12388 Warwick Boulevard, PO Box 6276, Newport News, VA 23606.)

AVIATION RESEARCH AND DEVELOPMENT COMMAND

Technical Report No. 72-36 - Design Guide for Load Suspension Points, Slings, and Aircraft Hard Points.

(Requests for copies should be addressed to the Director, Applied Technology Laboratory, US Army Research and Technology Laboratories (AVRADCOM), Fort Eustis, VA 23604.)

US ARMY TEST AND EVALUATION COMMAND

Test Operations Procedure 1-2-500 - Transportability.

(Requests for copies should be addressed to the US Army Test and Evaluation Command, Technical Information Center, Aberdeen Proving Ground, MD 21005.)

NONGOVERNMENTAL

National Railway Publication Co.

Official Railway Register.

(Requests for copies should be addressed to the National Railway Publication Co., 424 West 33rd Street, New York City, NY 10001.)

American National Standards Institute (ANSI)

ANSI-MH5.1.1 - 1979	- Basic Requirements for Cargo Containers
ANSI-MH10.1 - 1972	- Unit Load Sizes for Dimensioning Transport-Packages Sizes

(Requests for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

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US Naval Oceanographic Office Publication

No. 150 - World Port Index, Ports of the World.

(Requests for copies should be addressed to John de Graff, Inc., 34 Oak Avenue, Tuckahoe, NY 10707.)

Association of American Railroads

Outline Diagram for Single Loads, Without End Overhang, on Open-Top Cars.

(Requests for copies should be addressed to the Association of American Railroads, 59 East Van Buren Street, Chicago, IL 60605.)

American Association of State Highway and Transportation Officials (AASHTO)

Legal Maximum Dimensions and Weights of Motor Vehicles Compared with AASHTO Standards, 12/31/78.

Questionnaire on Oversize, Overweight Movement of Military Cargo by Government-Owned Vehicle or by Commercial Carrier.

(Requests for copies should be addressed to The American Association of State Highway and Transportation Officials, 444 N Capitol Street, NW, Suite 225, Washington, DC 20001.)

International Road Federation

Limits of Motor Vehicle Sizes and Weights.

(Requests for copies should be addressed to the International Road Federation, 1023 Washington Building, Washington, DC 20005.)

American Society for Testing and Materials

ASTM E 380-76 - Standard for Metric Practice.

(Requests for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

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International Air Transport Association (IATA)

Air Cargo Guide.

(Requests for copies should be addressed to the Official Airline Guides, Inc., 2000 Clearwater Drive, Oak Brook, Illinois 60521.)

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3. DEFINITIONS

3.1 Airborne operation. An operation involving the air movement of combat forces and their logistic support into an objective area for execution of a tactical or a strategic mission. The means employed may be any combination of airborne units, air transportable units, and types of transport aircraft, depending on the particular situation.

3.2 Airdrop (aerial delivery). An operation involving the delivery of supplies or equipment of combat forces wherein those items are removed from an aircraft during flight.

3.3 Air transportable. Denotes equipment and cargo items that can be carried in an aircraft with only minor sectionalization and reassembling within the capabilities of user units.

3.4 Clearance limits. The dimensions beyond which the size of, or projection on, a shipment may not extend in order to clear obstructions that restrict the handling or transport of such shipment. Such limits may be actual, or they may be prescribed by law or regulation.

3.5 Extraction system. System used to withdraw airdrop items from aircraft in flight.

3.6 Fragile item. An item of systems/equipment/munitions (S-E-M) that is susceptible to damage and/or loss of serviceability during transport and handling, and that requires special shipping procedures or equipment, environmental control, or special packaging for protection during transport.

3.7 Hazardous material. A substance or device that, as determined by the Secretary of Transportation, is capable of posing an unreasonable risk to health, safety, and property when transported in commerce and is so designated in the table set forth in 49 CFR 172.101. Included are explosives, flammable, combustible, and pyrophoric liquids; flammable solids, oxidizers, and organic peroxides; corrosive materials; compressed gases; poisons and irritating materials; etiologic agents; and radioactive materials.

3.8 Platform. A unit-load device, similar to a pallet, designed specifically for airdrop.

3.9 Retardation system. System used to retard and stabilize the descent of an airdrop item.

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3.10 Systems/equipment/munitions (S-E-M) materiel. All items and item components necessary for the equipment, maintenance, operation, and support of military activities without distinction as to their application for administrative or combat purposes, excluding ships.

3.11 Strategic airlift. The continuous or sustained air movement of units, personnel, and logistic support between CONUS and oversea areas and between area commands. The C-141 and C-5 are US Air Force aircraft used in strategic airlifts.

3.12 Tactical airlift. The means by which personnel, supplies, and equipment are delivered by air on a sustained, selective, or emergency basis to dispersed sites, at any level of conflict, throughout a wide spectrum of climate, terrain, and combat conditions. The C-130 is a US Air Force tactical airlift aircraft.

3.13 Technical order. An Air Force publication that gives specific technical directives and information with respect to the inspection, storage, operation, modification, and maintenance of given Air Force items and equipment.

3.14 Transportability. The inherent capability of an item to be moved by towing, self-propulsion, or carrier, via railway, highway, waterway, pipeline, ocean, and airway, using existing equipment or equipment that is planned for the movement of the item.

3.15 Transportability approval. A statement by the appropriate service transportability agent that an item of materiel, in its shipping configuration, is transportable by the mode(s) of transportation specified in development guides or materiel requirements, or meets amended transportability characteristics approved by higher authority.

3.16 Transportability criteria. The physical characteristics of the individual modes of transportation, together with legal and administrative requirements, that should be considered in the design of S-E-M items to assure that they can be moved efficiently by existing and proposed transportation systems.

3.17 Transportability engineering. The performance of those functions required to identify and measure the limiting constraints, characteristics, and environments of transportation systems; the integration of these data into design criteria to use operational and planned transportation capability effectively; and the development of technical transportability guidance.

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3.18 Transportability guidance. Published information concerning loading, securing, moving, and handling operations to insure safe and effective logistics transportation of an item of military equipment, or component thereof, by railway, highway, waterway, ocean, airway, and off-road route either as cargo, tow, or by self-propulsion. The information includes the technical and physical characteristics that affect transportability, such as: loading, blocking, bracing, tiedown, and anchoring; validated dimensions with metric equivalents; significant considerations for movement by air, land, and water transportation; sectionalization to conform to the limits of the various modes of transportation; center of gravity and distribution of load; shipping cube for both operational and sectionalized configurations, and transportation regulations, special procedures, and special permits for movement.

3.19 Transportability problem item. An equipment item in its proposed shipping configuration that, because of its size, weight, fragile, or hazardous characteristics, or lack of adequate means for lifting and tiedown, may be denied movement, or will require special permits or waivers and/or special equipment or handling, or may be unacceptably delayed when moving within existing or newly designed transportation systems.

3.20 Transportability report. A report, submitted on a transportability problem item during S-E-M development/acquisition, with all information necessary for a comprehensive transportability review.

3.21 Transportability review. An evaluation of the transportability of an S-E-M item or its components, to assess its ability to be transported by the modes specified in the materiel requirements documents.

3.22 Unit load. Assembly of one or more individual items into a single compact load, secured together. Skids and cleats may be provided for ease of handling.

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4. GENERAL REQUIREMENTS

4.1 DOD Engineering for Transportability program. This program is designed to provide for the inclusion of transportability requirements in the design of end-items of equipment that are obtained through the materiel acquisition program for adoption into the military services supply systems. In the materiel acquisition program, the transportability requirements for nondeveloped items should be considered in the definition and validation phase and, for developed items, in the concept phase. DOD Directive 3224.1, Engineering for Transportability, is a directed DOD-wide function. This directive assigns transportability functions to the three military departmental secretaries and the Director, Defense Logistics Agency. It was implemented by joint regulation AR 70-44/ OPNAVINST 4600.22B/AFR 80-18/MCO 4610.14C/DLAR 4500.25, DOD Engineering for Transportability. The designated transportability agents are: Commander, Military Traffic Management Command, Washington, DC 20315, for the Department of the Army; Commander, Naval Supply Systems Command, Deputy Commander for Transportation, Washington, DC 20376, for the Department of the Navy; AFSC/LGTT, Washington, DC 20334, for the Department of the Air Force; Commandant, US Marine Corps (LFT-3), Washington, DC 20380, for the US Marine Corps; and Director, Defense Logistics Agency, DLA-HT, Cameron Station, Alexandria, VA 22314, for the Defense Logistics Agency. This regulation designates the transportability agencies, promulgates policy, assigns responsibilities, and outlines procedures for conducting the DOD Engineering for Transportability program. MIL-STD-1319 contains a list of those characteristics that will affect transportability.

4.1.1 Transportability problem item. An item of equipment will be reviewed for approval by the appropriate service transportability agent(s), subject to the reporting requirements of the latest issue of AR 70-44, DOD Engineering for Transportability, when any one of the following conditions apply:

a. Materiel does not meet the transportability characteristics specifically imposed by a military service.

b. Item is wheeled or tracked and is to be towed or self-propelled on or off highways.

c. Materiel is fragile or hazardous. Hazardous materiel that is packaged in accordance with the requirements of Title 49 of the United States Code of Federal Regulations

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and of AFR 71-4/TM 38-250/NAVSUPPub 505/MCOP 4030.19/DSAM 4145.3 will not be considered as a potential transportability problem item solely because of its hazardous properties.

d. Materiel is suited for transport by van or stake body trucks or by cargo containers and exceeds any one of these conditions:

Length	18 ft 6 in. (5.639 m)
Width	7 ft 0 in. (2.134 m)
Height	6 ft 6 in. (1.981 m)
Weight	10,000 lb (4 536 kg)

e. Materiel design requirement includes transportability in US Air Force aircraft, and the item exceeds any one of the following conditions:

Length	20 ft 0 in. (6.100 m)
Width	8 ft 0 in. (2.438 m)
Height	8 ft 0 in. (2.438 m)
Weight	20,000 lb (9 072 kg)
Weight per linear foot	1,600 lb (726 kg)
Floor contact area pressure limit	50 psi (344 750 Pa or non-SI 3.515 kg/sq cm)
Maximum axle load (vehicle with pneumatic tires)	5,000 lb (2 268 kg)
Maximum wheel load (vehicle with pneumatic tires)	2,500 lb (1 134 kg)

Any item that requires special equipment or procedures for loading in an aircraft, such as nuclear weapons.

f. Materiel is not to be transported by either method stated in 4.1.1d or 4.1.1e above and exceeds one of these conditions:

Length	32 ft 0 in. (9.754 m)
Width	8 ft 0 in. (2.438 m)
Height	8 ft 0 in. (2.438 m)
Weight	11,200 lb (5 080 kg)

4.2 Worldwide transportability. The following criteria establish maximum cargo dimensions and weight constraints for worldwide transportability.

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4.2.1 All-mode transportability. Items that fall within parameters of paragraph 4.1.1d qualify for all-mode transportability. Items that exceed the criteria of paragraph 4.1.1d will require study to determine how the item is to be shipped.

4.2.2 Limited all-mode transportability. If an item is not to be transported by any of the methods described in paragraphs 4.1.1d and 4.1.1e, the criteria of paragraph 4.1.1d may be increased to that of paragraph 4.1.1f.

4.2.3 Other dimensional restrictions. It may be necessary to impose dimension or weight restrictions on the item lower than those contained in paragraph 4.1.1d or 4.1.1f because of the method(s) of delivery to tactical units. Attempts should be made to design items so that multiples can be loaded on a planned transport vehicle without exceeding the restrictive criteria for that vehicle.

4.3 Containerization. Cargo containers are intended to facilitate international and intermodal interchange and transportation of containerized cargo shipments between modes. Items should be transportable, where practicable, in International Organization for Standardization (ISO) Series 1 or American National Standards Institute (ANSI) containers.

4.3.1 Shelters and vans. Shelters and/or special-purpose vans developed to provide for operational requirements, such as automatic data processing vans, repair machine shops, communications vans, fire direction centers, and so forth, will conform to ANSI and ISO container specifications to the extent practical (DOD Instruction 4500.37).

4.4 Palletization. The two principal pallets in widespread use in DOD are the 40-inch (1.016-m) by 48-inch (1.219-m) general-purpose, and the 108-inch (2.743-m) by 88-inch (2.235-m) component of the HCU-6/E 463L aircraft loading system. Note, the usable space on the pallet for the 463L system is 104 inches (2.642 m) by 84 inches (2.134 m), due to the rail interlock feature of the pallet.

4.4.1 Unit loads. The 40-inch (1.016-m) by 48-inch (1.219-m) pallet is used extensively as a base for unit loads (MIL-STD-147). Nonstandard pallet sizes will be permitted for palletization of general supplies only where it would be technically and economically impractical to use the 40-inch (1.016-m) by 48-inch (1.219-m) pallet. Nominal plan dimensions that will permit efficient utilization of the floor area in cargo containers, truck vans, and closed railcars include: 54 inches (1.372 m) by 45 inches (1.143 m), 45 inches (1.143 m) by 45 inches (1.143 m), 49 inches

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(1.245 m) by 41 inches (1.041 m), 45 inches (1.143 m) by 36 inches (0.914 m), and 45 inches (1.143 m) by 34 inches (0.864 m) (ANSI MH 10.1 1972). Maximum height of unit loads that will allow double tiering is 41 inches (1.041 m) for cargo containers 8 feet (2.438 m) high, 43 inches (1.092 m) for containers 8 feet 6 inches (2.591 m) high, 46 inches (1.168 m) for containers 9 feet (2.743 m) high, and 49 inches (1.245 m) for containers 9 feet 6 inches (2.896 m) high.

4.4.2 General-purpose unit loads for shipment to NATO forces. For shipment of unit loads of general cargo to NATO forces, the 40-inch (1.016-m) by 48-inch (1.219-m) pallet with a maximum height of 41 inches (1.041 m) and a maximum weight of 2,500 pounds (1 134 kg) must be adhered to (STANAG 2828).

4.4.3 Ammunition unit loads. Ammunition unit loads shall conform to the dimensional limitations and weight of MIL-STD-1660.

4.5 Restrictions on open-top equipment. Wherever practical, ammunition, explosives, and other hazardous and/or classified materiel should be designed for transport in closed-top carrier equipment.

4.6 Sectionalization. Sectionalization refers to the characteristic of an item of materiel/equipment that makes possible removal or reorientation of sections or subsystems to reduce weight or shipping dimensions. This may be necessary to assure that the item can be accommodated within normal transportation envelope dimensions. For smaller items, sectionalization should be minimized to facilitate rapid deployment.

4.6.1 Water movement. The item must fit into the hatch opening for below-deck stowage. If a helicopter is to be used by both the Army and the Navy, operational compatibility with an aircraft-carrier deck elevator is an important sectionalization consideration.

4.6.2 Design considerations. Any sectionalization should be accomplished in a manner that will permit reassembly in the field, using a minimum of special tools, personnel, and handling equipment. In all designs, quick-disconnections, electrical plugs, and receptacles and other easy-to-separate and easy-to-reconnect devices should be incorporated at separation or folding points and should be designed to be connected in only one way.

4.6.3 Preparation for shipment. Most items of materiel/equipment require some sort of preparation for shipment except in those cases

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where such items will be delivered under their own power. Depending on the cargo item, certain subassemblies require folding, removal, or reorientation. Typical subassemblies for trucks include bows, tarpaulins, exhaust pipe extensions, windshield mirrors and arms, and cargo-body side racks. For combat tanks, the turret traverse and gun-elevating mechanism must be in the travel position, locked, and wire-tied to prevent rotation; antennae must be tied down or removed; the cupola must be fastened in closed position and loose gear secured; and the machine gun and mount must be removed. All removable items, especially those that are subject to pilferage, should be stowed and secured within the confines of the vehicle or vehicle cargo compartment. Exposed machined metal surfaces will require a protective coating to prevent corrosion. Dissimilar metals will set up a galvanic action, and must be insulated from each other in the original design or as part of the preparation for shipment. Fuel systems may have to be drained, purged, and preserved. These features, and other similar ones, should be considered when the item is designed to reduce the time required to accomplish these tasks, ease their performance, and reduce the possibility of damage (MIL-STD-1366 and FED-STD-101).

4.7 Tiedown and lifting points. The design of some items of materiel/equipment limits the points at which attachments may be placed for lifting and tieing down. Certain tiedown practices may cause extensive damage to the item. Therefore, consideration should be given to the design, location, and marking of attachments for lifting, ground handling, and restraining for surface and air transportation. MIL-STD-209 provides guidance for the design and location of lifting and tiedown provisions. All devices used as hoist or lift equipment for hazardous materials should have an allowable design stress below one-fifth the yield stress of the material.

4.8 Restraint. Restraint criteria should be met for all applicable modes of shipment. MIL-STD-209, MIL-STD-814, MIL-A-8421, and Air Force System Command (AFSC) Design Handbook (DH) 1-11, Air Transportability, provide appropriate guidance for restraint of cargo. However, the designer should consider also the tiedowns and any other appurtenances or possible attachment points, from the viewpoint of the person charged with moving the item, to determine how an inexperienced person would restrain it. Parts of the item that could be damaged if they are used as restraining points should be marked to preclude improper use, and suitable parts should be marked for use as restraining points. Typical lifting and tiedown provisions available on equipment, and diagrams for shipment of equipment by the different modes, are shown in figures 1 through 16.

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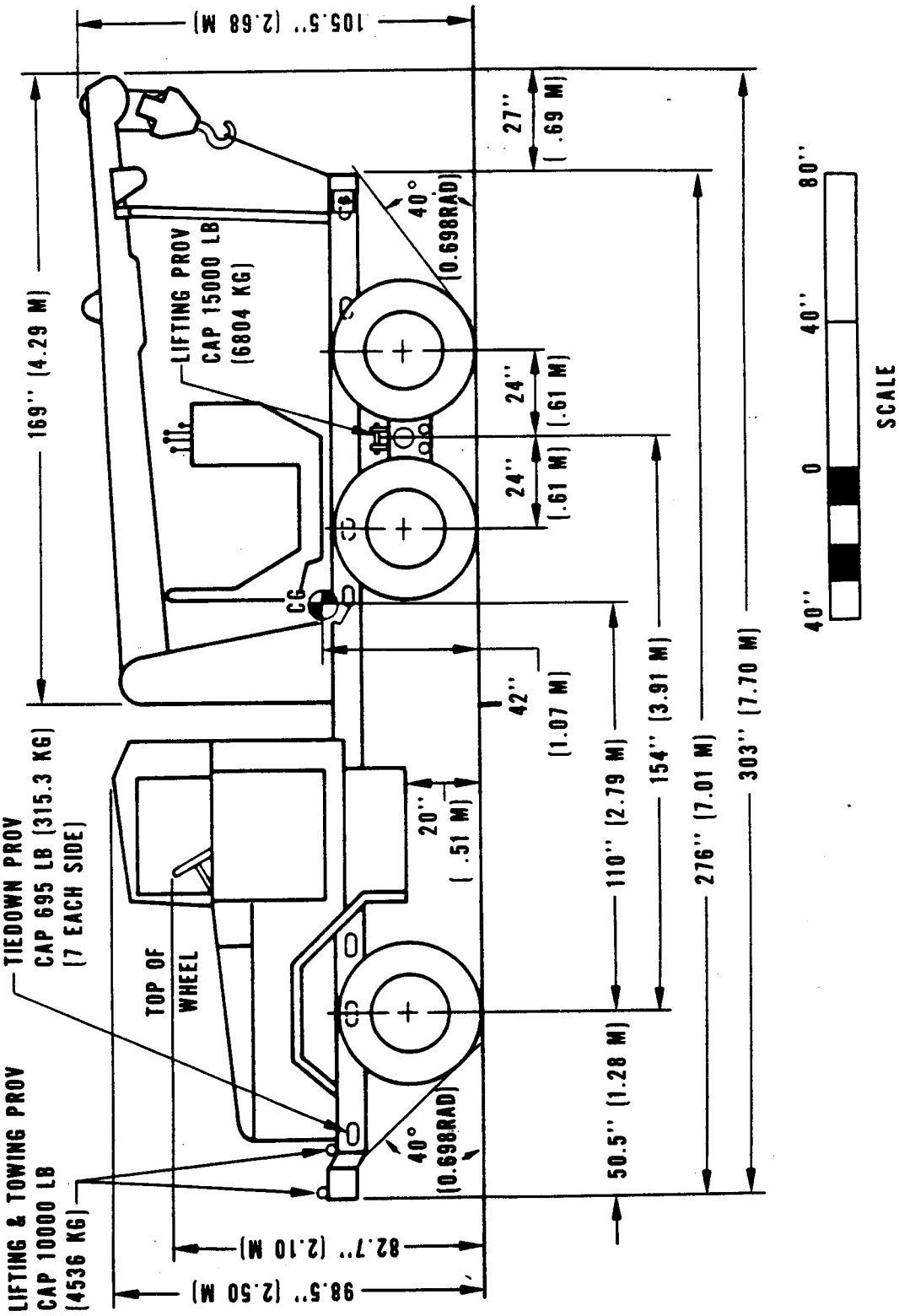


FIGURE 1. Diagram with transportability information for wrecker truck, side elevation.

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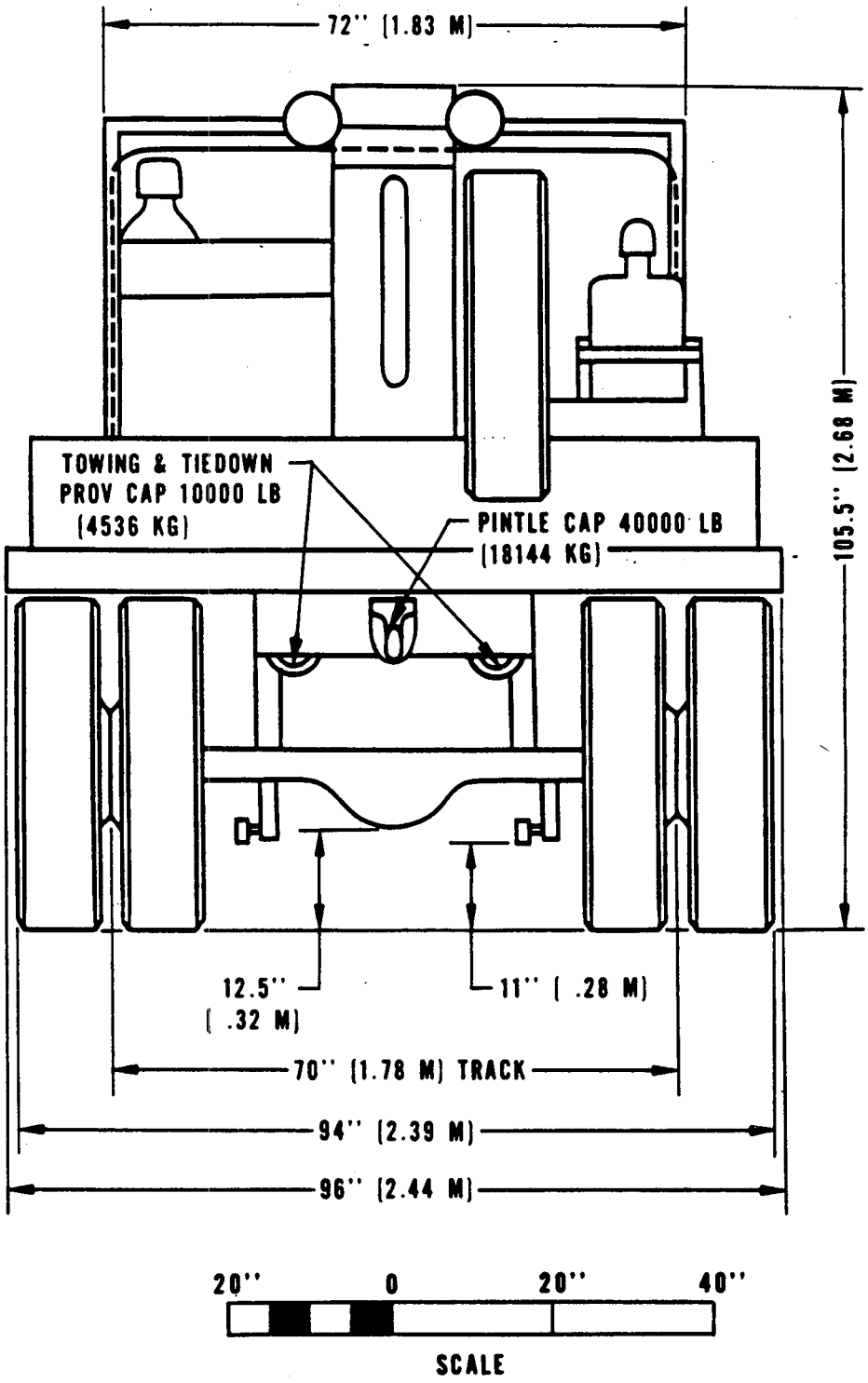


FIGURE 2. Diagram with transportability information for wrecker truck, rear elevation.

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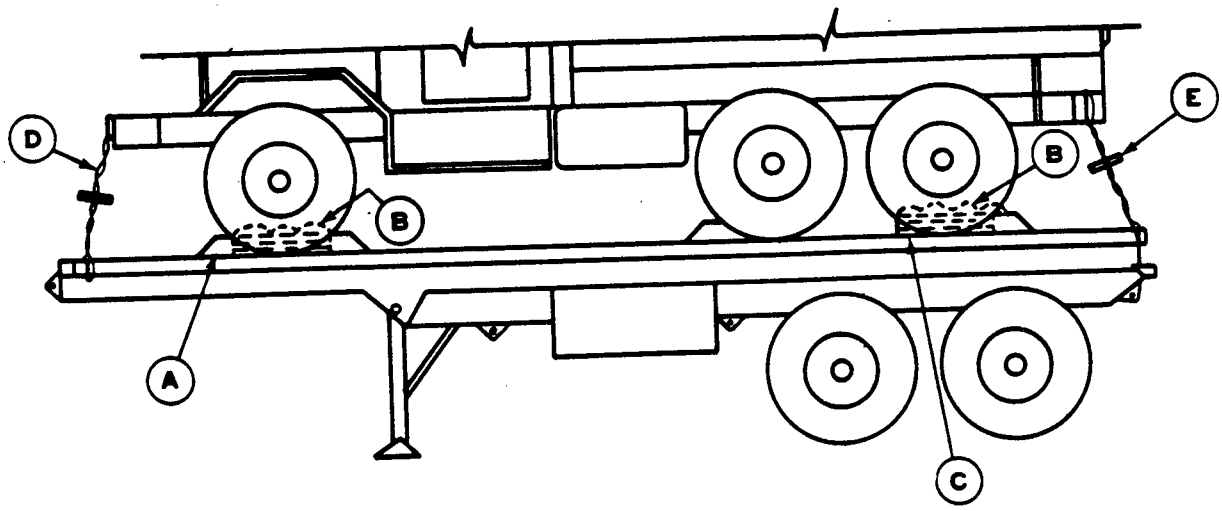


FIGURE 3. Blocking and tiedown of typical 2-1/2-ton (2 268-kg), 6x6 truck on 12-ton (10 884-kg) or larger semitrailer.

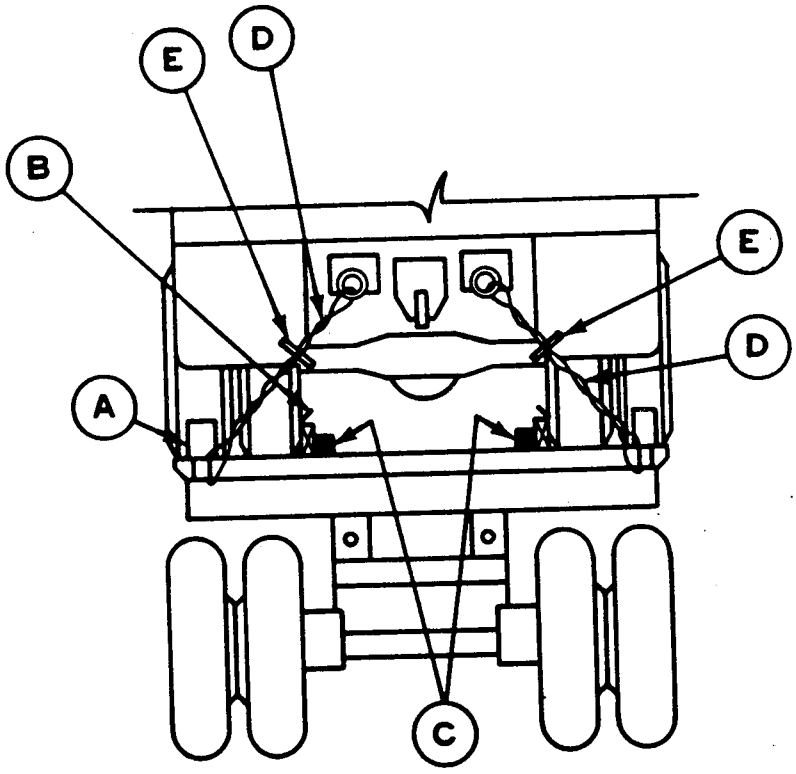


FIGURE 4. Rear view of blocking and tiedown of typical 2-1/2-ton (2 268-kg) 6x6 truck on 12-ton (10 884-kg) or larger semitrailer.

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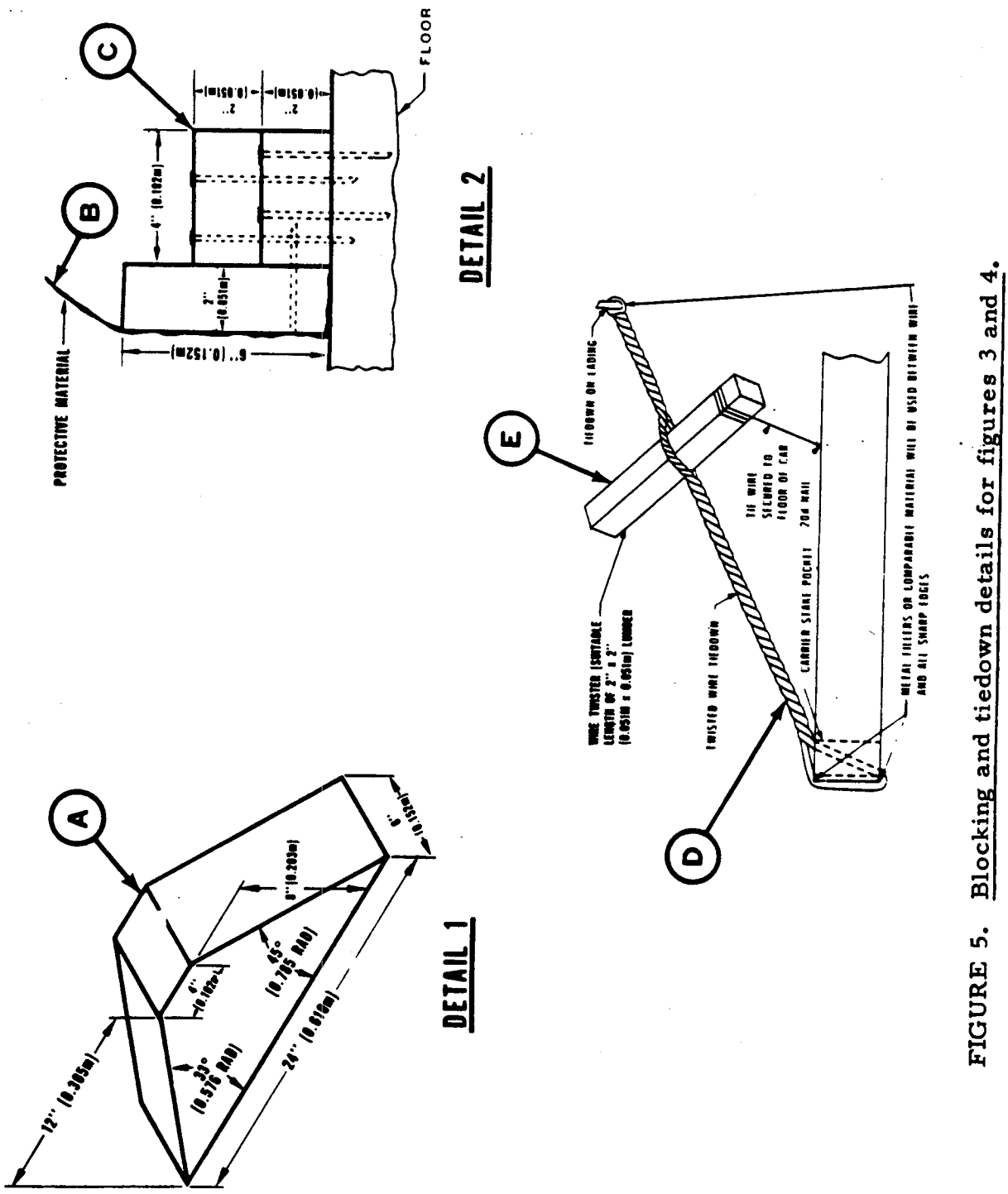


FIGURE 5. Blocking and tiedown details for figures 3 and 4.

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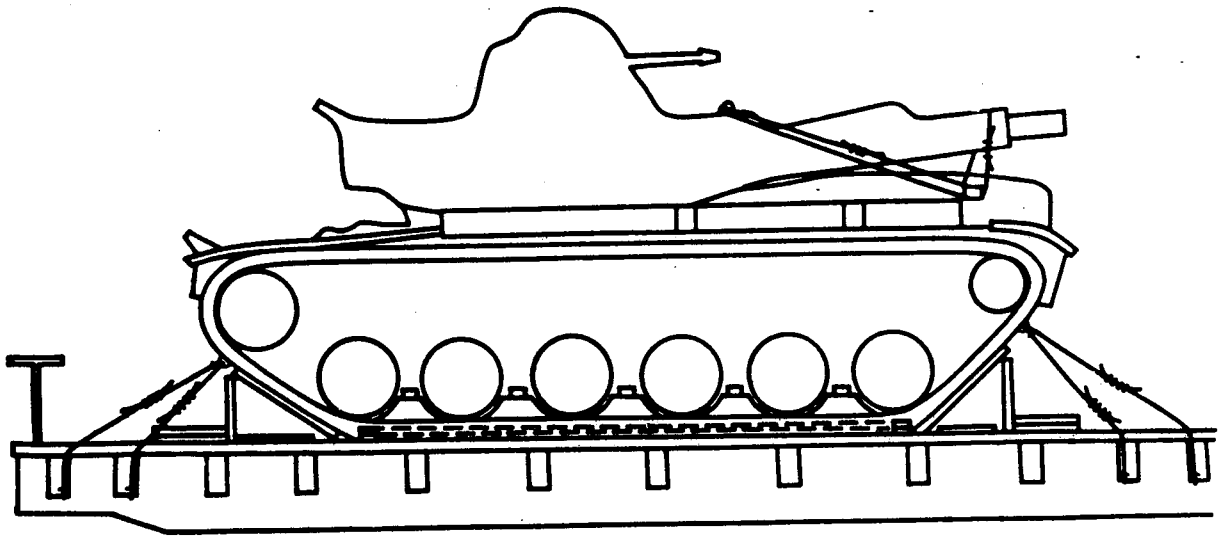


FIGURE 6. Blocking and tiedown for tank on general-purpose flatcar (side view).

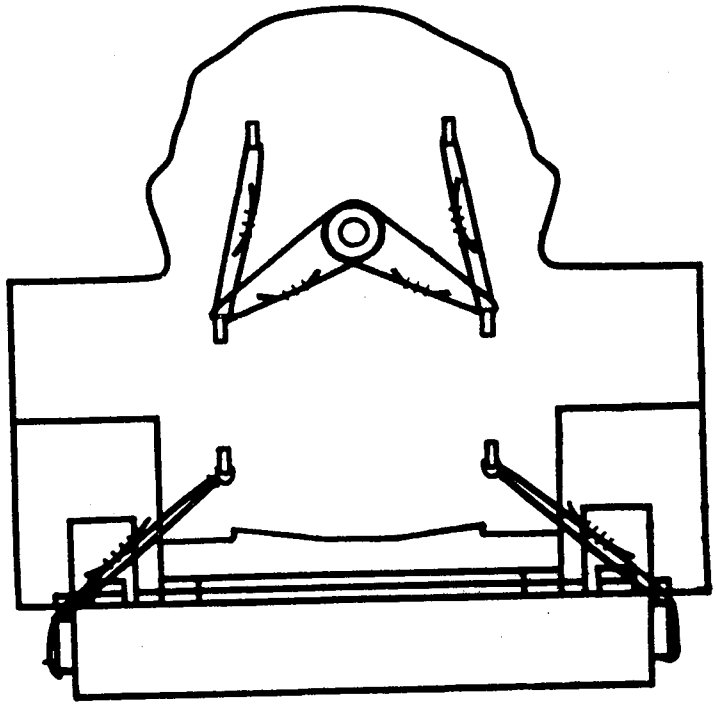
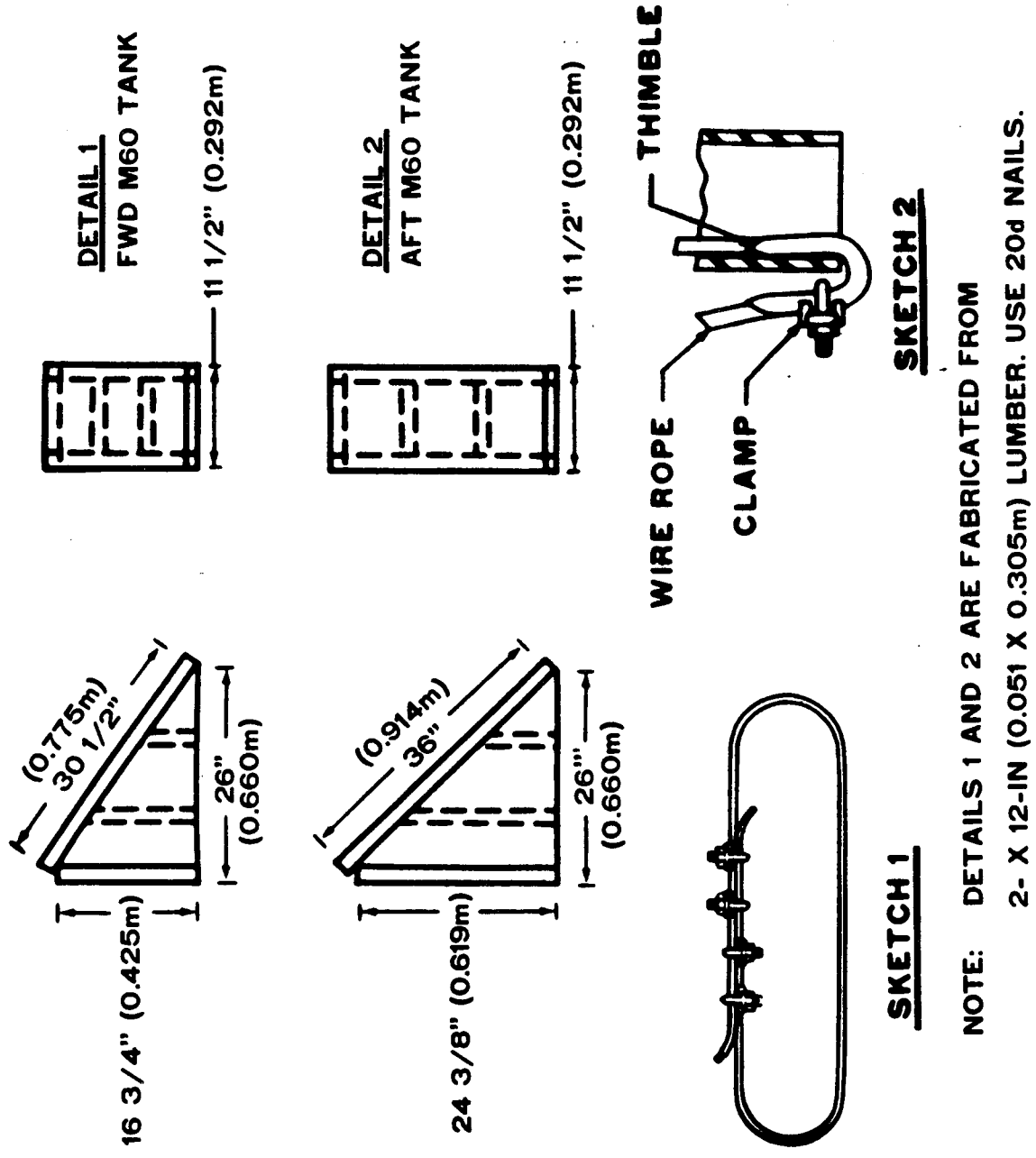


FIGURE 7. Blocking and tiedown for tank on general-purpose flatcar (end view).

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NOTE: DETAILS 1 AND 2 ARE FABRICATED FROM 2- X 12-IN (0.051 X 0.305m) LUMBER. USE 20d NAILS.

FIGURE 8. Blocking detail diagram.

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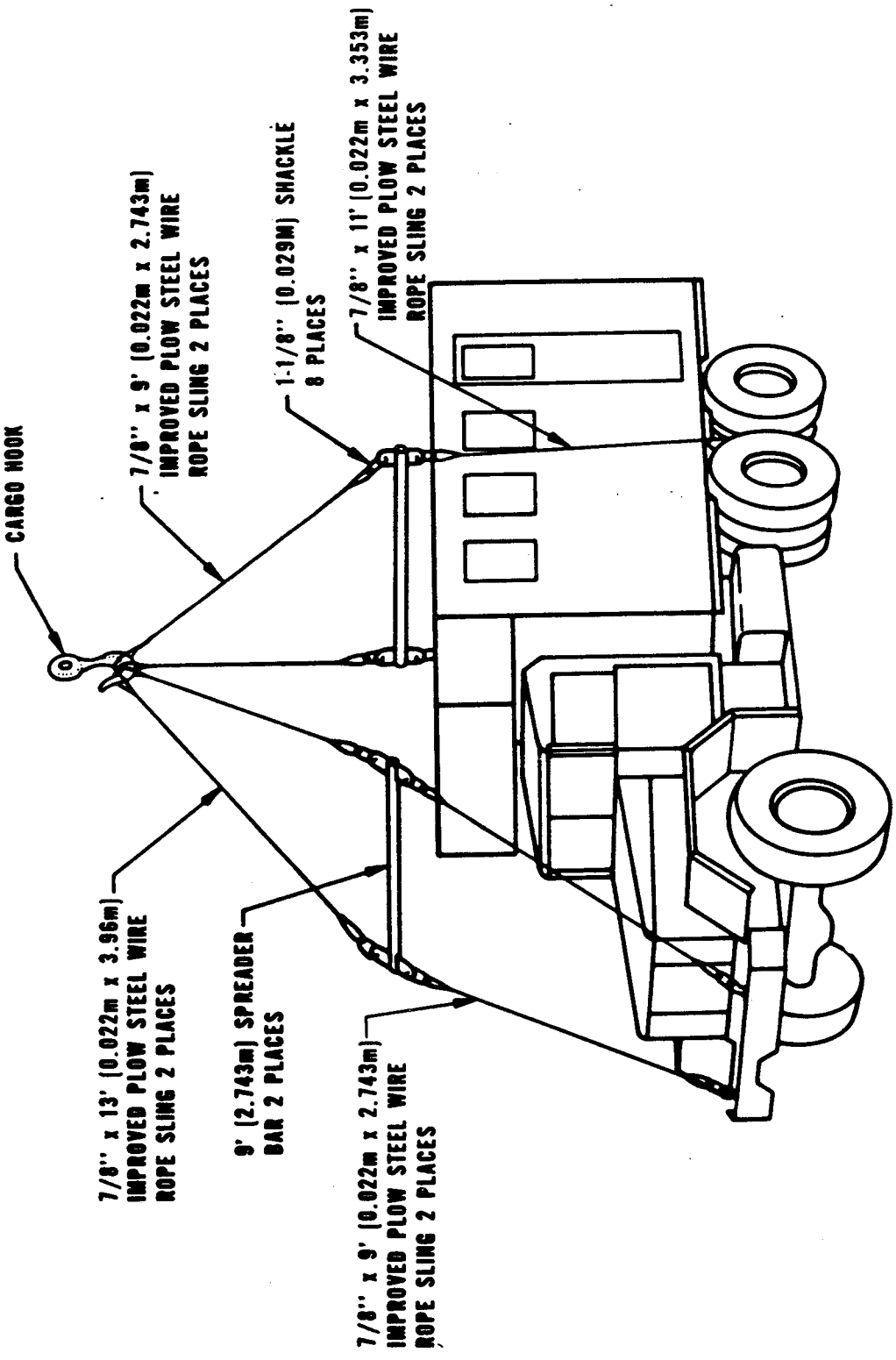


FIGURE 9. Lifting diagram for 2-1/2-ton (2 268-kg) van truck using an eight-wire sling.

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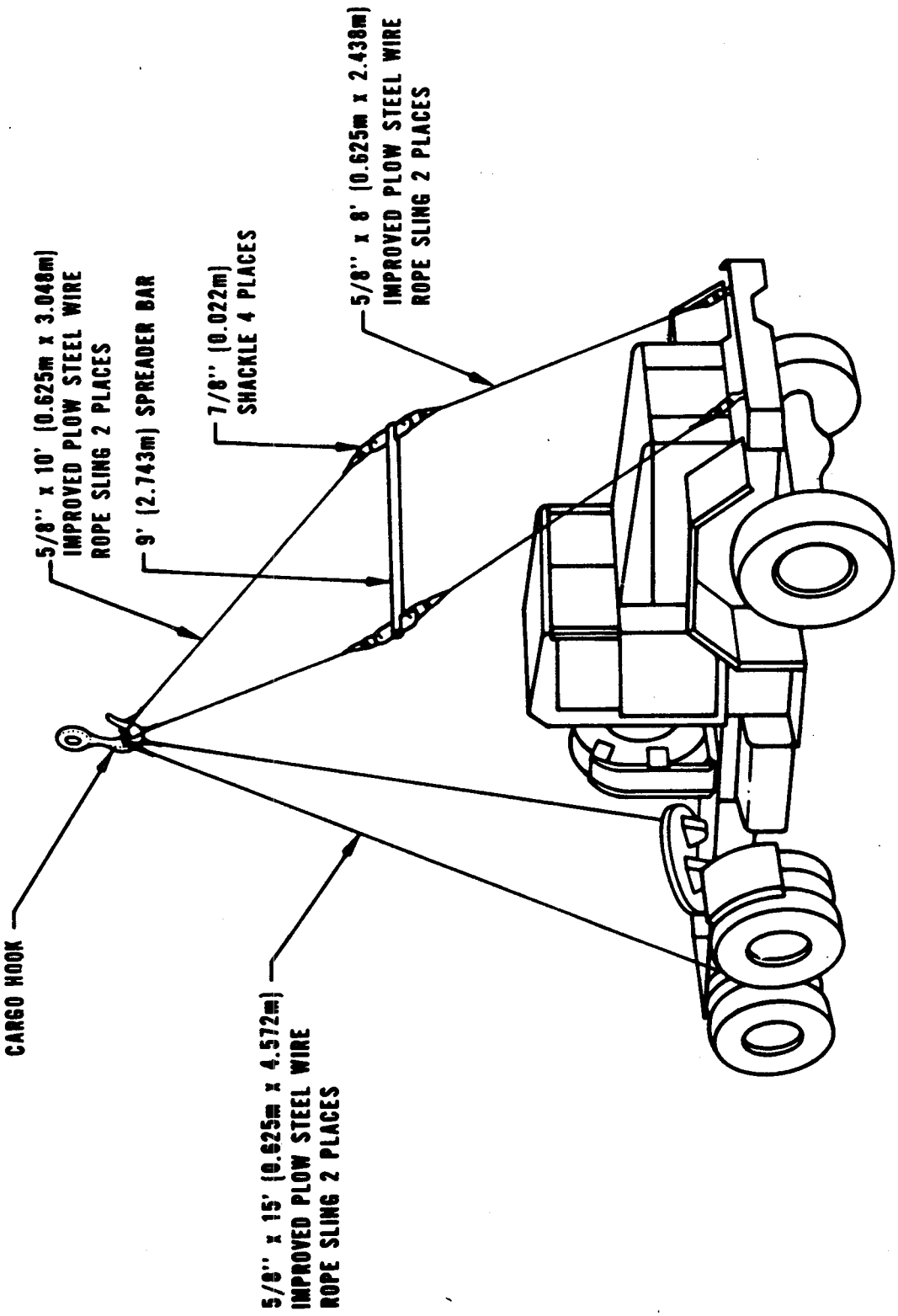


FIGURE 10. Lifting diagram for 2-1/2-ton (2 268-kg) tractor truck using a six-wire sling.

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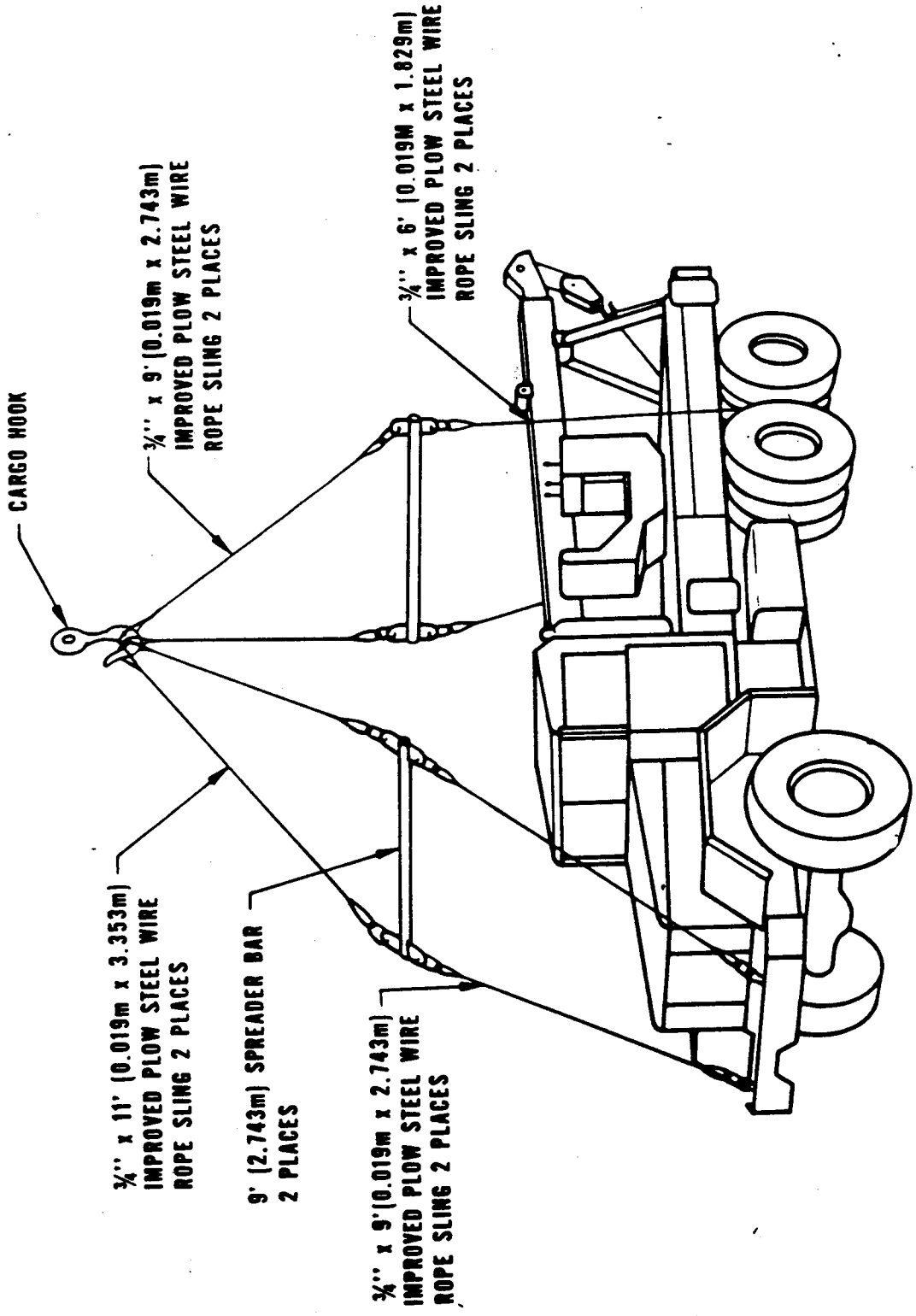


FIGURE 11. Lifting diagram for 2-1/2-ton (2 268-kg) wrecker truck using an eight-wire sling and two spreader bars.

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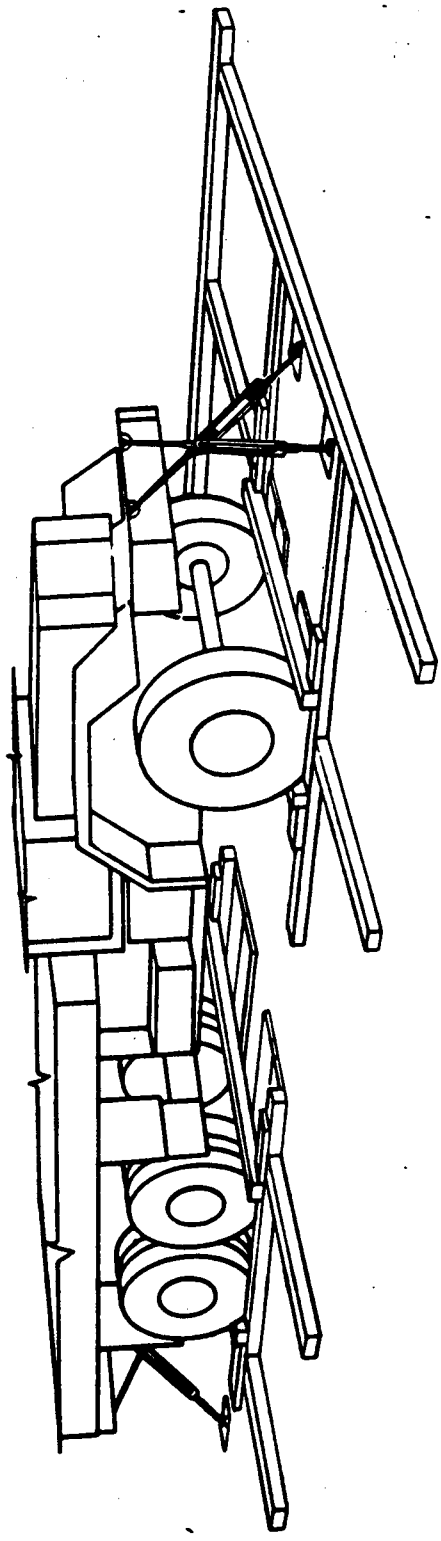


FIGURE 12. Typical blocking and tiedown of 2-1/2-ton (2 268-kg), 6x6 truck in general-cargo vessel.

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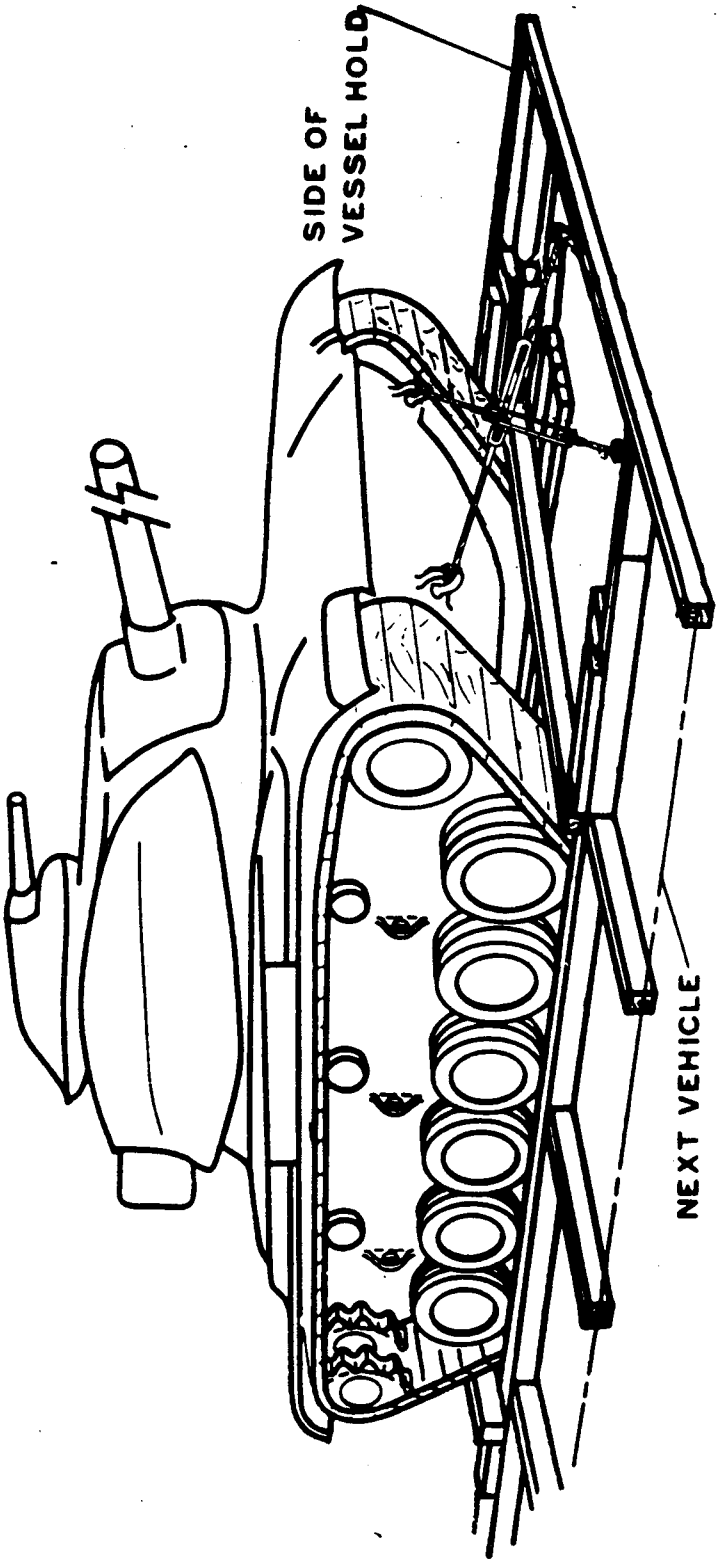


FIGURE 13. Typical blocking and tiedown for tank in hold of general-cargo vessel.

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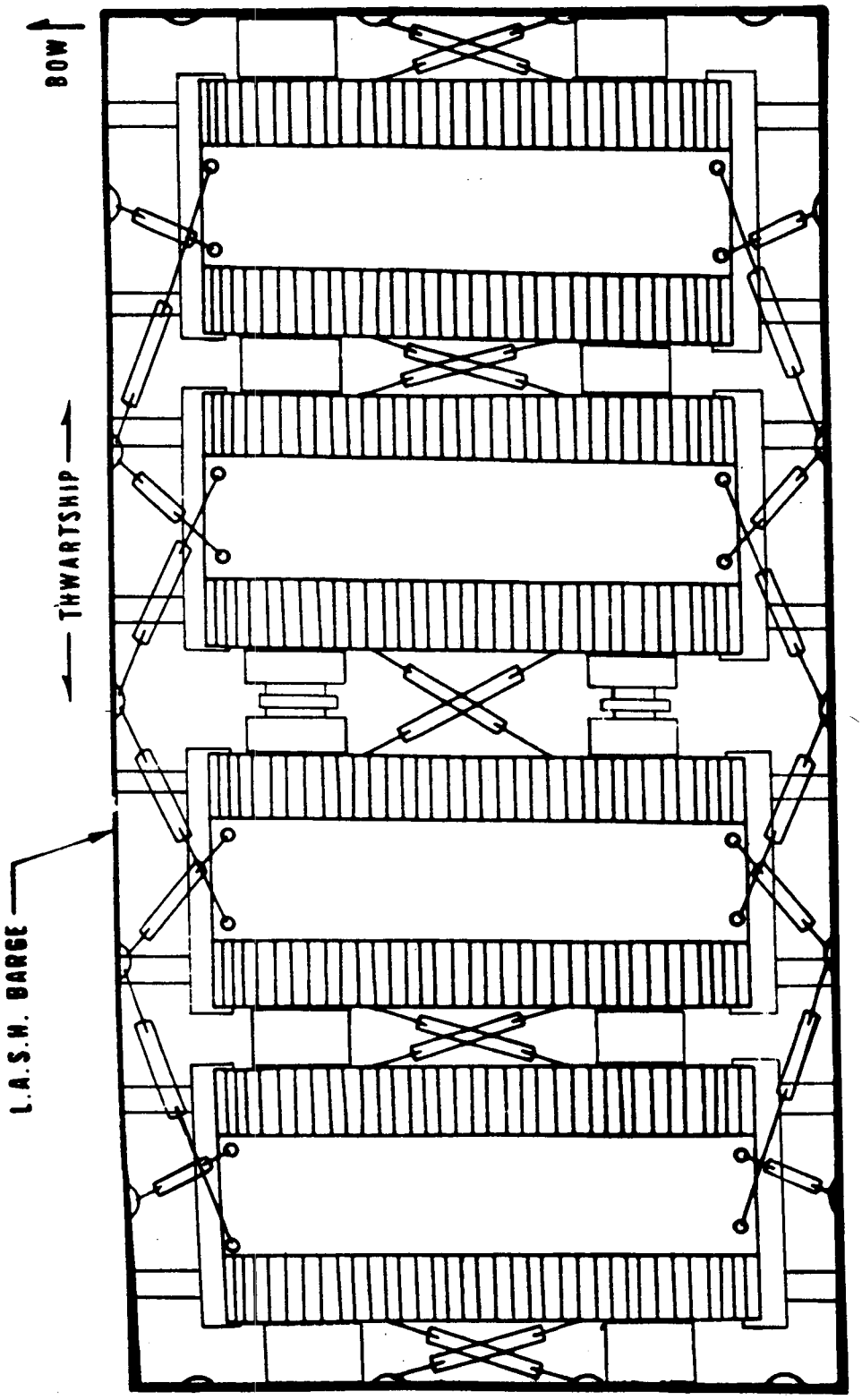


FIGURE 14. Typical loading of four tanks on a LASH (59.9-foot by 29.5-foot or 18.25-m by 8.992-m) barge using wire rope, cable clips, and turnbuckles, with blocking between tanks and tanks to hull.

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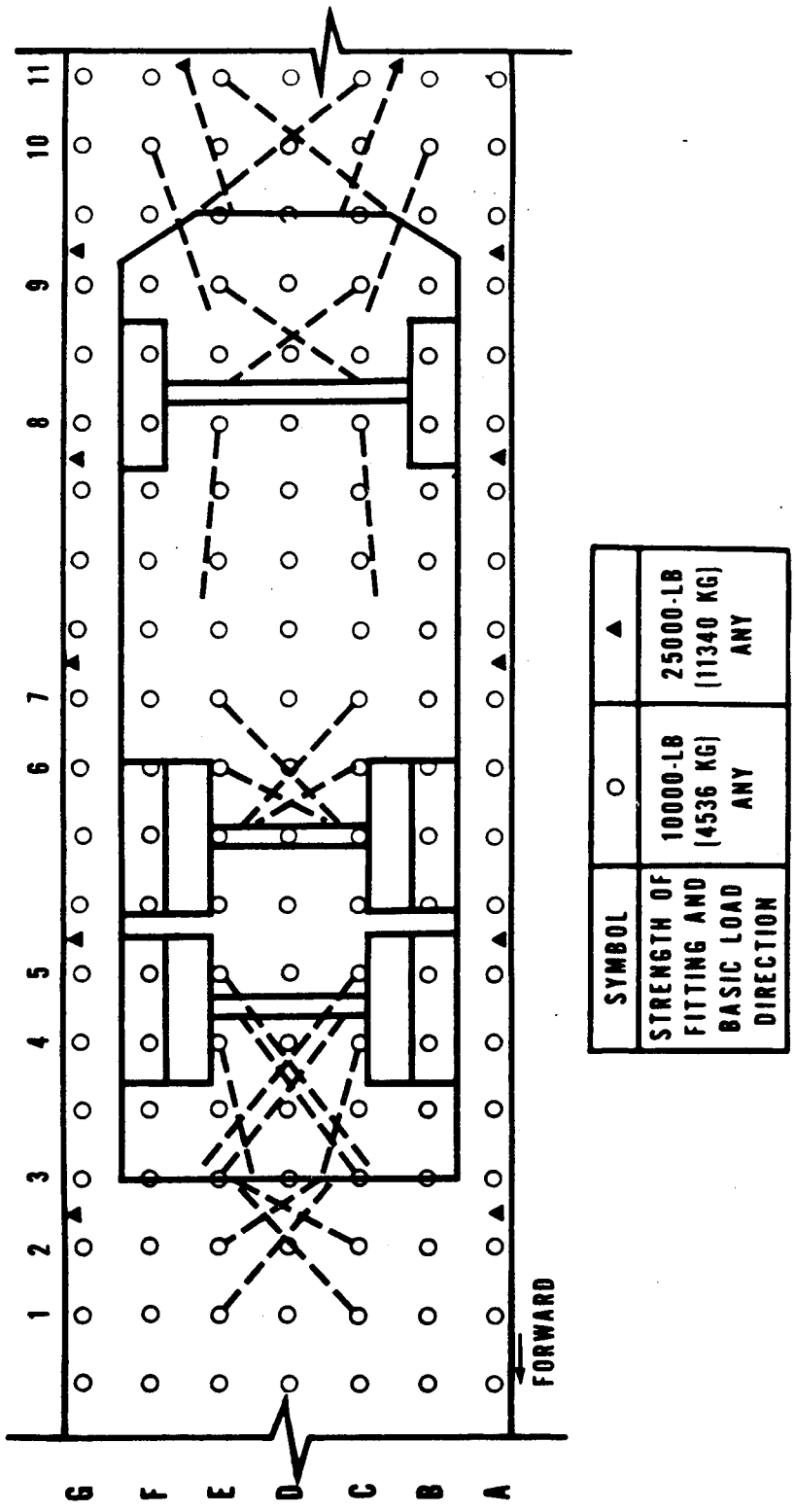
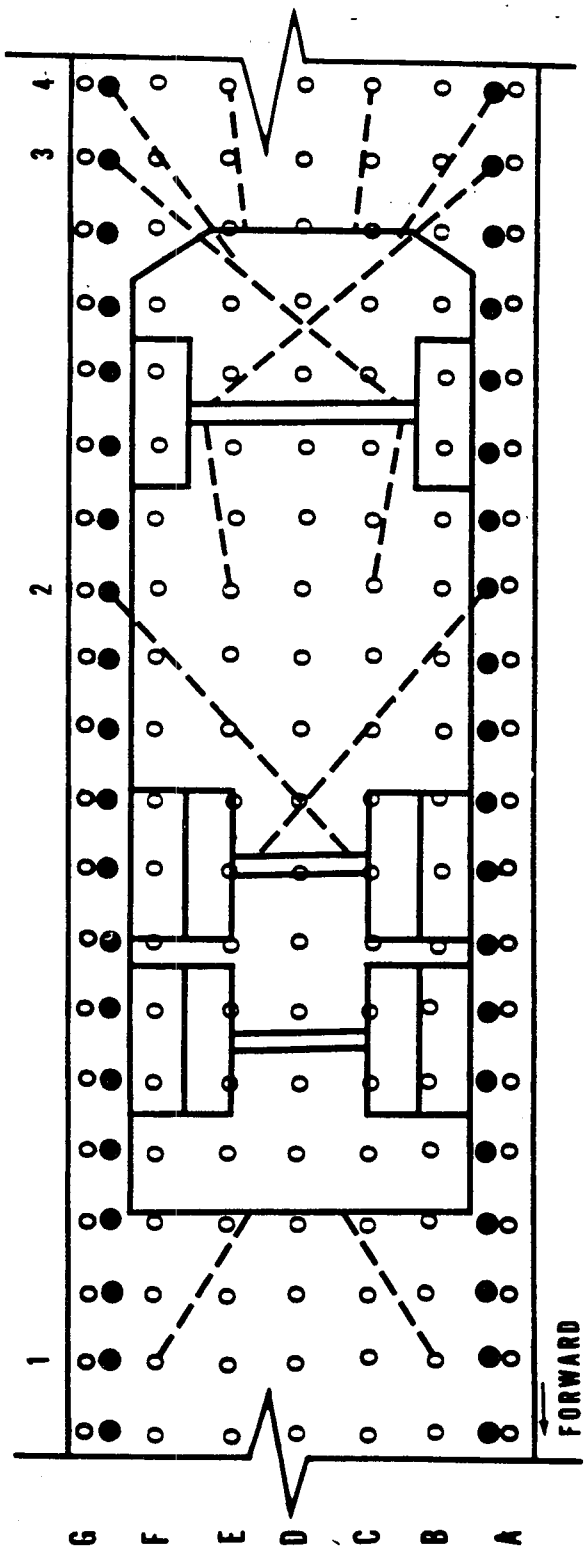


FIGURE 15. Tiedown diagram for 2-1/2-ton (2 268-kg) cargo truck in C-130 aircraft.

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SYMBOL	○	●
STRENGTH OF FITTING AND BASIC LOAD DIRECTION	10000-LB (4536 KG) ANY	25000-LB (11340 KG) ANY

FIGURE 16. Tiedown diagram for 2-1/2-ton (2 268-kg) cargo truck in C-141 aircraft.

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5. DETAILED REQUIREMENTS

5.1 Highway transportation.

5.1.1 General. Highway transport vehicles developed for over-the-road movement should meet the limitations imposed by the physical, legal, and administrative characteristics of roadways, bridges, and other structures. The most significant physical, structural, legal, and administrative characteristics and constraints of CONUS highway systems, along with the legal, dimensional, and weight criteria for highway vehicles in various foreign countries, are presented herein. These criteria establish guidelines to insure that new military highway transport equipment is compatible with the capabilities and limitations of CONUS and foreign highway systems.

5.1.2 Highway systems. CONUS highway systems are divided into the following functional classifications:

a. Principal arterials - an integrated network of highways that provide a high level of service for statewide or interstate travel, with more emphasis on mobility than on access. The network connects all urban areas of 50,000 or more population and includes the National System of Interstate and Defense Highways (NSIDH) and other freeways, expressways, and principal arterials not on the interstate system.

b. Minor arterials - an integrated network of highways that connect cities, larger towns, and principal arterials, with more emphasis on access than on mobility.

c. Collectors - provide intracounty and city connections, connecting arterials with local roads, streets, and land access, at a lower level of service than arterials.

d. Locals - all highways not otherwise classified that provide land access and feed higher-order systems at the lowest level of service.

These functional classifications include all the traditional classifications such as Federal-Aid Primary and Secondary, State, and local systems.

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5.1.3 Dimensional restrictions for vehicles. Items transported on open-top highway vehicles should be designed so that the vehicle/load combinations do not exceed the weight and clearance limits for the highway systems on which they travel.

5.1.4 Highway transportability. The limits shown in figure 17 represent limiting length, width, and height of vehicles and vehicle/cargo combinations capable of legal movement without requiring special permit (paragraph 5.1.5.2) in all but a few states and most foreign countries. The criteria are based upon the dimensional and weight recommendations of the American Association of State Highway and Transportation Officials (AASHTO), Federal-Aid Highway Acts, physical capabilities generally expected in oversea highway systems, and capability of commercial highway equipment. Details on current legal limits are available from the Commander, Military Traffic Management Command, ATTN: MT-SA, Washington, DC 20315.

a. Maximum single-axle load is 20,000 pounds (8 165 kg). A single axle is an assembly of two or more wheels whose centers are in one transverse plane or that may be included between two transverse vertical planes 40 inches (1.016 m) apart.

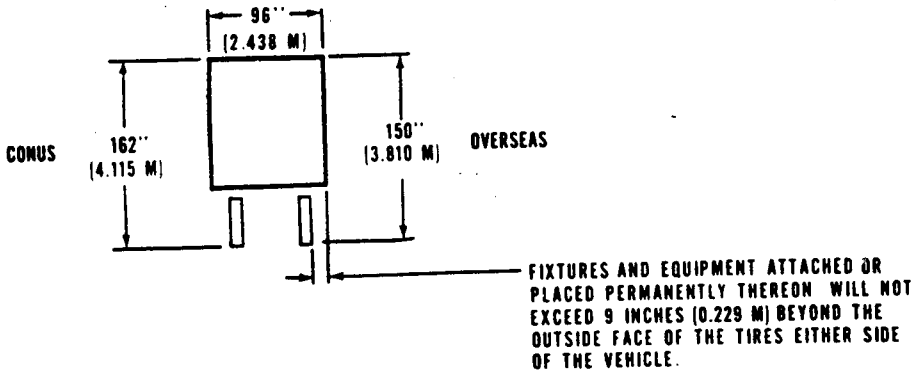
b. Maximum tandem-axle load is 34,000 pounds (14 505 kg). A tandem axle consists of any two or more axles whose centers are more than 40 inches (1.016 m), but less than 96 inches (2.438 m), apart.

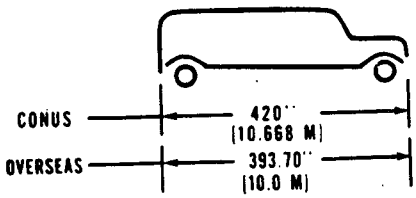
c. Vehicle gross weight is determined by the number of axles, axle spacing, practical weight limit on front axle, and the manner in which the load is distributed to each axle and to each combination of axles.

5.1.4.1 Tracked vehicles. Tracked vehicles normally are not operated on public highways and, with some exceptions, their normal highway transport is on a standard or special wheeled vehicle. Therefore, the combined tracked vehicle/transporter load should be considered when evaluating the highway transportability of tracked vehicles.

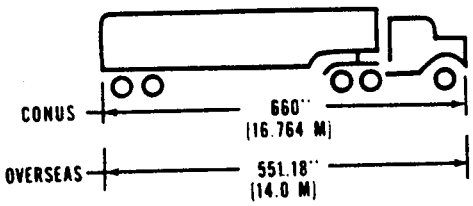
5.1.5 Categories of oversize/overweight movements. Oversize or overweight highway movements in CONUS can be categorized as follows:

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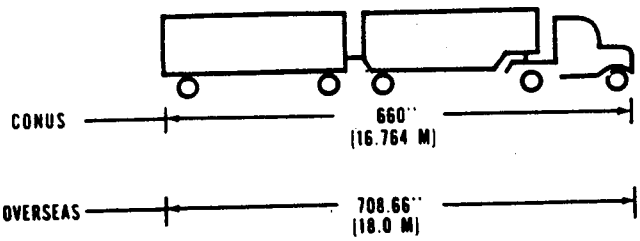




SINGLE VEHICLE



TRACTOR WITH SEMITRAILER



OTHER COMBINATIONS

NOTE: THESE DIMENSIONS WILL ALLOW LEGAL MOVEMENT, WITHOUT REQUIRING SPECIAL PERMIT, IN ALL BUT A FEW STATES AND MOST FOREIGN COUNTRIES.

FIGURE 17. Vehicle dimensional limitations.

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a. Movements that a state will allow under its permit procedures, but are not certified as essential to national defense.

b. Movements that have been certified as essential to national defense.

5.1.5.1 Movement of oversize/overweight vehicles in CONUS. When overweight or oversize highway movements are contemplated, vehicle designers should refer to paragraph 4.1.1.

5.1.5.2 Special permits. AASHTO surveyed all state highway departments to determine the size and weight limits they would permit without certification of the movement as being essential to national defense. The results of this survey are summarized in table I. These maximum values are subject to adjustment based upon specific circumstances at the time of a move. For example, frequency of use, distance, weather, traffic, or other conditions at a particular time may make it necessary for a state to make adjustments so as to improve highway safety. In view of the above, the data in table I are presented for guidance only and should be considered as design criteria and exceeded only when absolutely necessary to meet essential national defense needs (AR 55-162).

5.1.5.3 Subjection to unusual hazards. Oversize/overweight public highway movements, or other movements that subject normal highway users to unusual hazards, should be considered only when the movement is essential to national defense and no other mode of transportation would meet defense needs.

5.1.5.4 Sectionalization of loads. Oversize/overweight public highway movements will be made only after the load has been sectionalized to the maximum amount possible (paragraph 4.6).

5.1.6 Off-road mobility. During transport, occasions may arise in which wheeled or tracked vehicles will be required to leave the highway and travel cross country. A wheeled or tracked vehicle should be designed to meet appropriate cross-country mobility requirements.

5.1.7 Vehicle dimensional and weight limitations. Table II is a summary of 1977 CONUS legal maximum dimensions and weight of motor vehicles, reproduced through the courtesy of AASHTO, 341 National Press Building, Washington, DC 20004.

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TABLE I

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS

	LENGTH	WIDTH	HEIGHT	GROSS WEIGHT	SINGLE AXLE	TANDEM AXLE	CONDITIONS UNDER WHICH STATES ALLOW MOVEMENTS OVER THESE DIMENSIONS/WEIGHTS WITHOUT CERTIFICATION AS ESSENTIAL TO THE NATIONAL DEFENSE.
ALABAMA	75'	12'	14'	(1)	20,000	40,000	
ALASKA	80'	16'	18'	150,000	30,000	70,000 (8)(6)	Approval of State Maintenance Engineer.
ARIZONA	78'	12'	14'	(1)	28,000	46,000	
ARKANSAS	75'	12'	13'6"	73,280	18,000	32,000	Required to obtain permit from Permit Officer, Little Rock.
CALIFORNIA	135'	12'(5)	16'	115,000(1)	28,000	48,000	
COLORADO	100'	14'	14'6"	100,000	22,000	45,000	
CONNECTICUT	80'	14'	14'	120,000	22,000	36,000 (2)	Dual tandem - 45,000.
DELAWARE	100'	12'	13'6"	120,000	20,000	36,000	With escort weekdays, daylight hrs., certification essential to national defense nights, weekends & holidays.
DISTRICT OF COLUMBIA	80'	12'	13'	90,000	25,000	50,000	
FLORIDA	75'	12'	14'	100,000	22,000	44,000	Checks, escort required with stipulated routing, single trip with nondivisible load, specified routing determined by permit officer.
GEORGIA	75'	12'	13'6"	100,000	25,000	50,000	Night & weekend certification. Each load & route analyzed.
HAWAII	(3)	(3)	(3)	(3)	(3)	(3)	
IDAHO	100'	12'6"	14'	110,000(1)	25,000(1)	50,000 (1)	Routing restricted. Certification required for excessive loads or continuous travel.
ILLINOIS	(3)	14'	(3)	(3)	(3)	(3)	
INDIANA	85'	12'	14'6"	108,000	28,000	48,000	Certification, private carrier exceeding 14' in width.
IOWA	120'	12'5"	(15)	90,000	18,540	34,000	Escorts, approval of Road & Bridge Design & Exec. Branch.
KANSAS	126'(3)	14'(3)	18'(3)	110,000(3)	22,500(3)	45,000 (3)	
KENTUCKY	80'	14'	15'	120,000	22,000	45,000	Secretary of Transportation approval only.
LOUISIANA	90'	12'	14'	106,000(6)	24,000	48,000	Only if made by government-owned vehicle. Certification required.
MAINE	90'	14'	14'	159,000(11)	24,200	46,000	Prior approval for excesses in axle weight.
MARYLAND	85'	14'	15'6"	110,000	30,000	30,000/axle	
MASSACHUSETTS	70'	12'	14'	100,000	22,400	36,000	Nonreducible loads only w/escort vehicle, and/or certification and/or additional restriction.
MICHIGAN	150'	14'	15'	none	60,000(4)	45,000	
MINNESOTA	85'	14'6"	14'	108,000	20,000	36,000	Each move considered separately. Special route checks.
MISSISSIPPI	55'	12'	13'6"	73,280	18,000	32,000	Each request considered on its own merit.
MISSOURI	65'	12'4"	13'6"	98,000	19,600	38,000	Extensive analysis of routing.
MONTANA	85'	15'	14'6"	110,000	22,500	48,000	
NEBRASKA	(16)	12'	14'6"	110,000	20,000	40,000 (14)	Permits could be issued for selected routes and times.
NEVADA	78'	11'	14'	110,000	22,000	40,000	
NEW HAMPSHIRE	90'	12'	13'6"	120,000	22,400	40,000	Beyond legal height-route analysis.
NEW JERSEY	(17)		(9)	(9)	(9)	(9)	
NEW MEXICO	78'	12'	14'	130,000	21,600	40,000	Military cargo exempt from regulations. Motor transportation to be notified.
NEW YORK	78'	12'	14'	110,000	22,400	40,000	
NORTH CAROLINA	80'	14'	14'(10)	120,000(1)	25,000	50,000	Special engineering study.
NORTH DAKOTA	100'	14'	15'	87,000(12)	20,000	38,000	None
OHIO	(13)	13'	(10)	120,000	29,000	46,000	(48,000 triple)
OKLAHOMA	80'	12'	13'6"	(18)	20,000	34,000	
OREGON	60'	12'	14'	96,000	21,500	43,000	Subject to bridge division check.
PENNSYLVANIA	85'	12'	14'	120,000	22,400	36,000	
PUERTO RICO	50'	8'	13'6"				Military owned, operated vehicles exempt.
RHODE ISLAND	78'	12'	14'	110,000	22,000	40,000	Special permit and safety procedure.
SOUTH CAROLINA	75'	12'	13'6"	120,000	20,000	40,000	(a) Under hardship cases.
SOUTH DAKOTA	85'	16'	14'	120,000	22,000	40,000	
TENNESSEE	85'	14'	14'6"	(1)	18,000	36,000	See footnote # 1.
TEXAS	95'	12'	14'	(1)	25,000(7)	45,000	Dependent upon route and hour of travel.
UTAH	78'	12'	14'	100,000	25,000	42,500	Prior authorization from DOT. Nondivisible loads only.
VERMONT	84'	10'6"	14'6"	120,000	23,520	36,000	Military certification necessary.
VIRGINIA	90'	12'	14'	115,000(1)	24,000	44,000	Lengths to 125' dependent upon routing & length of move.
WASHINGTON	100'	14'	14'	200,000(1)	22,000	43,000	None.
WEST VIRGINIA	78'	12'	14'	110,000	28,000	45,000	
WISCONSIN	99'	12'	13'6"	125,000	35,000	65,000	
WYOMING	90'	15'	15'	135,000(10)	22,500(10)	55,000 (10)	Nondivisible loads only. Height allowance to 16' on non-interstate/freeway routes.
GUAM	65'	8'6"	13'6"	76,800(1)	20,000	36,000	Load & route analysis & approval for overweight loads. Escorts for width of 14'.

(1) Dependent on number of axles and spacing.
(2) Two tires per axle.
(3) Engineering analysis required on all permits.
(4) Special earth moving equipment based on 850 lb/inch of tire width. Limited moves only.
(5) Moves over 50 miles.
(6) Based on four or more load carrying axles.
(7) On 850 lb/inch of tire width.
(8) Tire loading maximum 750 lb/inch of tire width.
(9) Restricted to 800 lb/inch of tire tread.
(10) Controlled by route.
(11) Seven axles.
(12) Five axles, seven axles to 147,000 lbs.
(13) 75 feet over 8 feet wide, 100 feet less 8 feet wide.
(14) 50,000 triple.
(15) No limit.
(16) Unlimited
(17) No regulation.
(18) Not specified.

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5.1.8 Cargo dimensional and weight limitations for highway transport. The following general cargo dimensional and weight limitations are for unrestricted highway transport:

5.1.8.1 Limitations based on closed-top vehicles in oversea areas. These limitations are for cargo to be placed in indigenous closed-top vehicles in oversea areas:

Length	9 ft 0 in. (2.743 m)
Width	5 ft 6 in. (1.676 m)
Height	5 ft 0 in. (1.524 m)
Weight	4,400 lb (1 996 kg)

5.1.8.2 Limitations based on a 2-1/2-ton (2 268-kg military truck). These cargo limitations are based upon a combination of the materials handling equipment, a 2-1/2-ton (2 268-kg) military cargo truck, and the physical limitations of worldwide highways:

Length	12 ft 3 in. (3.734 m)
Width	6 ft 8 in. (2.032 m)
Height	6 ft 7 in. (2.007 m)
Weight	5,000 lb (2 268 kg)

5.1.8.3 Limitations based on van-type vehicles in CONUS. These cargo limitations are based on the door openings and should be considered when material or equipment is designed to be transported in van-type vehicles within CONUS:

Length	18 ft 6 in. (5.639 m)
Width	7 ft 0 in. (2.134 m)
Height	6 ft 6 in. (1.981 m)
Weight	10,000 lb (4 536 kg)

5.1.9 Lifting and tiedown provisions. Consideration must be given to the design, location, and marking of lifting and tiedown points for equipment or materiel. Equipment weighing more than 150 pounds (68 kg) shall be provided with lifting and tiedown fittings. MIL-STD-209 provides guidance and criteria for these provisions. Restraint criteria must be met for all applicable modes of shipment.

5.1.10 Safety. For unrestricted movement on public highways, reference should be made to safety, lighting, brake, and stopping-distance specifications currently required by the Department of Transportation for commercial vehicles. The requirements of Titles 10, 39, and 49, Code of Federal Regulations, are applicable.

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5.1.11 Highway design standards. Highway design standards shall be considered in the development of new transport equipment. Standards provide guidance on structural and dimensional characteristics for components such as bridges and pavement. Highway standards prepared and published by AASHTO are in common use in CONUS and many foreign countries.

5.1.12 Bridge design loadings. AASHTO bridge specifications provide for two general types of bridge design loadings:

5.1.12.1 H loadings. The H loadings (figure 18) are based on the configuration of the two-axle truck. The first number following the H indicates the gross weight in tons of the standard truck, and the second number indicates the year of adoption of the loading designation.

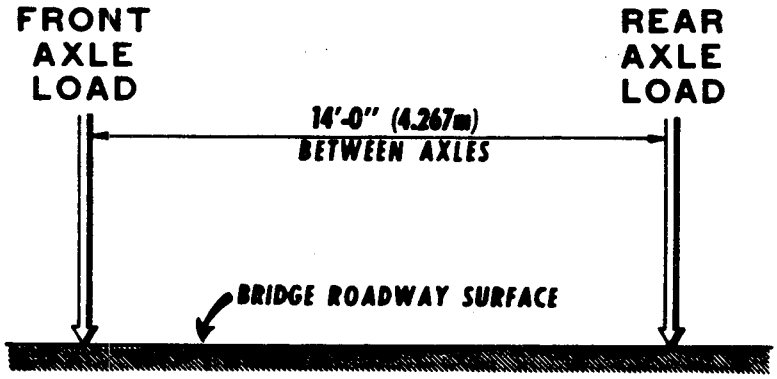
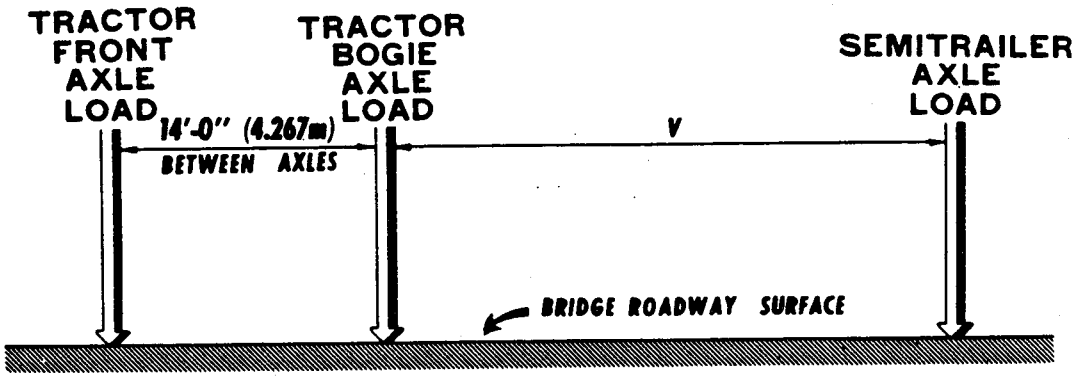


FIGURE 18. Standard H-design load.

5.1.12.2 HS loadings. The HS loadings (figure 19) are based on the configuration of a two-axle tractor truck and a one-axle semitrailer. The HS loadings are designated by the letters HS, followed by a number indicating the gross weight in tons of the simulated two-axle tractor truck, and the second number indicates the year of adoption of the loading designation. The variable axle spacing has been introduced so that the spacing of axles may approximate more closely the tractor-trailers now in use. A modification of the HS design load is used for short spans on the National System of Interstate and Defense Highways (NSIDH) bridges to better accommodate the heavier military-type loads. The H and HS loadings do not reflect the capabilities of the structure in terms of axle loads and are for design purposes only.

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8,000 LB (3629 KG)	32,000 LB (14515 KG)	H20-S16-44	32,000 LB (14515 KG)
6,000 LB (2722 KG)	24,000 LB (10886 KG)	H15-S12-44	24,000 LB (10886 KG)

V—Variable-spacing 14 feet (4.267m) to 30 feet (9.144m) inclusive. Spacing to be used is that which produces maximum stresses.

NOTE: Bridges supporting highways shall be designed in accordance with the current standard specifications for highway bridges of the American Association of State Highway Officials, using the H20-S16-(44) loading. However, to overcome deficiencies for systems of bridges designed for such loading, all bridges and floor systems with spans under 40 feet (12.192 m) shall be designed using the alternate limitations of 2 axles, 4 feet (1.219 m) apart, with each axle weighing 75 percent of the rear loading of the H20-S16-(44) loading.

FIGURE 19. Standard HS-design load.

5.1.12.3 Special vehicles. Relationships for a number of special vehicles have been computed. Ratings of a lowboy type of vehicle and a construction-equipment type of vehicle, tested on HS20-44 and HS15-44 bridges, as shown in figure 20, illustrate the criticality of loadings relative to span length and axle-load distribution.

5.1.13 Interstate system bridges.

5.1.13.1 Load design. Design loadings for all bridges carrying the main line traffic is AASHTO HS20-44.

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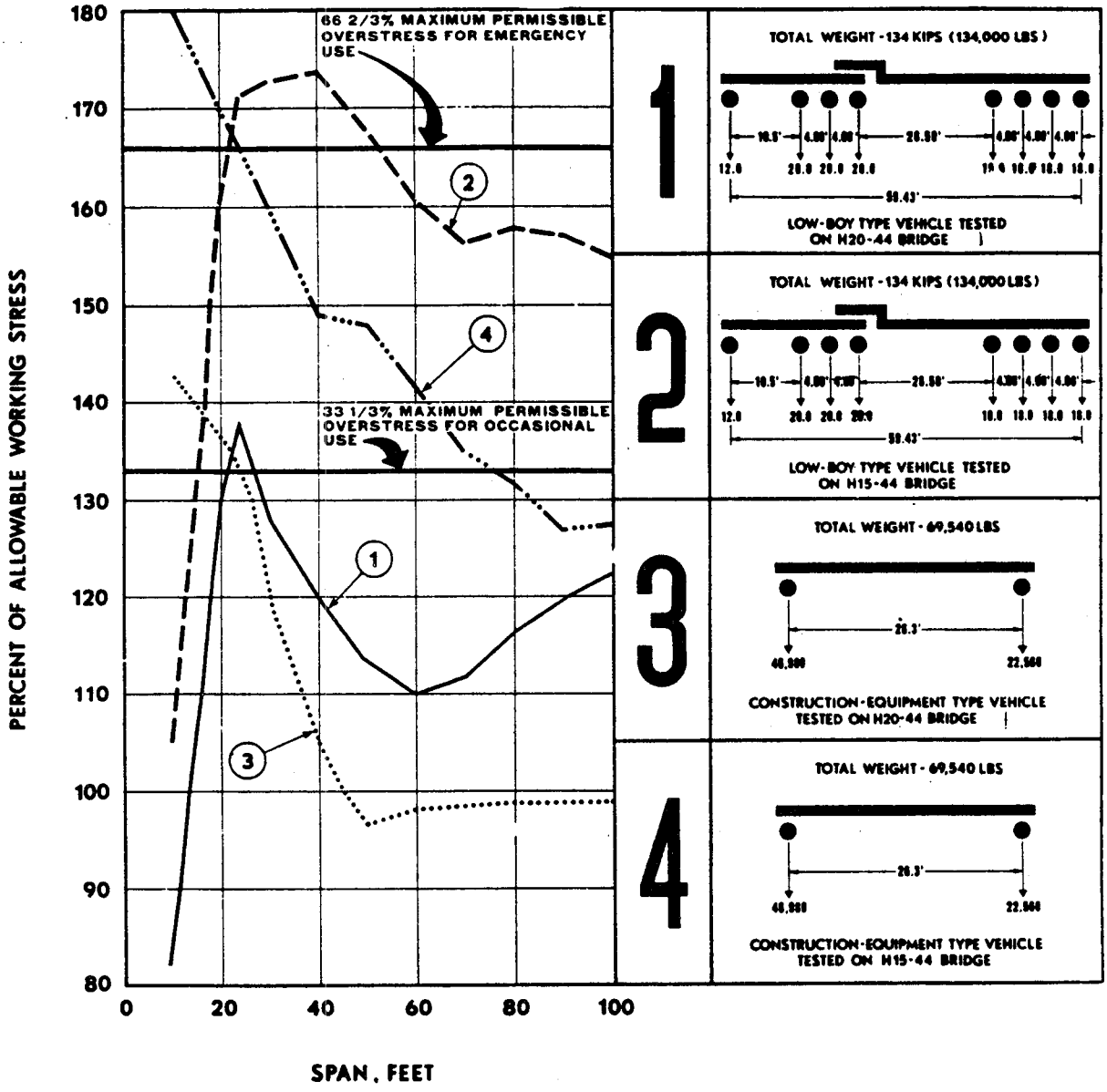


FIGURE 20. Ratings of a lowboy type of vehicle and a construction-equipment type of vehicle tested on HS20-44 and H15-44 bridges.

5.1.13.2 Lateral clearance.

a. Lateral or horizontal clearance limits normally occur at structures; however, on the interstate system, these limits may be encountered at ramps leading on

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and off the system. Ramps on interstate systems can be expected to provide 24-foot (7.315-m) clearances. In the primary highway system, some pavement is under 20 feet (6.096 m) wide, with shoulders ranging from the widths specified in paragraph 5.1.14.1 to no shoulder at all. In the secondary highway system, bridges with as little as an 11-foot (3.353-m) lateral clearance may be encountered.

b. Lateral bridge railings or the face of the parapet is offset at least 3 feet 6 inches (1.067 m) from the edge of the traffic lane. Bridge barrier curbs are offset at least 2 feet (0.610 m). Lateral clearance, from the edge of through traffic lanes to the face of walls or abutments and piers at underpasses, is the width of the usable shoulders; it is not less than 8 feet (2.438 m) on the right or less than 4 feet 6 inches (1.372 m) on the left.

5.1.13.3 Vertical clearance. Most structures on the principal highways of the United States have a vertical clearance of 14 feet (4.267 m); however, some bridges, underpasses, and tunnels with a vertical clearance of less than 14 feet (4.267 m) do exists. Some urban freeways are parts of systems or routes for which a minimum vertical clearance of 16 feet (4.877 m) has been established; however, this clearance may not be a reality in all cases. Interstate system standards call for such clearance for at least one route either passing through or around each urban area, but for other routes, the lower vertical clearance has been acceptable.

5.1.14 Interstate highway characteristics.

5.1.14.1 Lane and shoulder widths. Traffic lanes on all freeways within the United States generally are 12 feet (3.658 m) wide. Shoulder widths are important from the standpoint of transport operations. On principal arterial highways, these widths generally range from 4 feet (1.219 m) to 6 feet (1.829 m) for left shoulders and from 8 feet (2.438 m) to 10 feet (3.048 m) for right shoulders. On freeways with six or more lanes, left shoulder widths of 10 feet (3.048 m) and right shoulder widths up to 12 feet (3.658 m) can be found.

5.1.14.2 Intersection and ramp considerations. Design of intersections and interchange ramps is based on the characteristics of the largest type of commercial vehicle combinations. Use of these characteristics

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results in larger pavement-edge radii and pavement widths than would be required for passenger vehicles.

a. Present intersection design normally will accommodate large vehicle combinations, although it may be necessary to use the entire roadway width and to predetermine the placement of the vehicle. Older designs may not be adequate to operate these vehicles without extensive maneuvering, with the subsequent loss of capacity. Turning radii at 90-degree (1.57-rad) intersections may be expected to equal 25 feet (7.620 m) in each direction.

b. While 90-degree (1.57-rad) turns are a normal requirement at intersections, changes in direction of as much as 270 degrees (4.71 rad) at interchanges are often required. Ramps are normally designed for speeds of 25 miles per hour (40.234 km/h) or greater.

5.1.15 Vehicle turning characteristics. Because of space limitations on many roadway intersections within CONUS, transporters should be designed with sufficient turning capability to negotiate turns without undue danger or inconvenience to other highway users. Turning characteristics of transporters should be determined as early as possible during design. Upon request, MTMCTEA will prepare tracking diagrams (figures 21 and 22) for vehicles to illustrate vehicle turning characteristics so that an analysis of potential turning difficulties can be made.

5.1.16 Overseas size and weight limits of motor vehicles. Size and weight limitations for the movement of motor vehicles in various countries are provided in tables III through VI, through the courtesy of the International Road Federation (IFR). Figure 23 shows coded silhouettes of basic commercial vehicle types in regular operation as used in tables III through VI.

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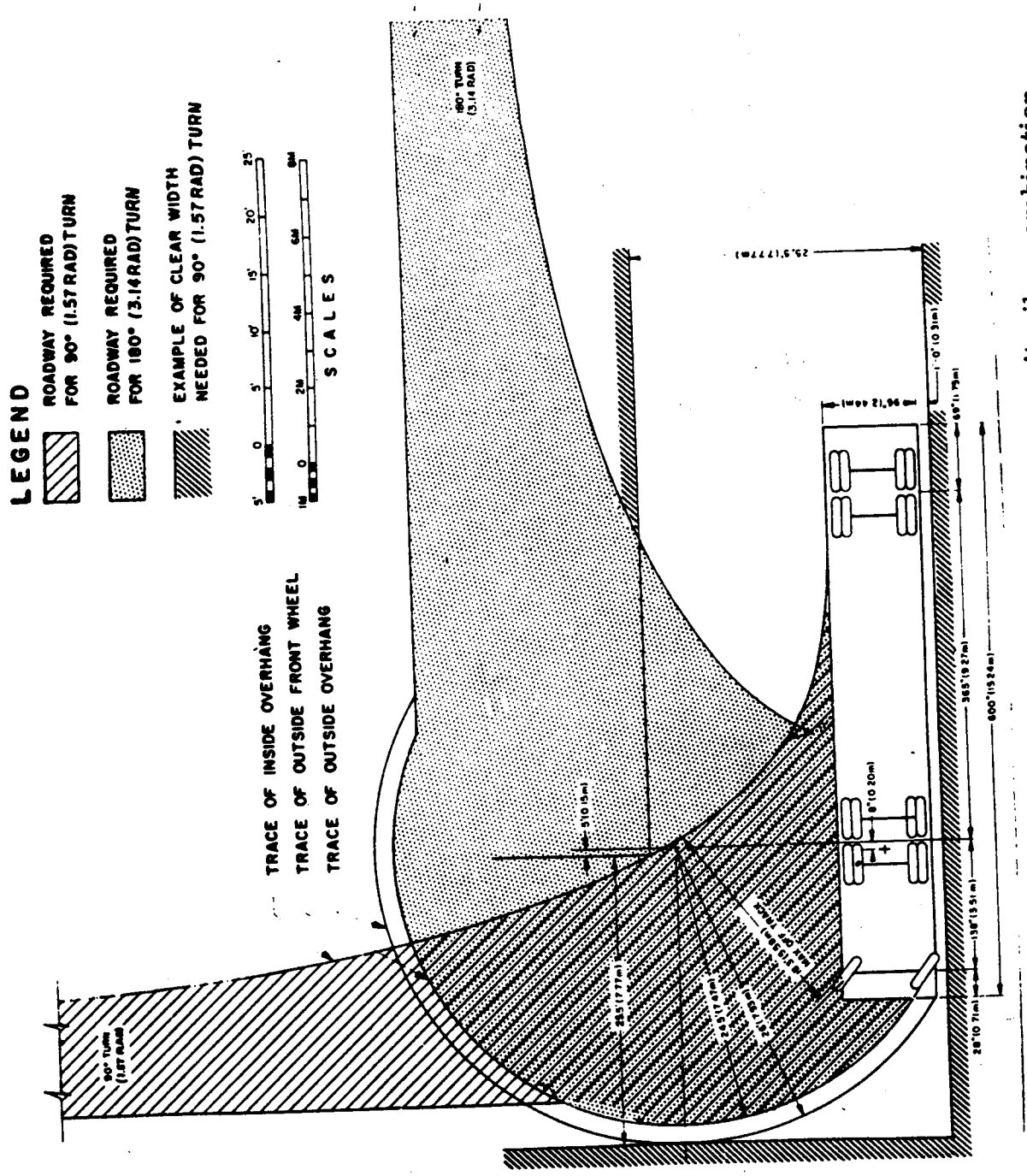


FIGURE 21. Turning diagram for a tractor-semitrailer combination.

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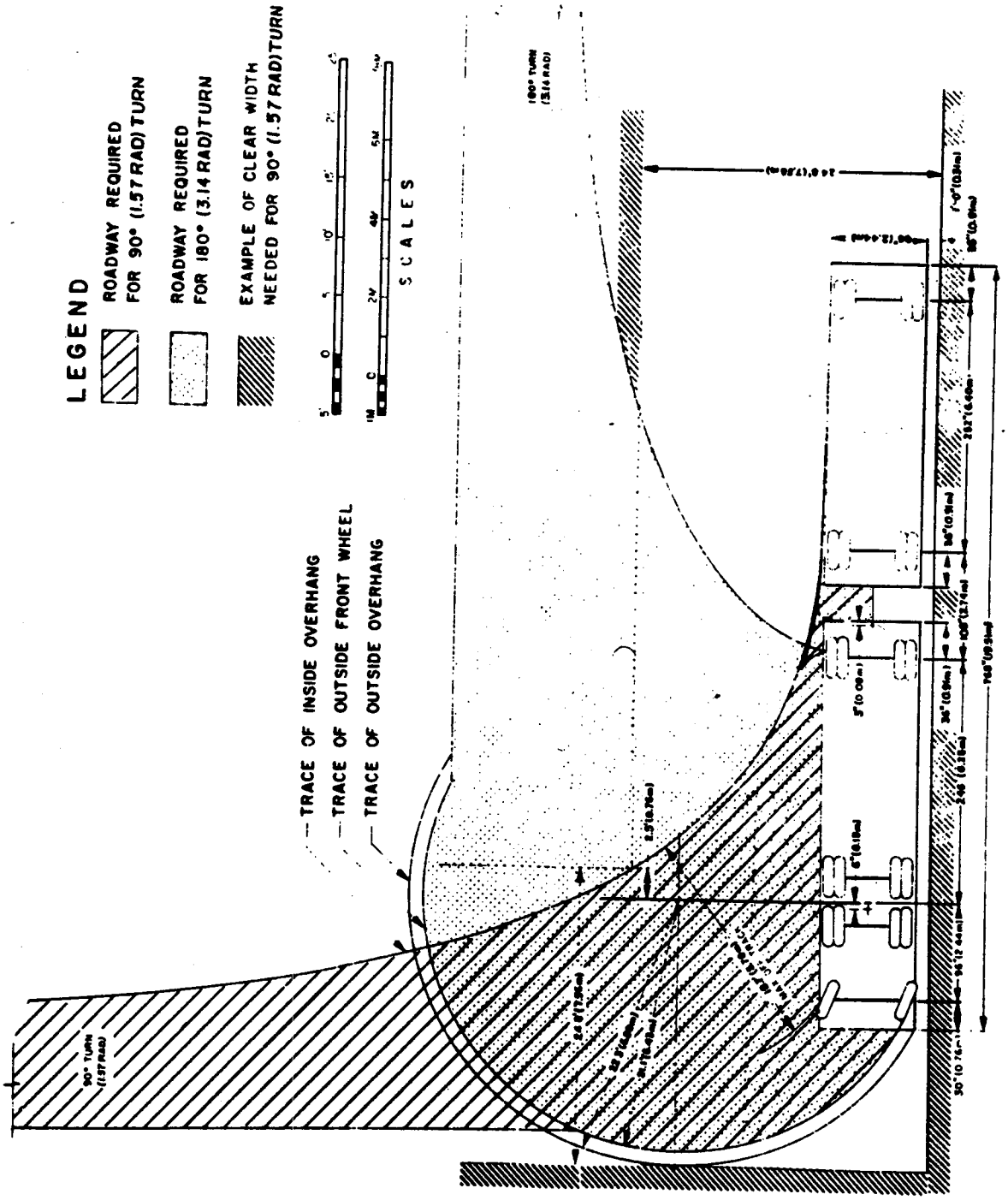


FIGURE 22. Turning diagram for tractor-semitrailer-trailer combination.

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TABLE III. LIMITS OF MOTOR VEHICLE SIZES AND WEIGHTS - AFRICA.

COUNTRY	WIDTH	HEIGHT	LENGTH				AXLE LOAD		MAXIMUM GROSS WEIGHT (1)									
			SINGLE UNIT		TRUCK SEMI-TRAILER	OTHER COMBINATIONS	SINGLE	TANDEM	2	3	2-S1	2-S2	3-S2	3-S2	2-2	2-3	3-2	3-3
			TRUCK	BUS														
	meters	meters	meters	meters	meters	meters	metric tons	metric tons	metric tons	metric tons	metric tons	metric tons	metric tons	metric tons	metric tons	metric tons	metric tons	metric tons
Algeria	2,5	–	11,0	12,0	15,0	18,0	13,0	(2)	19,0	26,0	35,0	35,0	35,0	35,0	35,0	35,0	35,0	35,0
Angola	2,5	4,0	12,0	12,0	15,0	18,0	10,0	16,0	16,0	22,0	26,0	32,0	32,0	38,0	32,0	38,0	38,0	38,0
Benin (3) (4) (5)	2,5	–	11,0	12,0	15,0	18,0	9,5 11,5(4)				35,0	35,0	35,0	35,0	35,0	35,0	35,0	35,0
Botswana	2,5	4,1	12,5	12,5	17,0	20,0	8,2	18,0	Dependent on Axle Spacing									
Cameroon (3)	2,5	4,0	11,0	11,0	15,0	20,0	13,0	14,7 21,0(15)	19,0	26,0	35,0	35,0	35,0	35,0	35,0	35,0	35,0	35,0
Central African Empire	2,5	–	11,0	12,0	14,5	18,0	13,0	28,7	16,0	22,0	38,0	38,0	38,0	38 0	38,0	38,0	38,0	38,0
Chad (19) (20)	2,5	–	11,0 12,0(6)	12,0	14,0	18,0(21)	10,0	20,0	16,0	22,0	30,0	35,0	35,0	35,0	35,0	35,0	35,0	35,0
Egypt (31)	2,6	3,5	12,0	12,0	15,0	20,0	10,0	16,0										
Ethiopia	2,4	3,8	11,0	11,0	14,0	18,0	8,0		Dependent upon Axle Spacing									
French Territory of Afars and Issas (9)																		
Gabon	2,5	4,8	11,0	12,0	15,0	20,0	10,0	16,0	16,0	22,0	26,0	32,0	32,0	38,0	36,0	38,0	38,0	38,0
Gambia	2,3	–	8,4	8,4			(14)	(14)	Dependent on number of wheels									
Ghana	2,5	3,4	11,0	11,0	13,0	13,0	10,2	–	16,3	24,4	24,4	32,5	32,5	32,5	32,5	32,5	32,5	32,5
Ivory Coast	2,5		11,0	12,0	15,0	18,0	10,0	17,0	16,0	23,0	25,0	32,0	32,0	41,0	41,0	41,0	41,0	41,0
Kenya	2,5	3,8	10,0 11,0(6)	11,0	15,0	18,0	8,0(7)	11,3(7)	22,5(7)	22,5(7)	22,5	38,0	38,0	38,0	38,0	38,0	38,0	43,0
Lesotho	2,5	3,8	10,0 11,0(6)	11,0	15,2	21,9	8,1	–	Dependent on Axle Spacing									
Liberia (3)	2,5	4,0	11,0	11,0	14,5	18,0	10,0	15,0	15,0	23,0	25,0	32,0	32,0	40,0				
Libya	2,5	4,0	12,0	12,0	15,0	18,0	10,0	16,0	16,0	22,0	26,0	36,0	36,0	38,0	38,0	38,0	38,0	38,0
Malagasy (19)	2,5	4,0	12,0	12,0	15,0	18,0	10,0	16,0	16,0	22,0	26,0	32,0	36,0	36,0	36,0	38,0	38,0	38,0
Malawi (3)	2,5	4,3(13)	11,0	11,0	15,2	18,3	8,0	16,0	16,0	24,0	24,0	32,0	32,0	40,0	32,0	40,0	40,0	40,0
Mali	2,5	4,0	11,0	11,0	15,0	18,0	11,0	16,0	16,0	23,0	25,0	38,0	41,0	41,0	35,0	38,0	38,0	38,0
Mauritius (23) (24)	2,7	4,0	10,0	12,0	10,0	14,0	10,0	10,0	15,0	15,0	15,0		15,0					
Morocco (20)	2,5	4,0	11,0	12,0	15,0	18,0	13,0	–	19,0	26,0	35,0	35,0	35,0	35,0	35,0	35,0	35,0	35,0
Mozambique	2,5	4,0	12,0	12,0	15,0	18,0	10,0	16,0	16,0	22,0	26,0	32,0	32,0	38,0	32,0	38,0	38,0	38,0
Niger	2,5		11,0		14,0	18,0 22,0(18)	11,5	(8)	17,0	22,0	35,0	35,0	35,0	35,0	35,0	35,0	35,0	35,0

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TABLE III. LIMITS OF MOTOR VEHICLE SIZES AND WEIGHTS - AFRICA. - Continued.

COUNTRY	WIDTH	HEIGHT	LENGTH				AXLE LOAD		MAXIMUM GROSS WEIGHT (1)									
			SINGLE UNIT		TRUCK SEMI-TRAILER	OTHER COMBI-NATIONS	SINGLE	TANDEM	2	3	2-S1	2-S2	3-S1	3-S2	2-2	2-3	3-2	3-3
			TRUCK	BUS														
	meters	meters	meters	meters	meters	meters	metric tons	metric tons	metric tons	metric tons	metric tons	metric tons	metric tons	metric tons	metric tons	metric tons	metric tons	metric tons
Nigeria	2,5	4,0	10,5	10,5	14,0	18,5	10,0	16,0	Not to exceed 32,0 tons									
Reunion (9)							(25)											
Rhodesia (3)	2,5	4,6	12,5	12,5	17,0	22,0	8,2	16,0	Dependent on Axle Spacing									
Rwanda	2,5	4,0	12,0	12,0	14,0	22,0	–	–	12,0	18,0	20,0	24,0	24,0	28,0	32,0	32,0	32,0	32,0
Senegal	2,5	–	11,0 12,0(6)	12,0	18,0	22,0	10,0	(26)	16,0	22,0	32,0	32,0	32,0	32,0	32,0	32,0	32,0	32,0
Seychelles Is. (10)	2,3	3,0	7,0	7,0	7,0	(10)	–	–	10,0	10,0	10,0	10,0	10,0	10,0	10,0	10,0	10,0	10,0
Sierra Leone	2,9	3,3	9,1	10,04	13,56	19,29	5,92(11)	25,4	13,0	18,0	30,48	30,48	35,56	50,0	35,56	35,56	40,64	35,56
Somalia (3)	2,5	4,5	10,0 11,0(6)	11,0	14,0	18,0	10,0	16,0	14,0	18,0	18,0	28,0	28,0	32,0	28,0	32,0	32,0	36,0
South Africa	2,5	4,1	12,5	12,5	17,0	20,0	8,2(12)	16,4(12)	15,0	24,0	31,0	31,0	31,0	39,0	31,0	39,0	39,0	49,0
Swaziland (27)																		
Tanzania	2,5	4,4	10,0 11,0(6)	11,0	14,0	18,0	8,0	14,5	13,0	19,5	21,0	27,5	27,5	34,0	29,0	35,0	35,0	35,0
Tunisia	2,5	4,0	11,0	12,0	15,0	18,0	11,0	(22)	19,0	26,0	35,0	40,0	40,0	48,0	45,0	52,0	52,0	54,0
Uganda (7) (29)	2,5	4,0	11,0	11,0	15,0	18,0	8,0(30)	14,5	13,0	18,0	18,0	27,5	27,5	32,0	27,5	35,0	35,0	35,0
Upper Volta	2,5	–	11,0	12,0	15,0	18,0	11,0	(22)	18,0	25,0	28,0	35,0	35,0	40,0	35,0	40,0	40,0	40,0
Zaire (19)	2,5	4,0	12,0	12,0	14,0	22,0	8,0	12,0	12,0	18,0	20,0	24,0	24,0	28,0	32,0	32,0	32,0	32,0
Zambia (18)	2,5	3,8	11,0	11,0	15,5	22,0	(16)	(17)	Dependent on Number of Wheels and Axle Spacing									

1. See vehicle type nomenclature, page 8.
2. The maximum weight for tandem axles is obtained from the formula $t = 7,35 + 0,35\left(\frac{d - 90}{5}\right)$, where 90 cm < d < 135 cm.
3. Latest available report, 1974.
4. Maximum axle load limits vary in accordance with different roads of the network.
5. Vehicle maximum gross weight figures are applicable to specific road sections.
6. Three axle vehicles have the highest limit.
7. Subject to other regulations related to axle distances and number of wheels.
8. If d = distance between axles,
   for: d>1,5 m, the maximum load = 9,5 tons.
   for: d<1,5 m, the maximum load = 6,5
   $$t = 6,5 + 0,25\ \frac{d - 0,9}{0,05}$$
9. Follows the legislation of France.
10. Trailers may be used only with special permission of the Commission of Police.
11. Special permits are issued by the Ministry of Works for overweight vehicles. The limits in practice is 9 tons gross weight.
12. For dual tire arrangement.
13. Two-deck buses are permitted 4,57 m height.
14. No more than 3/4 of the gross weight of a motor vehicle shall be transmitted on the road surface by any two wheels.
15. 14,7 tons for tandem axles separated 0,90 m. For tandem axles separated 1,35 m or more: 21,0 tons.
16. Two-tire axle maximum load: 8,0 tons.
    Four-tire axle maximum load: 10,0 tons.
17. Single-tire arrangement; maximum axle load: 14,5 tons.
    Dual-tire arrangement; maximum axle load: 16,33 tons.
18. Maximum gross vehicle weight up to 45 tons is allowed on paved roads provided the maximum axle loads are not exceeded. Vehicles on unpaved roads require permission for over 6,5 tons gross weight. Granting of permit dependent on strength of roads and structures.
19. Latest available report: 1971.
20. All self-propelled vehicles or trailers must not exceed 5 tons per meter of length measured between the first and the last axle.
21. 20 meters for vehicle types 3-2 and 3-3.
22. 13,0 tons; for tandem axles separated 0,9 m. For tandem axles separated 1,35 m or more: 19,0 tons.
23. Double deck bus, 5.0 m height.
24. 25 tons for trucks, restricted use.
25. 10 tons on some routes.
26. $t = 7 + 0.25\left(\frac{d - 90}{5}\right)$ where 90 cm < d < 150 cm.
27. Identical with South Africa.
28. Tractor and 2 trailers, not to exceed 22.0 m.
29. Special permits for special sizes and widths
30. Maximum wheel load 2.5 tons.
31. Truck, semi-trailers and truck and trailer combinations with load capacity greater than 13.0 tons need specific permission from Ministry of Transport. Also special permits for axle loads exceeding noted limits.

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TABLE IV. LIMITS OF MOTOR VEHICLE SIZES AND WEIGHTS - ASIA, MIDDLE EAST, AND OCEANIA.

COUNTRY	WIDTH	HEIGHT	LENGTH				AXLE LOAD		MAXIMUM GROSS WEIGHT (1)									
			SINGLE UNIT		TRUCK SEMI-TRAILER	OTHER COMBI-NATIONS	SINGLE	TANDEM	2	3	2-S1	2-S2	3-S1	3-S2	2-2	2-3	3-2	3-3
			TRUCK	BUS														
	meters	meters	meters	meters	meters	meters	metric tons	metric tons	metric tons	metric tons	metric tons	metric tons	metric tons	metric tons	metric tons	metric tons	metric tons	metric tons
Afghanistan	2,5	4,0	10,0	11,0	14,0	18,0	13,0	18,0	18,0	24,0	30,0	34,0	36,0	36,0	36,0	36,0	36,0	36,0
China (Taiwan)(27)	2,5	3,8	11,0	12,0	18,0	–	8,0	14,5	12,0	21,0	25,0				42,0			
Hong Kong (3) (33)	2,5	3,0(2)	11,0	10,0	17,0		10,0	11,0	Dependent on axle spacing and tire size									
India	2,4	3,4(4)	8,5(5)	9,5	15,3	23,0	10,0		15,0		22,2							
Iran (27) (28)	2,5	4,0	10,0	11,0	14,0	18,0	13,0	20,0	19,0	26,0	–	34,0	34,0	36,0				
Iraq (27)	2,6	3,8	10,7 12,2(6)	10,7	15,2	20,0	11,0	17,0	16,0	22,0	27,0	33,0	33,0	35,7	38,0	44,0	44,0	50,0
Israel	2,5	4,0	10,0 12,0(6)	11,0	15,0(7)	18,0(7)	12,0	17,5	16,0	22,5	30,0	30,0	30,0	30,0	38,0	38,0	38,0	38,0
Japan (8)	2,5	3,8	12,0	12,0	12,0(9)	12,0	10,0		Total weight not to exceed 20 tons (10)									
Jordan (33)	2,5	3,8	11,0	11,0	14,0	22,0	12,0	–	19,0	24,0	30,0	35,0	35,0	35,0	40,0	40,0	40,0	40,0
Korea (8)	2,5	3,5	12,0	12,0	12,0	12,0	10,0	–	Total weight not to exceed 20 tons									
Laos (27)	2,5		11,0		14,0		7,4		18,0	25,0	30,0	30,0	30,0	30,0	30,0			
Lebanon (11) (33)	2,5	3,8	11,0	12,0	14,0	18,0	14,0	14,7(11)	19,0	26,0	35,0	35,0	35,0	35,0	35,0	35,0	35,0	35,0
Malaysia (12) (13)	2,3	3,2(2)	9,1	9,1	11,2	12,2(14)	8,1	12,2	18,3	18,3	18,3	18,3	18,3	18,3	18,3	18,3	18,3	18,3
Philippines (27)	2,5	4,0	10,0	11,0	11,0	14,0	8,0	14,5	16,0	22,5	24,0	30,5	30,5	37,0	32,0	38,5	38,5	45,0
Saudi Arabia (15) (16) (33)	2,5	4,0	12,0	12,0	17,0	20,0	10,0	16,0	14,0	20,0	24,0	30,0	24,0	36,0	34,0	34,0	40,0	40,0
Singapore	2,5	3,2	10,0	11,9	11,0	–	8,0	16,0	16,0	24,0	24,0	24,0	24,0	24,0	24,0	24,0	24,0	24,0
Sri Lanka (30)	2,3	3,6 4,6 (29)	8,2		10,0													
Syria (27)	2,5	3,8	10,0 11,0(6)	11,0	14,0	18,0	12,0		18,0	22,0	24,0	26,0	26,0	35,0	26,0	–	30,0	
Thailand (17) (33)	2,5	(18)	10,0	10,0	14,0	20,0(14)	8,0	14,4	14,0	20,4	22,0	28,4	28,4	32,4	(19)	(19)	(19)	(19)
Australia (20)	2,5	4,0 4,4	9,5 20,1	10,7 20,1	13,7 20,1	15,2 45,0	8,1 9,0	9,2 16,4	Dependent upon axle spacing and other regulations									
Fiji Islands	2,4(21)	3,4	10,1	10,1	10,1	10,1	8,0	14,5	16,0	22,5	23,9	30,5	30,5	30,8	30,8	30,8	30,8	30,8
New Caledonia	2,5		11,0	12,0	15,0	18,0	13,0	(31)	19,0	26,0	38,0	38,0	38,0	38,0	38,0	38,0	38,0	38,0
New Zealand	2,5(22)	4,3	12,3(23)	12,3(23)	(24)	19,0(25)	8,2(26)	15,5(26)	Dependent upon axle spacing and road conditions									
Papua New Guinea	2,5	4,3	9,5	10,0(32)	13,7	15,3	8,2	16,2	12,0		Dependent upon axle spacing and other regulations							

1. See vehicle type nomenclature, page 8.
2. 4,57 m for double-deck buses.
3. Except for the height in buses, the Commissioner for Transport may authorize an increase in all other limits.
4. For double-deck buses, this limit is 4,75 m.
5. For rigid frame vehicles with more than two axles, this limit is 9,5 m.
6. Three axle vehicles have the highest limit.
7. Each component vehicle must not exceed 12,0 m of length.
8. To exceed these limitations requires approval of the Ministry of Transport (Korea), of the Road Administrator (Japan)
9. This limit is 16,5 m for vehicles operating in the National Motorway System.
10. Special vehicles up to 34,0 tons in total weight can be operated in the National Motorway System.
    Special vehicles up to 27,0 tons in total weight can be operated in other roads of the National System.
11. Each component of the tandem axle is allowed an increase of 0,35 tons for every 5 cms over 0,80 m up to a maximum of 14 tons, provided that the maximum gross weight of the vehicle is not exceeded.
12. Maximum axle loads and maximum gross weight of vehicles vary on different routes of the highway network.
13. Maximum gross weight regulations exclude special purpose vehicles and trailers for indivisible loads.
14. Not including length of draw-bar.
15. There is no legislation.
16. Maximum gross weight limitations are based on the maximum design loads for the bridges of the country.
17. The maximum gross vehicle weight limitations are the ones permitted only on primary and secondary roads.
18. No regulations.
19. Requires permission from the Department of Highways.
20. Regulations vary by States. Figures indicate range of variation in maximum limits.
21. Vehicles carrying sugar cane only 3,05 m.
22. Agricultural vehicles may be up to 3,7 m subject to travel conditions.
23. Distance ahead of rear axle limited to 8,3 m; rear overhang limited to 4 m.
24. Limits based on distance ahead of rear axle to the "turntable".
25. Subject also to the restriction in Note (24) and ability to drive around inside a circle of 25 m diameter.
26. Proposed limits to become effective in mid 1974 for twin tired axles. Lower limits apply on many roads.
27. Latest available report: 1971.
28. Current legislation provides for automatic yearly reductions in vehicle weights only, up to the year 1975
29. Double deck buses 4.6 m.
30. Weights are authorized by annual permits and licenses.
31. See note 7, page 6.
32. 11.3 m for approved buses.
33. Latest available report, 1974.

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TABLE V. LIMITS OF MOTOR VEHICLE SIZES AND WEIGHTS - EUROPE.

COUNTRY	WIDTH	HEIGHT	LENGTH				AXLE LOAD		MAXIMUM GROSS WEIGHT (1)									
			SINGLE UNIT		TRUCK SEMI-TRAILER	OTHER COMBINATIONS	SINGLE	TANDEM	2	3	2-S1	2-S2	3-S1	3-S2	2-S	2-S	3-S	3-S
			TRUCK	BUS														
	meters	meters	meters	meters	meters	meters	metric tons	metric tons	metric tons	metric tons	metric tons	metric tons	metric tons	metric tons	metric tons	metric tons	metric tons	metric tons
Austria (5)	2,5	4,0	12,0	12,0	15,0	18,0	10,0	16,0	16,0	22,0	–	38,0	38,0	38,0	38,0	38,0	38,0	38,0
Belgium	2,5	4,0	11,0	12,0	15,5	18,0	13,0	20,0	19,0	26,0	38,5	38,5	38,5	38,5	38,0	40,0	40,0	40,0
Bulgaria	2,5	4,0	12,0	12,0	16,5	22,0	10,0	–	16,0	26,0	35,0	35,0	35,0	35,0	40,0	40,0	40,0	40,0
Cyprus (3)	2,4	3,4	9,1	9,1	14,0	14,0	5,0	12,0	12,0	12,0	12,0	12,0	12,0	12,0	NP(2)	NP(2)	NP(2)	NP(2)
Czechoslovakia	2,5	4,0	12,0	18,0	15,5	18,0	10,0	11,5 to 16,0	16,0		20,0		24,0	32,0	Not to exceed 48,0			
Denmark (4)	2,5	3,6	10,0 12,0(6)	12,0	14,0	18,0	8,0	14,5	16,0	22,0	24,0	32,0	32,0	32,0	32,0	32,0	32,0	32,0
Finland	2,5	4,0	12,0	12,0	16,0	22,0	10,0	16,0	Depends on distance between outer axles (G = 20 + 1.60 (a - 2.5) Not to exceed 42.0 tons									
France	2,5	–	11,0	12,0	15,0	18,0	13,0	(7)	19,0	26,0	38,0	38,0	38,0	38,0	38,0	38,0	38,0	38,0
Germany	2,5	4,0	12,0	12,0	15,0	18,0	10,0	16,0	16,0	22,0	26,0	36,0	32,0	36,0	36,0	36,0	36,0	36,0
Gibraltar	2,1	3,7	6,1	6,1	6,7	6,7	9,1	11,2	14,2	20,3	20,3	20,3	20,3	20,3	20,3	20,3	20,3	20,3
Great Britain	2,5	4,6(8)	11,0	12,0	15,0	18,0	(9)	(9)	(9)	22,4	(9)	(9)	(9)	(9)	32,5	32,5	32,5	32,5
Greece	2,5	4,0	12,0	12,0	15,0	18,0	10,0 13,0(21)	18,0	19,0	26,0	29,0	38,0	Except for oil tankers to Yugoslavia border, 40 tons.					
Hungary	2,5	4,0	12,0	12,0	16,0	18,0 22,0	10,0	16,0	20,0	24,0	28,0	32,0	32,0	38,0	32,0	38,0	38,0	38,0
Iceland	2,4(16)	3,8	12,0	12,0	15,0	18,0	6,0(17)	8,0(17)	See Note No. 18									
Ireland	2,5	4,6(8)	11,0	12,0	14,9	16,5	10,2	16,3	16,3	22,4	22,4	25,4	25,4	25,4	32,5	32,5	32,5	32,5
Italy	2,5	4,0	12,0	12,0	15,5	18,0	12,0	19,0	18,0	24,0	30,0	40,0	36,0	44,0	40,0	43,2	44,0	44,0
Luxembourg	2,5	4,0	10,0 12,0(6)	12,0	15,5	18,0	13,0	19,0	19,0	26,0	38,0	38,0	38,0	38,0	38,0	40,0	40,0	40,0
Malta (15)	2,5	3,2	8,7 9,1(6)															
Monaco	2,5	4,0	11,0	12,0	15,0	18,0	13,0	14,0	19,0	26,0	38,0	38,0	38,0	38,0	38,0	38,0	38,0	38,0
Netherlands	2,5	4,0	11,0	12,0	15,0	18,0	8,0	16,0	16,0	24,0	24,0	32,0	32,0	40,0	32,0	40,0	40,0	48,0
Northern Ireland	2,5	4,6(8)	11,0	12,0	15,0	18,0	(9)	(9)	(9)	(9)	(9)	(9)	(9)	(9)	32,5	32,5	32,5	32,5
Norway (14)	2,5	–	12,4	12,4	15,5	18,0	6,0 10,0	9,0 16,0	Dependent solely on axle loading									
Poland	2,5	4,0	11,0	12,0	15,0	18,0	8,0(19)	14,5(19)	16,0	24,0	32,0	32,0	32,0	38,0	32,0	40,0	40,0	40,0
Portugal	2,5	4,0(13)	12,0	12,0	15,0(11)	18,0	10,0	18,0	16,0	22,0	26,0	32,0	32,0	38,0	32,0	38,0	38,0	38,0
Romania (22)	2,5	4,0	12,0	12,0	15,0	22,0	10,0 7,5	16,0 12,0	16,0 12,0	22,0 16,0	26,0 19,5	32,0 21,0	32,0 21,0	38,0 27,0	32,0 21,0	38,0 27,0	38,0 27,0	38,0 27,0
Spain	2,5	4,0	11,0 12,0(6)	12,0	16,5	18,0	13,0	14,7	20,0(7)	26,0(12)	38,0(12)	38,0(12)	38,0(12)	38,0(12)	38,0	38,0	38,0	38,0
Sweden	2,5	–	–	–	–	24,0	10,0	16,0	16,5	22,5	26,5	32,5	32,5	38,5	36,5	42,5	42,5	42,5
Switzerland	2,3(13)	4,0	10,0	12,0	14,0	18,0	10,0	14,0	16,0	19,0	26,0	26,0	28,0	28,0	26,0	26,0	28,0	28,0
Turkey	2,5	4,0	10,0 11,0(6)	11,0	15,0	18,0	8,0	14,5	Dependent upon axle spacing									
Yugoslavia(15)(20)	2,5	4,0	11,0 12,0(6)		15,0	18,0	10,0	16,0										

1. See vehicle type nomenclature, page 8.
2. NP = Not permitted
3. Special permission may be granted for vehicles exceeding these limits.
4. There are special regulations for commercial vehicles engaged in international road transport of goods
5. Latest available report, 1974.
6. Three axle vehicles have the highest limit.
7. Maximum weight for a tandem axle is 7,35 tons for the most heavily loaded axle, where the distance in the two axles is 0,9 m. The weight increases by 350 kg per 0,05 m of distance beyond 0,9 m, but must not exceed 10,5 tons for a distance of 1,35 m or more between the axles.
8. Only for buses.
9. Dependent on axle spacing and other conditions.
10. Urban buses may exceed 4.0 m.
11. Truck semitrailer is allowed to have 15,5 m when used for transportation of containers.
12. Provided it does not exceed 5.0 tons per meter of length measured between the first and the last axle.
13. On some main roads: 2,50 m.
14. Exceptions may be made, allowing vehicles to have a maximum weight of 10,0 tons on single axle and 16,0 tons on tandem axles (more than 2,5 m apart) and/or a total length of 22,0 m, by means of a special permit valid for certain main roads. Tandem axles are regarded as a single axle if the component axles are less than 2,50 m apart.
15. Latest available report 1971.
16. A maximum width of 2,50 m is allowed on 10% of the state roads and 2,45 m on 65% of the state roads.
17. A maximum of 10,0 tons for single axle and 16,0 tons for tandem axle is allowed on 65% of the state roads.
18. A maximum gross weight of 38,0 tons is allowed on 65% of the state roads and 36,0 tons on the remaining 35%.
19. A maximum weight for single axle of 10 tons and per tandem axle 16 tons is allowed in certain roads.
20. The maximum gross weight for all types and categories of vehicles, with or without trailer, is limited only by the maximum axle loads permitted.
21. 10 tons, no power, 13 tons with power.
22. Romania permitted weight depends on road quality.

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TABLE VI. LIMITS OF MOTOR VEHICLE SIZES AND WEIGHTS - NORTH, CENTRAL, AND SOUTH AMERICA.

COUNTRY	WIDTH	HEIGHT	LENGTH				AXLE LOAD		MAXIMUM GROSS WEIGHT (1)									
			SINGLE UNIT		TRUCK SEMI-TRAILER	OTHER COMBINATIONS	SINGLE	TANDEM	2	3	2-S1	2-S2	3-S1	3-S2	2-2	2-3	3-2	3-3
			TRUCK	BUS														
	meters	meters	meters	meters	meters	meters	metric tons	metric tons	metric tons	metric tons	metric tons	metric tons	metric tons	metric tons	metric tons	metric tons	metric tons	metric tons
Argentina	2,5	(2)	11,0	12,0	18,5	16,5 20,5(3)	10,6	18,0	Dependent on Axle Spacing									
Bahamas	2,4	3,7	9,1	9,1	9,1	9,1	(4)	(4)	No Regulations									
Barbados	2,3	(4)	7,6	9,1	7,6		(4)				11,2		11,2					
Bermuda	2,3	(2)	6,6	9,3	11,9	(2)	9,0	(4)	No Regulations									
Bolivia (6)	2,5	3,8	10,0		15,3		8,0	12,0	No Regulations									
Brazil (6)	2,6	4,0	12,0	12,0	16,0	18,0	10,0	17,0	15,0	22,0	25,0	32,0	32,0	39,0	35,0	40,0	40,0	40,0
Canada (5)	2,6	4,1 4,4	10,7 12,2	10,7 12,2	18,3 22,3		8,2 10,0	14,5 18,1	12,7 18,6	19,0 25,8	20,9 25,4	27,2 32,7	27,2 32,7	33,6 41,7	29,0 35,4	33,6 42,6	33,6 50,8	32,7 49,9
Chile	2,5	3,8	10,0	11,0	14,0	18,0 22,0	6,0 12,0	21,0	Dependent on Axle Spacing									
Colombia (6)	2,5	4,0	12,5	10,7 12,2(7)	16,8	20,0	8,2	14,5	Dependent on tire size and axle spacing									
Costa Rica	2,5	3,95	10,0 11,0(7)	12,0	15,5	18,3	9,0	16,5	13,8	23,0	23,0	30,5	30,5	37,4	32,2	–	37,4	–
Dominican Republic	2,5	4,0	10,0 11,0(7)	11,0	14,0 15,3(8)		9,0(9)		Dependent on tire size and axle spacing									
Ecuador (6) (16)	2,6	3,8	10,7		13,7	18,3	11,0	19,0	16,5	24,5	27,5	35,5	35,5	38,5	38,5	43,8	43,8	46,4
Falkland Islands (11)																		
French Guiana	2,5	–	11,0	12,0	15,0	18,0	10,0	(13)	19,0	26,0	38,0	38,0	38,0	38,0	38,0	38,0	38,0	38,0
Guadeloupe	2,5	4,0	11,0	12,0	11,0	18,0	13,0	13,0	19,0	26,0	38,0	38,0	38,0	38,0	38,0	38,0	38,0	38,0
Guatemala	2,5	4,0	10,0 11,0(7)	11,0	16,8	18,3	9,0	14,5	14,0	19,5	23,0	28,5	28,5	34,0	32,0		34,0	
Guyana (14)																		
Haiti (19)	2,5	3,5	11,0	11,0	15,0	18,0	12,0	20,0	No Regulations									
Honduras	2,5	3,8	10,0 11,0(7)	11,0	14,0	18,3	8,0	14,5	12,0	18,5	20,0	26,5	26,5	33,0	28,0	34,5	34,5	41,0
Jamaica (15)	2,5	3,2	9,1	9,8	12,8	12,8	8,9	8,9	13,5	13,5	22,5	22,5	22,5	22,5				
Martinique	2,5	–	11,0	12,0	15,0	18,0	10,0	11,3 16,2	15,0	20,0	27,0	27,0	27,0	27,0	27,0	27,0	27,0	27,0
Mexico (16)	2,5	4,2	11,6	11,6	(17)	18,3	9,0	14,5	14,0	19,5	23,0	28,5	28,5	34,0	32,0		37,5	
Montserrat	2,1	–	6,7	6,7	–	–	–	–	10,0									
Netherland Antilles	2,5	3,5	10,0	11,0	14,0	18,0			No Regulations									
Nicaragua	2,5	3,8	10,0	11,0	17,0	17,0	8,0	14,5	12,0	18,5	20,0	26,5	26,5	33,0	28,0	34,5	34,5	41,0

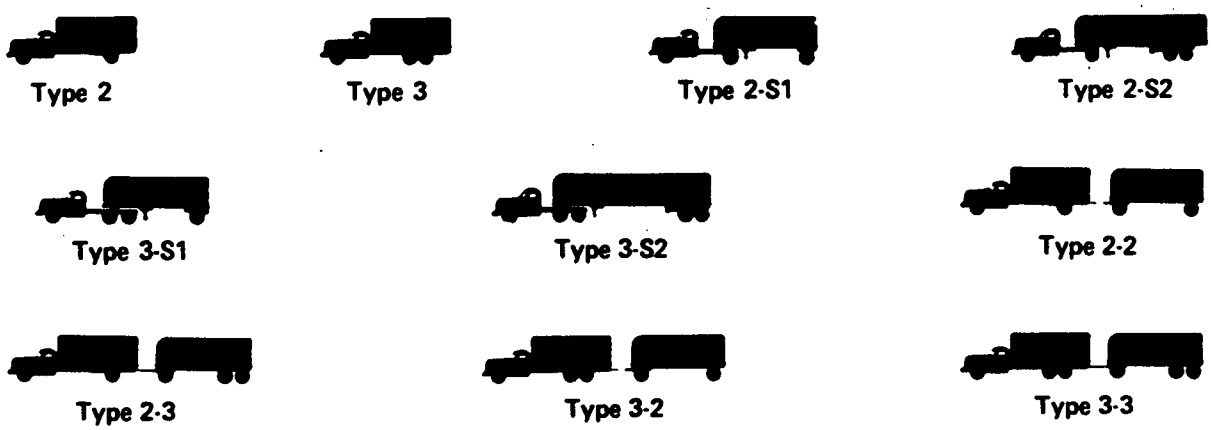
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TABLE VI. LIMITS OF MOTOR VEHICLE SIZES AND WEIGHTS - NORTH, CENTRAL, AND SOUTH AMERICA. - Continued.

COUNTRY	WIDTH	HEIGHT	LENGTH				AXLE LOAD		MAXIMUM GROSS WEIGHT (1)									
			SINGLE UNIT		TRUCK SEMI-TRAILER	OTHER COMBINATIONS	SINGLE	TANDEM	2	3	2-S1	2-S2	3-S1	3-S2	2-2	2-3	3-2	3-3
			TRUCK	BUS														
	meters	meters	meters	meters	meters	meters	metric tons	metric tons	metric tons	metric tons	metric tons	metric tons	metric tons	metric tons	metric tons	metric tons	metric tons	metric tons
Panama (18) (15)	2,5	4,1	11,0 12,0(7)		15,3	19,9	10,0	16,4	Dependent on Axle Spacing									
Canal Zone (18)	2,5	4,1	11,3 12,2(7)		15,2	19,8	10,0	16,4	Dependent on Axle Sapcing									
Paraguay	2,5	4,0	10,0		20,0	20,0	10,0	(18)	15,0	22,0	25,0	32,0	32,0	39,0	35,0	40,0	40,0	40,0
Peru (8)	2,5	4,0	10,5 12,0(7)	10,5	15,0	15,0	11,0	–	15,0	20,0	20,0	30,0	30,0	30,0	30,0	30,0	30,0	30,0
Puerto Rico	2,4	4,1	10,6 .	12,1	15,2	15,2			No Regulations									
Saint Christopher, Nevis, Kitts	2,3		6,7	6,7	7,6	–	–	–	5,1	–	5,1	–	–	–	–	–	–	–
Saint Pierre et Miquelon	2,5		8,0				5,0		Dependent solely on axle loading									
El Salvador	2,5	3,8	10,0 11,0(7)	11,0	14,0	18,3	8,0	14,9	12,0	18,9	20,0	28,9	26,9	33,8	28,0	33,8	33,8	–
Surinam (8)	2,5	3,8	11,0				8,0		Dependent on tire size and axle spacing									
Trinidad-Tobago							8,0	8,0	15,0									
United States (5)	2,4 2,6	3,8 4,3	18,7 17,2	18,7 17,2	15,2 22,9	18,7 24,4	8,2 10,9	13,6 20,0	13,6 20,3	18,1 29,5	21,8 30,5	27,2 36,3	27,2 36,3	32,2 38,8	NP 47,8	NP 47,8	NP 47,8	NP 47,8
Uruguay	2,5	4,0	18,7	18,7	15,3	18,3	10,0(20)	16,0	Dependent on tire size and axle spacing									
Venezuela	2,6	3,9	18,7 12,2(7)	18,7 12,2(7)	15,3	18,3	8,5	14,5	12,0	18,6	20,0			30,0	28,0			33,0
AASHTO POLICY – 1976	2,6	4,1	12,2	12,2	16,8	19,8	9,1	15,4	Dependent on axle spacing									
Pan American Highway Congress Policy	2,5	3,8	18,0 11,0	11,0	14,0	22,0	8,0	14,5	Dependent on axle spacing									

1. See vehicle type nomenclature, page 8.
2. Regulations vary with type of vehicle.
3. 18,50 meters for truck-tractor with semi-trailer. 20,50 meters for truck-tractor with semi-trailer and trailer.
4. No regulated.
5. Regulations vary by Province or State Figures indicate range of variation in maximum limits.
6. Latest available report, 1974.
7. Three axle vehicles have the highest limit.
8. Vehicle type 3-S2 has the highest limit.
9. Dependent on tire size and other conditions.
10. The gross weight depends on the distance between the outer axles. The figures given are the maximum allowed for each type of vehicle.
11. There is no legislation.
13. 2 axles, 15 t, 3 axles, 25 t.
14. Legislation under revision.
15. Special Permit may be granted for vehicles exceeding these limits.
16. Limits shown are for Class A highways. Class B highways have lower limits.
17. Regulations vary with type of vehicle. Type 2-S1 is 14,00 m; Type 2-S2 and 3-S1 is 14,50 m; 3-S2 is 15,25 m.
18. Dependent on distance between axles:
    a) 12,0 tons for distance less than 1,20 m.
    b) 16,0 tons for distance between 1,20 and 1,35 m.
    c) 17,0 tons for distance between 1,35 and 2,35 m.
19. Latest available report: 1971.
20. Two-tire axle maximum load: 5,0 tons. Four-tire axle maximum load: 10,0 tons.

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Notation

The figure shows silhouettes of most basic commercial vehicle types in regular operation as designated by code based on axle arrangement. The first digit indicates the number of axles of the truck or truck-tractor. The letter "S" indicates a semitrailer, and the digit immediately following an "S" indicates the number of axles on the semitrailer. Any digit other than the first in a combination, when not preceded by an "S", indicates a trailer and the number of its axles. For instance, a 2-S2 combination is a two-axle truck-tractor with a tandem-axle semitrailer. A 3-S1-2 combination is a three-axle truck-tractor with tandem rear axles, a semitrailer with a single axle, and a trailer with two axles.

FIGURE 23. Basic commercial vehicle types.

5.2. Rail transportation.

5.2.1 General. Railway systems throughout the world are composed of track gages of various widths, ranging from 23-5/8 inches (0.600 m) to 42 inches (1.067 m) for narrow-gage, 56-1/2 inches (1.435 m) for standard-gage, and from 56-7/8 inches (1.444 m) to 66 inches (1.676 m) for wide-gage. Approximately 65 percent of rail mileage consists of standard-gage track. Typical railcar dimensions are shown in tables VII and VIII. Description and exact configuration of cars, including numbers and types of cars available in North America, are listed in the Official Railway Equipment Register. There is a great variation in foreign railway equipment. Rail transportation criteria for individual countries and STANAG 2832 rail movement criteria may be obtained from the MTMC Transportation Engineering Agency (MTMCTEA), PO Box 6276, Newport News, Virginia 23606.

5.2.2 Open-top cars. Transportability problem items and other large and/or heavy items will normally require movement on open-top railcars. Overhead clearances and platform heights are measured from the top of the rail; side clearances, from the center line of the track.

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TABLE VII. TYPICAL CARGO-SPACE DIMENSIONS FOR RAILCARS WITHIN THE UNITED STATES.

Constraint	Type of car			
	Boxcar	Flatcar	Gondola	Drop center[1]
Length				
maximum	50 ft 6 in. (15.392 m)	89 ft 1 in.[2] (27.153 m)	65 ft 6 in. (19.964 m)	120 ft 6 in. (36.728 m)
minimum	40 ft 6 in. (12.344 m)	40 ft 8 in. (12.395 m)	52 ft 6 in. (16.002 m)	52 ft 2 in. (15.900 m)
Width				
maximum	9 ft 4 in. (2.845 m)	10 ft 6 in. (3.200 m)	9 ft 6 in (2.896 m)	10 ft 5 in. (3.175 m)
minimum	8 ft 5 in. (2.565 m)	8 ft 6 in.[2] (2.591 m)	7 ft 7 in. (2.311 m)	8 ft 4 in. (2.540 m)
Capacity (tons)	40 - 55[3] (36 288 kg-49 896 kg)	50 - 77 (45 360 kg-69 854 kg)	50 - 77 (45 360 kg-69 854 kg)	125 - 250 (130 400 kg-226 800 kg)
Door width				
maximum	16 ft 0 in. (4.877 m)	NA	NA	NA
minimum	5 ft 6 in. (1.676 m)	NA	NA	NA
Door height				
maximum	9 ft 10 in. (2.997 m)	NA	NA	NA
minimum	7 ft 11 in. (2.413 m)	NA	NA	NA
Well length				
maximum	NA	NA	NA	26 ft 0 in. (7.925 m)
minimum	NA	NA	NA	18 ft 0 in. (5.486 m)

[1] These cars are in short supply, and their use requires advance planning.
[2] Minimum width is compatible with maximum length factor.
[3] Larger-size cars of 80 ton (72 576 kg) capacity are reasonably common, and some 100 ton (90 720 kg) cars are available

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TABLE VIII
CHARACTERISTICS OF ORDINARY AND SPECIAL
FOREIGN FLATCARS, AS DEFINED BY STANAG 2175

Category	Floor length (m)	Usable floor width (m)	Wheel base	Height of loading plane (unloaded) (m)	Minimum load capacity (t)	Strength per unit length (t/m)
Ordinary Cars						
0	<9	≥2.47	≤8.0	1.24 - 1.30	15 - 20	>5/2.5
1	7-12	≥2.47	≤8.0	1.22 - 1.25	>20	>10/2.5
2	9-12	≥2.56	≤8.0	1.22 - 1.25	>20	>10/2.5
3	>12	≥2.77	8.0	1.22 - 1.25	>20	>18.5/2.5
Special Cars						
4	>12.5	≥2.74	8.5 - 9.0	1.22 - 1.27	≥40	>19/3
5	11.26 - 13.10	≥2.87	7.26 - 9.30	1.22 - 1.31	≥50	>46/3.5
6	8.80 - 9.95	3.11 - 3.15	5.80 - 8.15	1.24 - 1.31	≥48	>48/3.5
7	≤15	3.11	8.40	1.24 - 1.30	≥89	>62/4
8	≤11.90	3.15	7.20	1.26 - 1.31	≥81	>76/4
9	≥18.50	≥2.74	13.00	1.31	≥40	19/3

LEGEND
m - meters
t - metric tons

5.2.2.1 Height and width. For unrestricted movement, height and width of the loaded railcar shall remain within the limitations of the following clearance diagrams.

a. Association of American Railroads (AAR) outline diagram. Figure 24 is the Association of American Railroads outline diagram for single loads, without end overhang, on open-top cars. It provides for unrestricted movement on all railroad lines within CONUS, Canada, and Mexico except those lines listed on the AAR diagram in the Official Railway Equipment Register.

b. European rail system clearance envelopes A and B (STANAG 2832). Figures 25 and 26, designated envelopes A and B, respectively, are designed to facilitate the planning and movement of a vehicle or equipment classified as "exceptional transport" when

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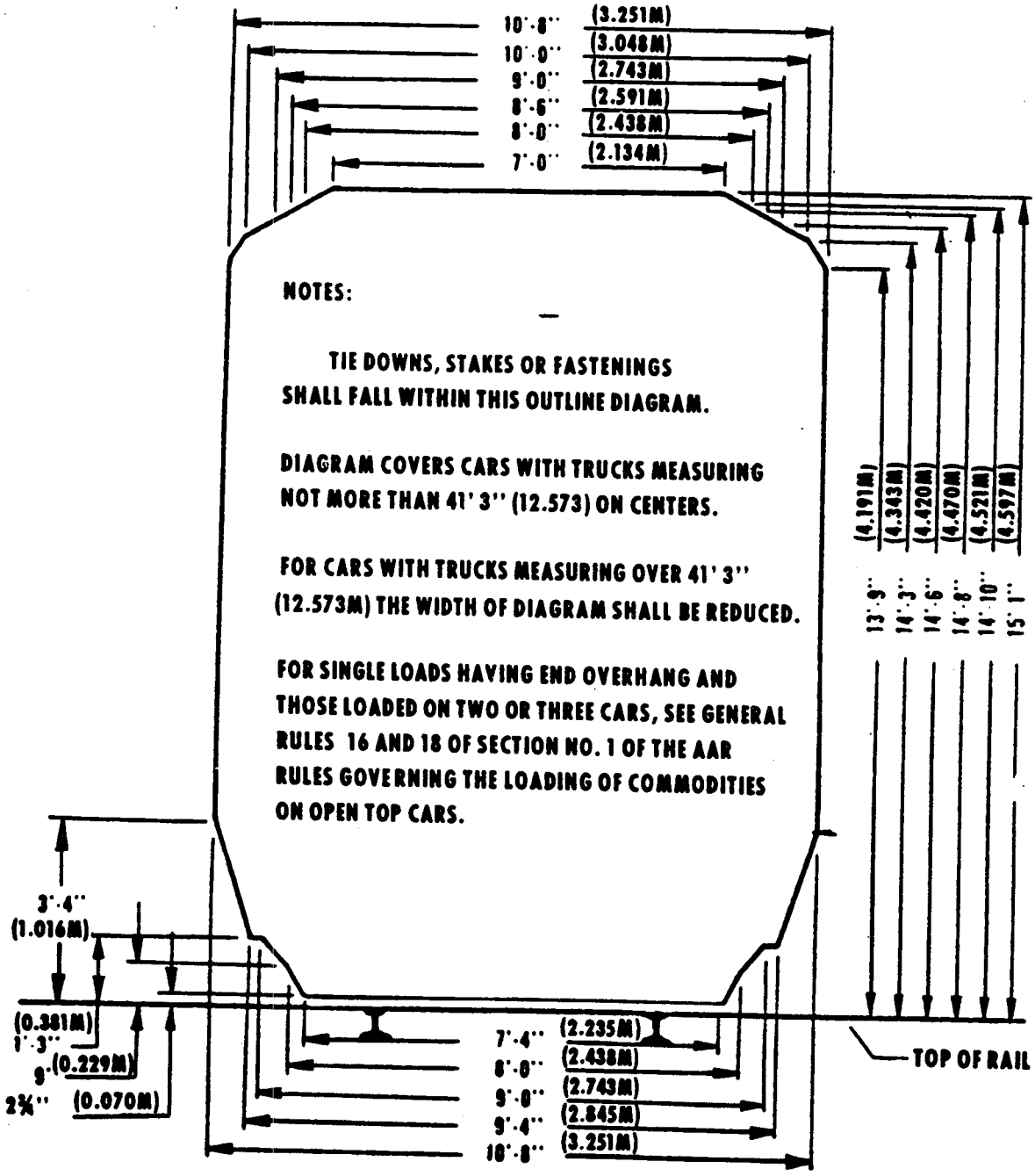
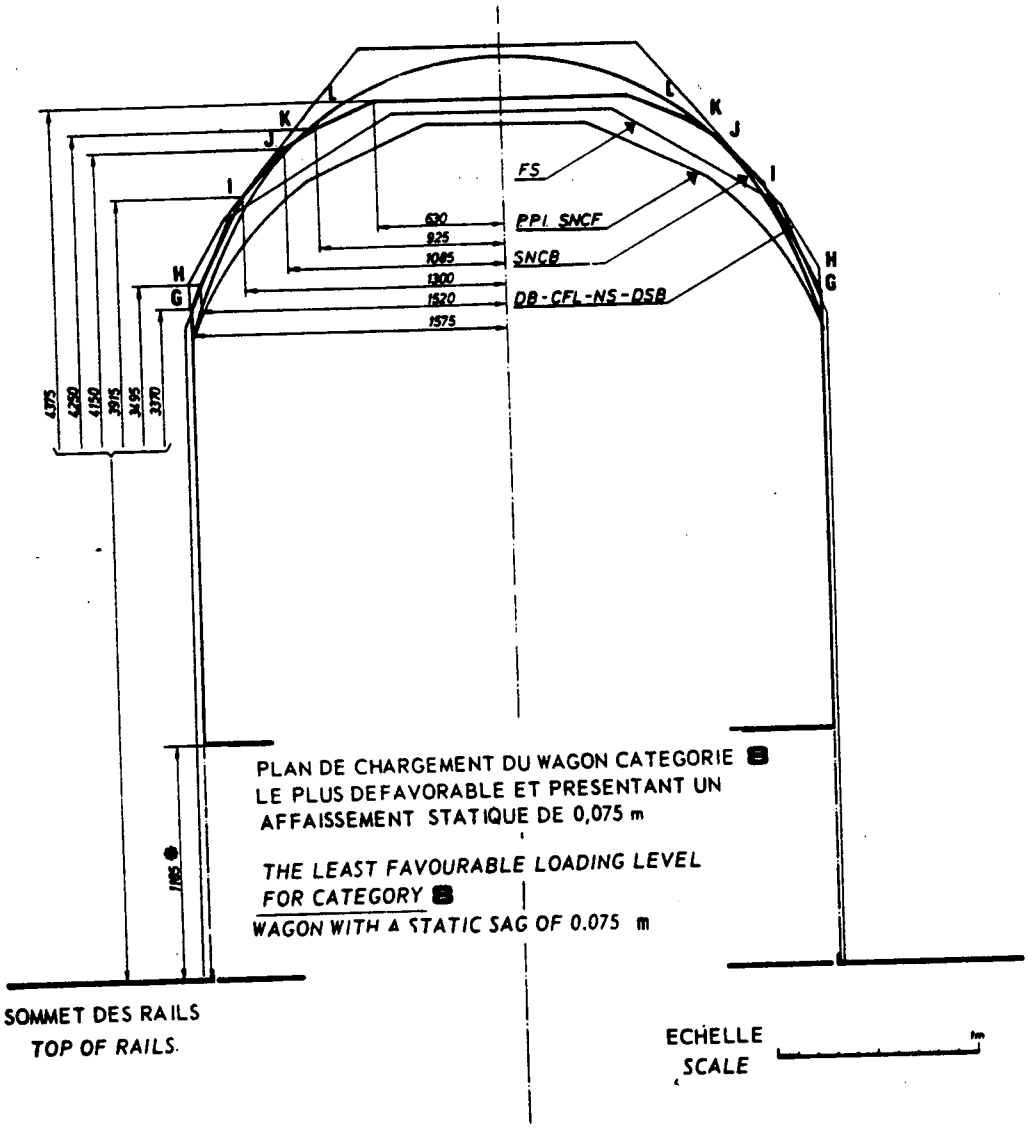


FIGURE 2. Outline diagram for single loads, without end overhang, on open-top cars.

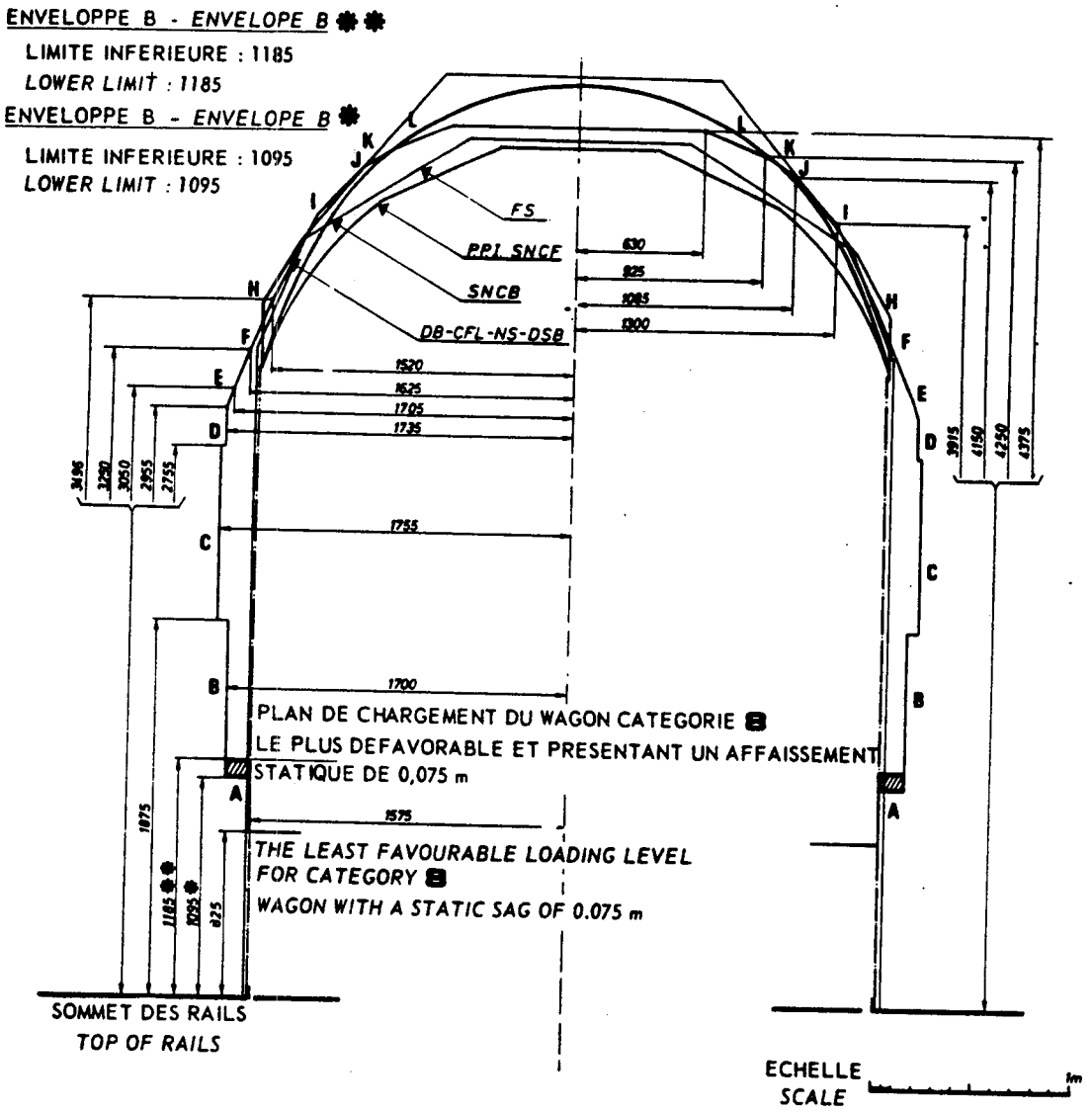
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✱ HAUTEUR MINIMALE SOUS CHARGE STATIQUE
MINIMUM STATIC LOAD HEIGHT

FIGURE 4. Diagram of envelope A.

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✽✽ 1185 = HAUTEUR MINIMALE SOUS CHARGE STATIQUE POUR LES TRANSPORTS AUTORISES A CIRCULER EN LIGNE ET SUR LES VOIES BORDEES DE QUAIS TYPE SNCF.
MINIMUM STATIC LOAD HEIGHT FOR CONSIGNMENTS PASSED FOR LINE TRAFFIC AND ON TRACKS RUNNING ALONGSIDE SNCF TYPE PLATFORMS.

✽ 1095 = HAUTEUR MINIMALE SOUS CHARGE STATIQUE POUR LES TRANSPORTS AUTORISES A CIRCULER EN LIGNE, MAIS INTERDITS SUR LES VOIES BORDEES DE QUAIS TYPE SNCF.
MINIMUM STATIC LOAD HEIGHT FOR CONSIGNMENTS PASSED FOR LINE TRAFFIC, BUT NOT ADMITTED ON TRACKS RUNNING ALONGSIDE SNCF TYPE PLATFORMS.

FIGURE 5. Diagram of envelope B.

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loaded on a NATO category 8 wagon (the wagon taken into consideration when establishing the envelope). The envelopes allow maximum flexibility of movement on the railway systems of Belgium (SNCB), Denmark (DSB), France (SNCF), Federal Republic of Germany (DB), Italy (FS), Luxembourg (CFL), and the Netherlands (NS). On the other hand, a study, which can be undertaken only by the Railway Administrations, is necessary to determine whether a vehicle or equipment loaded on a wagon other than a category 8 wagon will fit into envelopes A and B. Refer to STANAG 2832 when using these clearance envelopes.

c. Passe-partout international (PPI) gauge. Figure 27 is the standard accepted by NATO in STANAG 2832 for unrestricted rail movement in Europe. It provides a dimensional profile for generally unrestricted movement in all countries worldwide using standard- or wide-gage track.

d. Narrow-gage railroads. Figure 28 is a complete narrow-gage clearance diagram for gages of 36 inches (0.914 m), 39-3/8 inches (0.991 m), and 42 inches (1.067 m) for foreign railways.

e. Wide-gage railroads. Figure 29 is a composite wide-gage clearance diagram for gages of 56-1/2 inches (1.435 m) to 66 inches (1.676 m) for foreign railways.

5.2.3 Closed-top railcars in oversea areas. The following cargo height and width limitations for unrestricted movement are based upon the physical dimensions of the more restrictive doors on closed-top railcars.

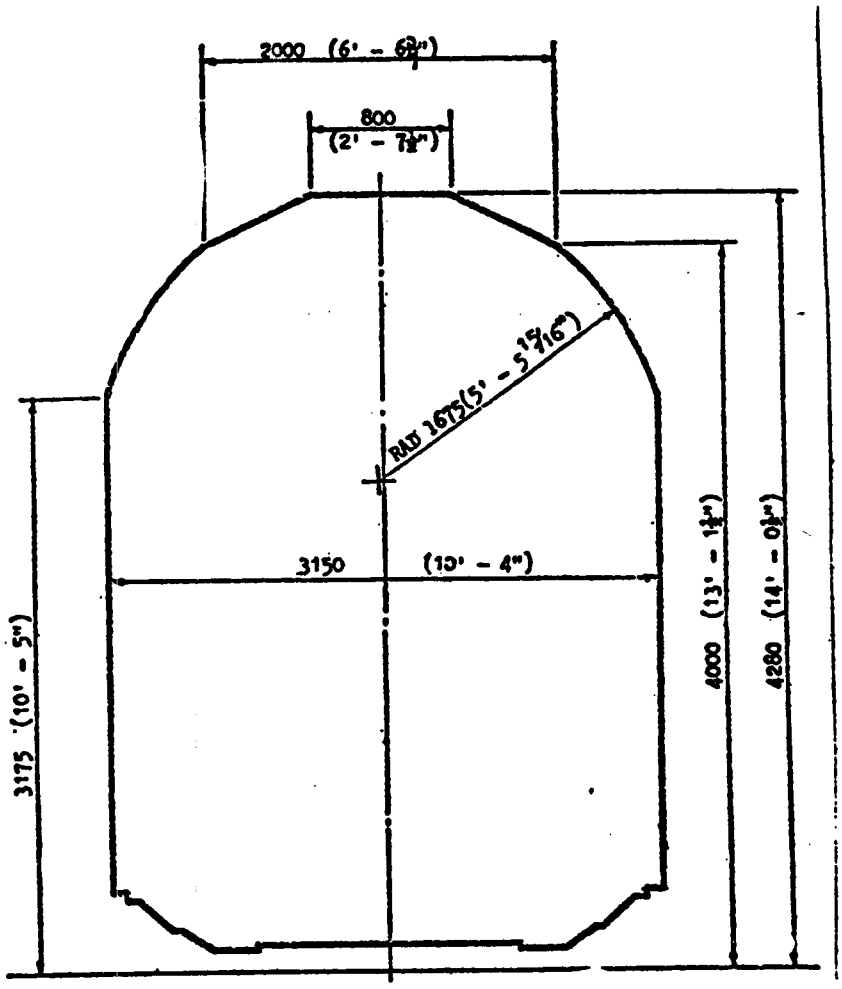
Width	4 ft 0 in. (1.219 m)
Height	5 ft 0 in. (1.524 m)

For larger items, refer to paragraph 5.2.1.

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PPI (PASSE-PARTOUT INTERNATIONAL) GAUGES/GABARIT PPI

Dimensions shown thus: millimetres (feet-inches)
Les dimensions sont indiquées comme suit:
millimètres (feet-inches)



Rail level/Plan de roulement

FIGURE 3. Passe-partout international (PPI) gauge.

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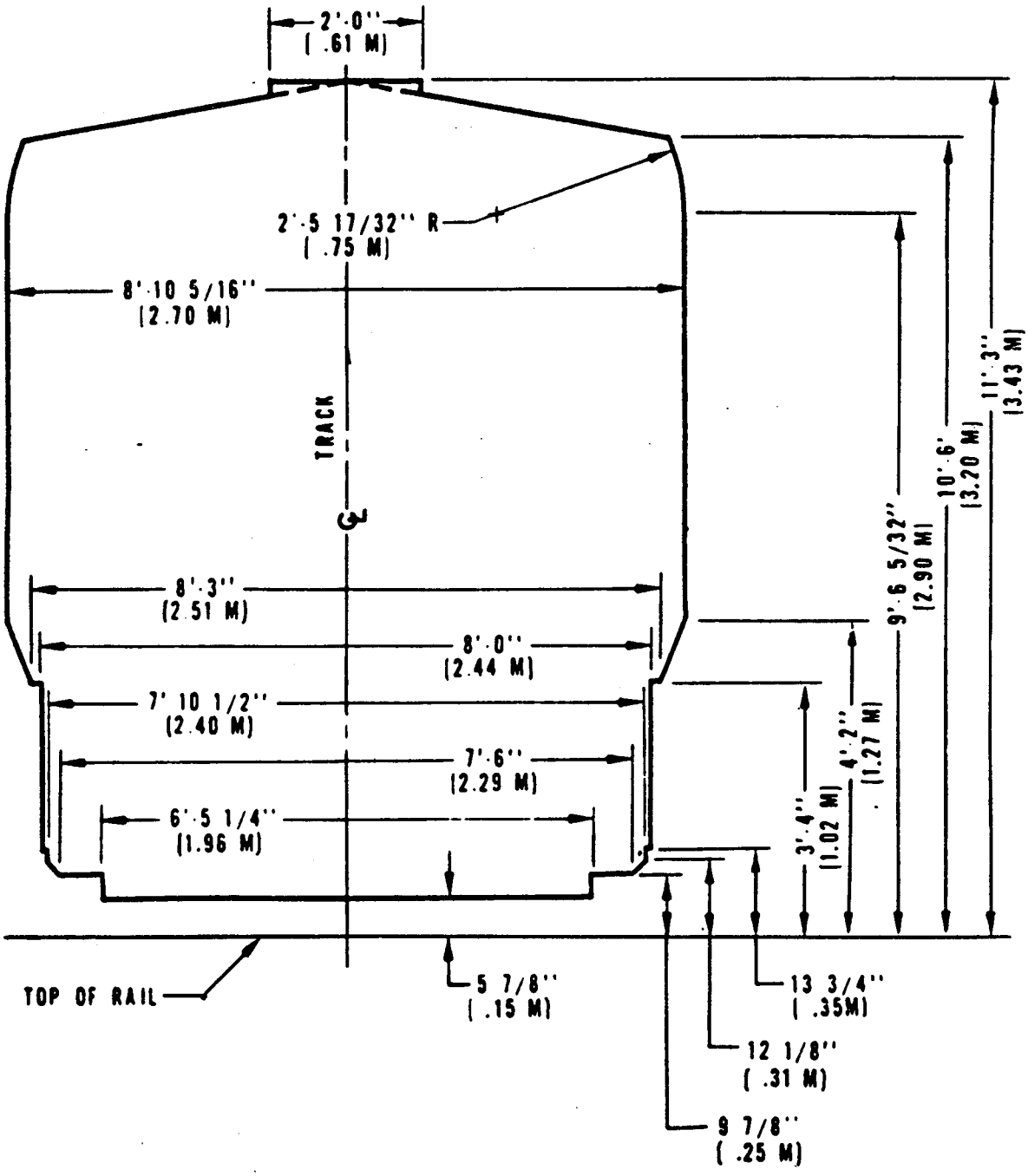


FIGURE 7. Diagram, clearance, composite 36-in. (0.914-m), 29-3/8-in. (0.991-m), and 42-in. (1.067-m) gage, narrow-gage foreign railways.

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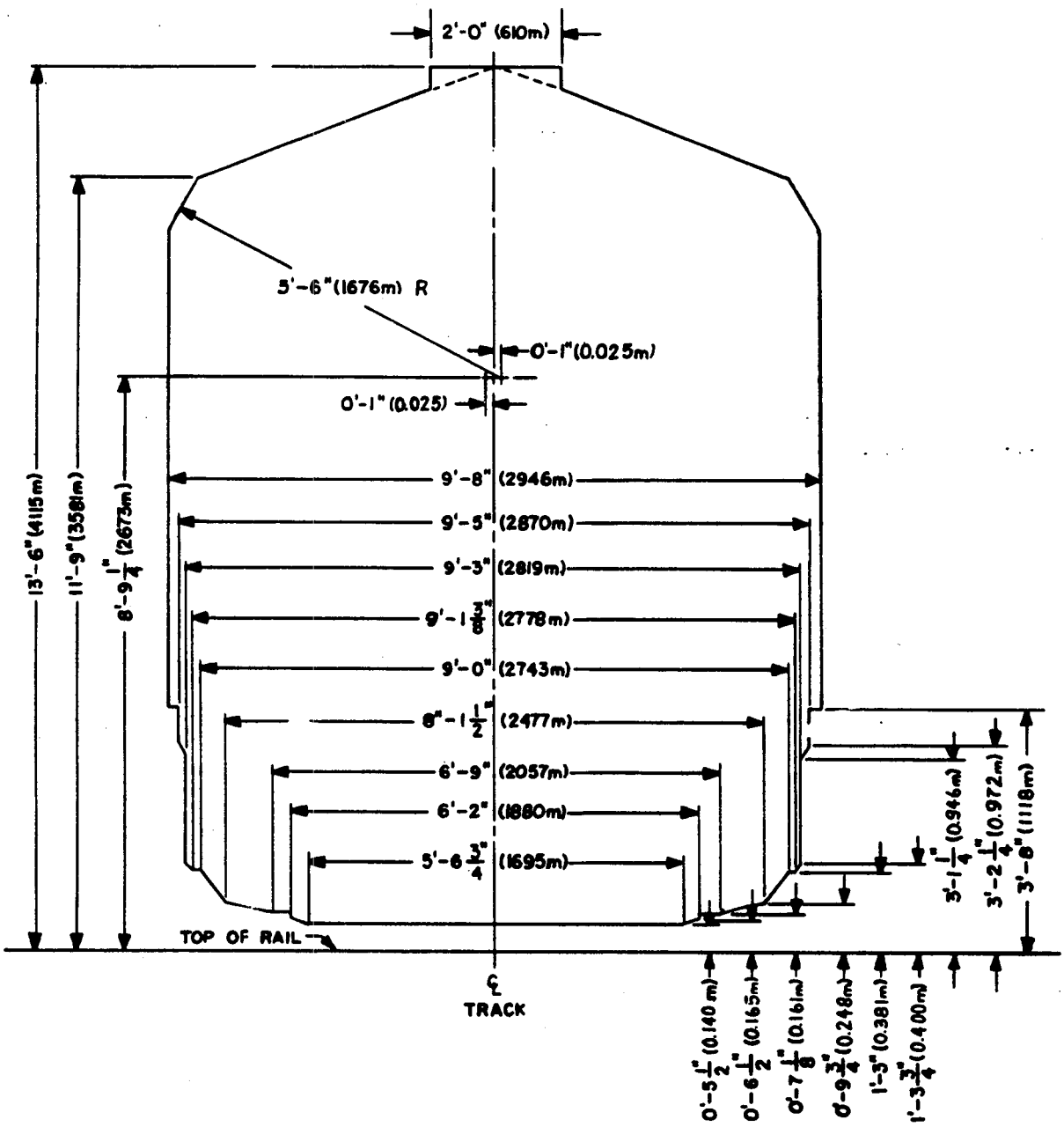


FIGURE 8. Diagram, clearance, composite 56-1/2-in. (1.435-m), 60-in. (1.524-m), 63-in. (1.600-m), and 66-in. (1.676-m) gage, wide-gage, foreign railways.

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5.3 Water transportation.

5.3.1 General. The United States fleet of oceangoing cargo ships consists of conventional breakbulk cargo ships, partial and full containerships, and a relatively small number of special-design ships, such as lighter aboard ship (LASH), barge carrying ships (SEABEE), and roll-on/roll-off ships (RORO). The trend of this fleet is toward containerships and other specialized ships; the conventional breakbulk fleet is steadily decreasing (paragraph 5.3.3.4). In view of the projected future composition of the United States cargo fleet, all material should, if possible, be designed for movement in standard intermodal cargo containers (paragraph 5.5). The data in paragraph 5.3.2 provide guidance for use when an item cannot be designed for shipment in a container.

5.3.2 Breakbulk general-cargo ships. Breakbulk cargo ships normally have a number of holds, often with different size hatch openings. Each hold is served by ship's gear, and one or more holds usually have heavy-lift gear. Table IX contains dimensional and weight capabilities of typical breakbulk vessels representative of the various Maritime Administration (MARAD) designs available in the US-flag fleet. Table X presents the distribution of vertical clearances among the below-deck cargo components for a fleet composed of one ship of each of the MARAD designs. These data allow determination of the maximum size and weight of a single item of cargo for below-deck stowage on a particular vessel. Breakbulk ships have limited container-carrying capability. These ships can carry a small number of containers, but only as conventional cargo, which means that for most vessels they are heavy lifts. Such vessels usually will not be employed to move military containers except to isolated, low-volume consignees.

5.3.3 Containerships.

5.3.3.1 General. Containers can move aboard most cargo ships; however, the full advantages of the system are realized only aboard containerships. Lift-on/lift-off is the more popular method for loading containers for movement on these vessels. This method may be accomplished by onboard cargo-lifting gear (self-sustaining vessel) or by lifting gear located ashore or on adjacent ships or barges (non-self-sustaining vessel). Containers usually are moved aboard these vessels without a chassis attached and are deposited in vertical cells; the container fits in the specially designed cell and is constrained at its four corners. Containers may also be stacked on deck and secured by special locking devices.

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TABLE IX. DIMENSIONAL AND WEIGHT CAPABILITIES OF TYPICAL BREAKBULK VESSELS

MARAD design	Hold no.	Hatch opening length x width	Height in hold minimum to maximum	Allowable deck load minimum to maximum lb/sq ft	Boom lift capacity (LTON)
C3-S-37c	1	31 ft 3 in. (9.525 m) by 17 ft 10 in. (5.436 m)	8 tt 8-3/8 in. (2.651 m) to 10 ft 7-3/8 in. (3.235 m)	495 (2 417 kg/m^2) to 1,790 (8 739 kg/m^2)	15 (15 240 kg)
	2	39 ft 9 in. (12.116 m) by 24 ft 10 in (7.569 m)	8 ft 9-3/8 in. (2.676 m) to 13 ft 1-3/4 in. (4.007 m)	468 (2 285 kg/m^2) to 2,060 (10 057 kg/m^2)	15 (15 240 kg) 60 (60 960 kg)
	3	34 ft 9 in. (10.592 m) by 24 ft 10 in. (7.569 m)	8 ft 7-5/8 in. (2.632 m) to 13 ft 1-1/8 in. (3.991 m)	452 (2 207 kg/m^2) to 2,060 (10 057 kg/m^2)	15 (15 240 kg)
	4	34 ft 9 in. (10.592 m) by 24 ft 10 in. (7.569 m)	8 ft 5-5/8 in. (2.581 m) to 13 ft 0 in. (3.962 m)	452 (2 207 kg/m^2) to 2,060 (10 057 kg/m^2)	15 (15 240 kg)
	5	29 ft 9 in. (9.068 m) by 24 ft 10 in. (7.569 m)	9 ft 5-3/4 in. (2.889 m) to 15 ft 10-1/2 in. (4.839 m)	452 (2 207 kg/m^2) to 1,360 (6 640 kg/m^2)	15 (15 240 kg)
C4-S-57a	1	27 ft 0 in. (8.230 m) by 16 ft 0 in. (4.877 m)	7 ft 10-3/8 in. (2.397 m) to 13 ft 2-1/2 in. (4.026 m)	400 (1 953 kg/m^2) to 1,000 (4 882 kg/m^2)	10 (10 160 kg)
	2	42 ft 6 in. (12.954 m) by 23 ft 0 in. (7.010 m)	10 ft 2-5/8 in. (3.115 m) to 12 ft 2 in. (3.708 m)	495 (2 417 kg/m^2) to 1,865 (9 105 kg/m^2)	10 (10 160 kg) 15 (15 240 kg)
	3	42 ft 6 in. (12.954 m) by 16 ft 0 in. (4.877 m)	8 ft 7-3/8 in. (2.625 m) to 14 ft 10-3/8 in. (4.530 m)	430 (2 099 kg/m^2) to 1,740 (8 495 kg/m^2)	10 (10 160 kg) 15 (15 240 kg) 70 (71 120 kg)
	4	42 ft 6 in. (12.954 m) by 16 ft 0 in. (4.877 m)	7 ft 11-1/4 in. (2.419 m) to 14 ft 10-3/8 in. (4.530 m)	390 (1 904 kg/m^2) to 1,740 (8 495 kg/m^2)	10 (10 160 kg) 15 (15 240 kg) 70 (71 120 kg)
	5	35 ft 0 in. (10.668 m) by 23 ft 0 in. (7.010 m)	8 ft 3-1/8 in. (2.518 m) to 12 ft 11-3/8 in. (3.946 m)	380 (1 855 kg/m^2) to 1,630 (7 958 kg/m^2)	10 (10 160 kg) 15 (15 240 kg)
	6	35 ft 0 in. (10.668 m) by 23 ft 0 in. (7.010 m)	7 ft 8-5/8 in. (2.353 m) to 8 ft 7-5/8 in. (2.632 m)	390 (1 904 kg/m^2) to 1,260 (6 151 kg/m^2)	10 (10 160 kg)
VC2-S-AP3	1	25 ft 0 in. (7.620 m) by 22 ft 4 in. (6,807 m)	6 ft 9 in. (2.057 m) to 17 ft 4 in. (5.690 m)	485 (2 368 kg/m^2) to 1,650 (8 055 kg/m^2)	5 (5 080 kg)
	2	24 ft 0 in. (7.315 m) by 22 ft 4 in. (6.807 m)	7 ft 0 in. (2.134 m) to 13 ft 0 in. (3.962 m)	540 (2 636 kg/m^2) to 1,650 (8 055 kg/m^2)	10 (10 160 kg)
	3	36 ft 0 in. (10.973 m) by 22 ft 4 in. (6.807 m)	7 ft 8 in. (2.337 m) to 13 ft 5 in. (4.089 m)	540 (2 636 kg/m^2) to 1,650 (8 055 kg/m^2)	10 (10 160 kg) 50 (50 800 kg)
	4	36 ft 0 in. (10.973 m) by 22 ft 4 in. (6.807 m)	8 ft 2 in. (2.489 m) to 12 ft 2 in. (3.708 m)	540 (2 636 kg/m^2) to 1,650 (8 055 kg/m^2)	10 (10 160 kg) 30 (30 480 kg)
	5	24 ft 0 in. (7.315 m) by 22 ft 4 in. (6.807 m)	10 ft 2 in. (3.099 m) to 14 ft 9 in. (4.496 m)	620 (3 027 kg/m^2) to 1,650 (8 055 kg/m^2)	10 (10 160 kg)

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TABLE X. DISTRIBUTION OF VERTICAL CLEARANCES BELOW DECK FOR TYPICAL BREAKBULK VESSELS.

Clearance	Percent of cargo compartments
17 ft (5.182 m)	3
16 ft (4.877 m)	6
15 ft (4.572 m)	13
14 ft (4.267 m)	18
13 ft (3.962 m)	23
12 ft (3.658 m)	27
11 ft (3.353 m)	33
10 ft (3.048 m)	44
9 ft (2.743 m)	60
8 ft (2.438 m)	82
7 ft (2.134 m)	98
6 ft (1.829 m)	100

5.3.3.2 Full containerships. These vessels are designed, constructed, converted, or equipped for carrying containers as their normal, full cargo load; they have little or nor capability to carry cargo in any manner except in the size of containers for which the ship is designed. Modern full containerships travel at 25 to 32 knots (12.86 to 16.46 m/s; 46.30 to 59.26 km/hr) and transport 900 or more containers in a full range of sizes. The latest full containerships are not self-sustaining (they have no onboard cargo-lifting gear), resulting in increased cargo capacity.

5.3.3.3 Partial containerships. Partial containerships are actually breakbulk general-cargo ships with one or two holds fitted with vertical container cells. In most configurations, additional holds can be converted to cell stowage if container traffic demands. These vessels are equipped to handle a minimum of fifty 20-foot (6.096-m) containers.

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5.3.3.4 Current trends. The growing use of containers in water transport has significantly increased ship productivity, resulting in a decline in shipping cost per measurement ton of cargo. This has resulted in the following trends in the US-flag cargo ship fleet:

a. Containerships have increased in number from 6, in 1957-58, to 104, in 1978.

b. The majority of cargo ships constructed in the past few years, currently on order, or planned are containerships. However, five new breakbulk general-cargo ships are now being constructed for US registry.

5.3.4 Barge-carrying ship systems.

5.3.4.1 General. With the advent of the barge-carrying ships (LASH and SEABEE systems), a new concept in ocean transportation was born. These highly versatile ships can carry bulk, liquid, baled, containerized, palletized, refrigerated, and large, single-piece cargoes. Most breakbulk and military equipment, not containerizable in conventional containers, can be loaded and discharged from barges and lighters. Some items, because of excessive width or height, pose transportability problems: aircraft, watercraft, railway equipment, cranes, and vehicles. Although such equipment cannot be loaded in the normal manner, various loading alternatives such as removal of barge hatch covers, disassembly, or top-deck stowage are possible.

5.3.4.2 LASH. Any LASH system has two basic components, a mother ship and a family of lighters or lighter/container combinations. Five United States shipping lines with LASH ship systems in operation are the Prudential-Grace Lines, Inc., Pacific Far East Lines, Inc.; Waterman Steamship Lines; Delta Steamship Lines; and Central Gulf Lines. Although there are minor variations in the ships and lighters of the different carriers, all are basically similar. Each lighter (barges that are actually floating cargo holds) has a capacity of approximately 20,000 cubic feet (566 m^3), or 370 LTON (376 000 kg) of cargo. A 500-LTON (508 000-kg) gantry crane, mounted on rails running the full length of the LASH ship, hoists each lighter out of the water, over the transom stern, and lowers it into one of the ship's holds. A ship can transport up to 89 lighters without containers, or a lesser number if containers are also carried. If containers are transported, container loading and unloading are handled by a 30- to 35-LTON (30 480- to 35 560-kg) onboard container crane that operates independently of the lighter gantry crane; otherwise, container

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handling is performed by shoreside cranes. Since LASH ships can handle their cargoes of lighters at any appropriate anchorage within the port harbor area, as well as alongside a pier, their cargoes are less likely to be delayed by port congestion than those of conventional ships. In addition to their lighter capacities, the ships are equipped to handle both 20-foot (6.096-m) and 40-foot (12.192-m) containers. A typical LASH vessel is shown in figure 30.

5.3.4.2.1 LASH ship characteristics. The following characteristics of the C8 ship are typical:

Length	820 feet (249.900 m)
Beam	100 feet (30.480 m)
Shaft horsepower	32,000 (32 448 mhp)
Service speed	22.5 knots (41.67 km/h) at designed draft of 28 feet (8.534 m)
Lighter capacity	73
Container capacity	1,498 standard 20-foot (6.096-m) containers
Cargo deadweight	27,000 long tons (LTON) (27 433 000 kg)
Grain cubic	1,650,000 cubic feet (46 728 m^3)
Bale cubic	1,423,500 cubic feet (40 314 m^3)
Bulk liquid	1,000 LTON (1 016 000 kg)
Gantry crane loading capacity	1,784 LTON (1 813 000 kg) an hour
Container crane loading capacity	360 LTON (365 760 kg) an hour
Combined crane loading capacity	2,144 LTON (2 178 000 kg) an hour

5.3.4.2.2 LASH lighter characteristics. LASH lighers are essentially floating ship holds. They are rectangular in shape, normally with double-skinned metal hulls and removable hatch covers. There is normally a vertical inside ladder in one corner of the lighter that protrudes approximately 9 inches (0.229 m) into the lighter interior. The lighters are designed with 36 tiedown fittings along the side walls and ends to assist in securing the cargo. Fittings are rated at a minimum capacity of 40,000 pounds (18 144 kg) each. There are two levels of fittings extending around the entire interior of the lighter. The first level, with five fittings on each side and four on each end, is 2 feet 6 inches (0.762 m) above the deck. The second level, also with five fittings on each side and four on each end, is 8 feet (2.438 m) above the deck. Interior characteristics of C-8 and C-9 lighters are given in table XI.

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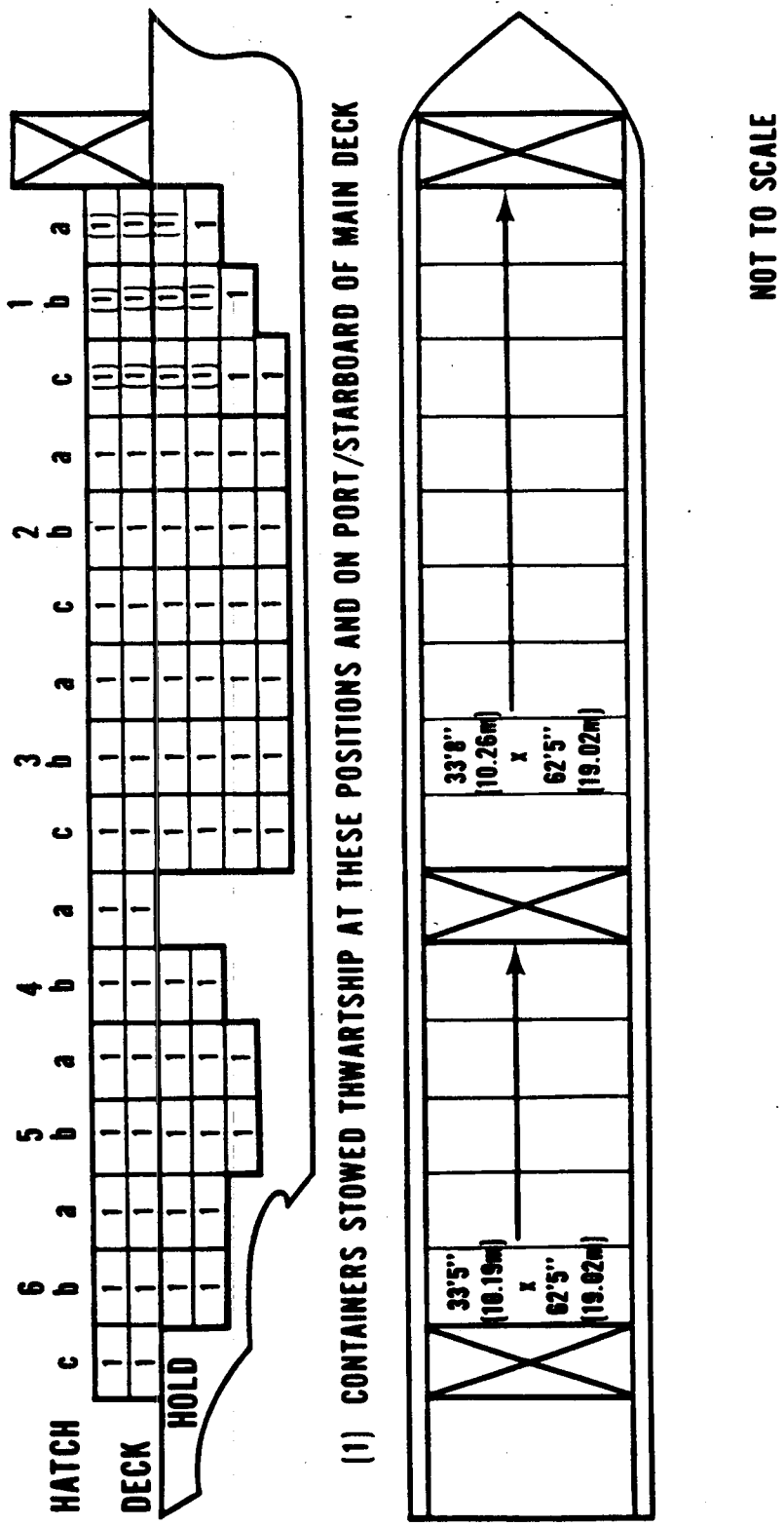


FIGURE 30. Container and lighter stowage plan, LASH typical vessel - LASH Italia.

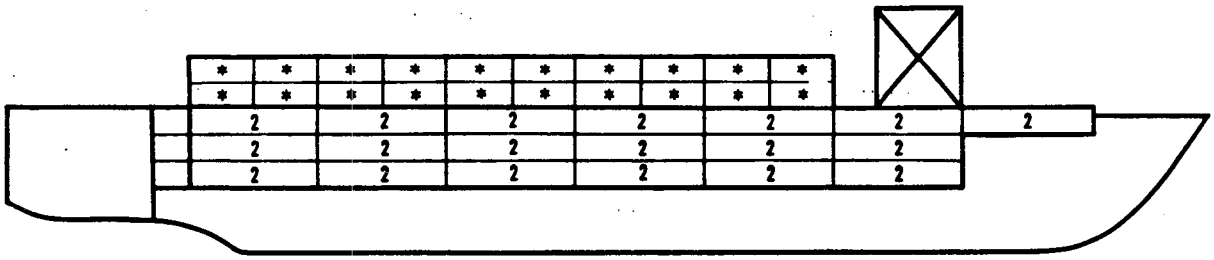
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TABLE XI. TYPICAL DIMENSIONS OF LASH LIGHTERS AND SEABEE BARGES.

Constraint	LASH C-8 lighter	LASH C-9 lighter	SEABEE barge
Inside dimensions			
length	59 ft 9 in. (18.212 m)	59 ft 11 in. (18.263 m)	90 ft 0 in. (27.432 m)
width	29 ft 5 in. (8.966 m)	30 ft 1 in. (9.169 m)	30 ft 3 in. (9.220 m)
Hatch			
opening	43 ft 11 in. (13.386 m)	44 ft 0 in. (13.411 m)	84 ft 0 in. (25.603 m)
width	29 ft 5 in. (8.966 m)	26 ft 0 in. (7.925 m)	30 ft 3 in. (9.220 m)
Height			
at forward and aft bulkheads	10 ft 0 in. (3.048 m)	10 ft 0 in. (3.048 m)	11 ft 3 in. (3.429 m)
under pontoons	11 ft 4 in. (3.455 m)	11 ft 9 in. (3.581 m)	14 ft 7 in. (4.445 m)

5.3.4.3 SEABEE. The sea barge is the largest barge-carrying ships. As with the LASH, the SEABEE system has two primary components, a mother ship and a family of barges. The SEABEE concept was conceived and developed by the Lykes Bros. Steamship Company, Inc. The Lykes SEABEE system consists of three barge-carrying vessels and approximately 250 SEABEE barges. Each SEABEE barge has a capacity of approximately 39,000 cubic feet (1 104 m^3) or 834 LTON (847 000 kg) of cargo. The transfer of barges from and to the water is by the lift-on/lift-off principle, employing a 2,000-LTON (2 032 000-kg) capacity submersible ship elevator. Barges are transferred from the elevator platform to one of three decks by two barge transporters. Each transporter is capable of lifting a 1,000-LTON (1 016 000-kg) barge 3 inches (0.076 m) off the supporting beams on the ship's elevator, transporting it to its assigned stowage location along the deck, and lowering it to the barge support pedestal. The three decks provide space for 38 barges, 12 on each of the two lower decks and 14 on the weather deck. A typical SEABEE vessel is shown in figure 31.

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* CONTAINERS 8 ABREAST MAY BE STOWED ABOVE BARGES

NOT TO SCALE

FIGURE 31. Container and barge stowage plan, SEABEE typical vessel - Doctor Lykes.

5.3.4.3.1 SEABEE ship characteristics. The SEABEE is slightly larger than the LASH ship. Following are typical characteristics of SEABEE ships:

Length overall	874 feet (266.4 m)
Beam	106 feet (32.3 m)
Shaft horsepower	36,000 (26 504 mhp)
Service speed	20.9 knots (33.64 km/h) at a draft of 28 feet (8.534 m)
Barge capacity	38
Container capacity	1,784 standard 20-foot (6.096-m) containers
Cargo deadweight	32,000 LTON (32 513 000 kg)
Grain cubic	1,960,000 cubic feet (55 507 m^3)
Bale cubic	1,500,000 cubic feet (42 480 m^3)
Elevator and transporter loading capacity	2,000 LTON (2 032 000 kg)

5.3.4.3.2 SEABEE barge characteristics. SEABEE ships carry barges larger than the LASH lighers. The barges are rectangular in shape, with a double hull and with interior cargo tiedown fittings. Each barge has seven watertight metal hatch cover panels that weigh approximately 5,800 pounds (2 631 kg) each and require a crane for emplacement and removal. A vertical ladder attached to the interior end wall of the barge facilitates access to the deck. Of particular interest is the provision for a removable tween deck. This is accomplished by a series of removable steel beams thwartship of the barge at approximately 8-foot (2.438-m) intervals and at a height of 5 feet 7 inches (1.702 m) above the

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deck. The beams are suspended in place by locking into a series of sockets built into the wall of the barge. The use of this tween deck arrangement permits cargo of moderate height to be loaded on two levels and the mixing of cargo to optimize weight and cube utilization.

The barges are designed with 90 tiedown fittings of which 68 are recessed "D" rings and 22 are sockets. The "D" rings have a capacity of 40,000 pounds (18 144 kg) each and the sockets, 24,000 pounds (10 866 kg). The first level of fittings is "D" rings located 14 inches (0.356 m) above the deck and spaced 8 feet 4 inches (2.540 m) apart on the barge sides and 8 feet (2.438 m) apart on the barge ends, for a total of 30 rings. This level of fittings may be eliminated on later barge models. The second level of fittings is positioned 5 feet 7 inches (1.702 m) above the deck and consists of 30 "D" rings, 22 on the sides and 8 on the ends, spaced 9 feet 2 inches (2.794 m) apart. Their interior characteristics are as shown in table XI.

5.3.5 Transportability data. For detailed data and transportability guidelines for cargo movement in LASH lighters and SEABEE barges, refer to MTMTS Pamphlet 55-3, titled Transportability Guidelines for Military Cargo in LASH Lighters and SEABEE Barges. Copies may be obtained from the Director, Military Traffic Management Command Transportation Engineering Agency (MTMCTEA), 12388 Warwick Boulevard, Newport News, Virginia 23606.

5.3.6 Roll-on/roll-off (RORO) ships.

5.3.6.1 General. The military value of RORO ships has long been recognized. Their quick turnaround, ability to handle easily a variety of cargo, container capability (containers on chassis), and their increasing numbers in commercial service make certain their desirability for use in follow-on support of future amphibious landings.

5.3.6.2 Current ships. At present, the potentially available American-owned RORO ships are the USNS Comet, the USNS Meteor, the GTS Admiral William M. Callaghan, four ships of the Defiance class, four ships of the Maine class, five ships of the Ponce de Leon class, two ships of the Lurline class, and three ships of the Great Land class. The USNS Comet, the USNS Meteor, the GTS Admiral William M. Callaghan, and the Defiance class ships have complete gear for conventional breakbulk cargo handling in addition to their RORO capability. The Defiance class ships are used primarily as container carriers; however, they have limited RORO capacity. The Maine class ships are primarily RORO container carriers (containers loaded by special forklift trucks), but tiedown

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provisions are provided for vehicular cargo. The Ponce de Leon, Lurline, a Great Land classes are trailer ships. In addition to these, it should be noted that about a dozen RORO ships of the Maine type (but smaller) and numerous conventional RORO ships are owned by allied countries.

5.3.6.3 RORO ship characteristics. RORO ships are designed to transport self-propelled and towed, wheeled, or tracked vehicles. These vessels are loaded and discharged through stern ramps and/or side ports. Cargo is positioned throughout the ship through the use of internal ramps between decks. Maximum dimensional and weight limitations for items of equipment are generally the same as for highway vehicles and for vehicles approved for use as off-the-road tactical vehicles, but shall be determined for specific vessels. In general, the only dimensional and weight limitations imposed by the vessel are the loading ramp/deck dimensions, the vessel ramp capability, and the prescribed deck load limitation. RORO ships' characteristics are shown in table XII.

5.3.7 Terminals. The controlling cargo dimensional limitations of a terminal are normally the rail, highway, or inland waterway capabilities. However, cargo weight limitations may be limited by the terminal and the vertical lift and carrying capabilities. In the case of ocean terminals, the lift capabilities of maritime ports range from 1 LTON (1 016 kg) to 600 LTON (610 000 kg) as shown in the world port index, Ports of the World (US Naval Oceanographic Office Publications No. 150). The range of lift capabilities is identified as follows:

Range	Percentage
Very low range (1 to 5 LTON or 1 016 to 5 587 kg)	1
Low range (6 to 49 LTON or 5 588 to 50 291 kg)	18
Low to medium range (50 to 99 LTON or 50 292 to 101 091 kg)	22
Medium range (100 to 299 LTON or 101 092 to 304 291 kg)	47
High range (300 to 600 LTON or 304 292 to 609 600 kg)	12

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TABLE XII. ROLL-ON/ROLL-OFF SHIPS' CHARACTERISTICS.

Constraint	USNS Comet	USNS Meteor	GTS Adm. Wm. M. Callaghan	Defiance class[1/]	Maine class	Ponce/ Lurline classes	Great Land class
No. of ships	1	1	1	4	4	6	4
Type/Design	C3-ST-14a	C4-ST-67a	NA	C5-S-78a	C7-S-95a	NA	NA
Length (ft)	499 (152.095 m)	540 (164.592 m)	695 (211.836 m)	602 (183.490 m)	684 (208.483 m)	700 (213.360 m)	791 (241.097 m)
Beam (ft)	78 (23.774 m)	83 (25.298 m)	92 (28.042 m)	90 (27.432 m)	102 (31.090 m)	105 (32.004 m)	105 (32.004 m)
Draft (ft)	27 (8.230 m)	29 (8.839 m)	29 (8.839 m)	34 (10.363 m)	32 (9.754 m)	28 (8.534 m)	28 (8.534 m)
Speed (knots)	18 (9.260 m/s)	20 (10.289 m/s)	25 (12.861 m/s)	24 (12.347 m/s)	23 (11.832 m/s)	25 (12.861 m/s)	24 (12.347 m/s)
Cargo deadweight (LTON)	8,085 (8 214 360 kg)	9,592 (9 745 472 kg)	9,247 (9 394 952 kg)	13,589 (13 806 424 kg)	15,107 (15 348 712 kg)	10,323 (10 488 168 kg)	12,555 (12 755 880 kg)
Measurement ton (STON)	17,096 (15 509 230 kg)	24,456 (22 186 110 kg)	49,200 (44 633 489 kg)	32,500 (29 483 504 kg)	45,000 (40 823 313 kg)	19,300 (17 508 665 kg)	30,000 (27 215 542 kg)
RORO deck parking (ft^2)							
Below deck	69,389 (6 446.45 m^2)	79,046 (7 343.61 m^2)	118,800 (11 036.88 m^2)	33,919[2/] (3 151.18 m^2)	123,504 (11 473.89 m^2)	95,250 (8 849.01 m^2)	113,300 (10 525.91 m^2)
On deck	11,404 (1 059.47 m^2)	19,510 (1 812.54 m^2)	45,275 (4 206.18 m^2)	NA	27,844 (2 586.79 m^2)	56,950 (5 290.83 m^2)	76,900 (7 144.24 m^2)
Other areas	5,685 (528.15 m^2)	714 (66.33 m^2)	3,362 (312.34 m^2)	NA	NA	7,680 (713.50 m^2)	9,600 (891.87 m^2)
Total	86,478 (8 034.07 m^2)	99,270 (9 222.48 m^2)	167,537 (15 564.69 m^2)	33,919 (3 151.18 m^2)	151,348 (14 060.68 m^2)	152,280 (14 147.27 m^2)	199,800 (18 562.02 m^2)
Deck height							
Hold number[3/]							
Spar	NA	NA	NA	NA	NA	NA	15 ft 0 in. (4.572 m)
1	6 ft 6 in. (1.981 m) to 15 ft 1 in. (4.597 m)	8 ft 6 in. (2.591 m) to 14 ft 8 in. (4.470 m)	14 ft 7 in. (4.445 m)	NA	8 ft 2 in. (2.489 m) to 9 ft 11 in. (3:023 m)	14 ft 0 in. (4.267 m)	NA
2	6 ft 3 in. (1.905 m) to 12 ft 11 in. (3.937 m)	7 ft 11 in. (2.413 m) to 14 ft 0 in. (4.267 m)	8 ft 7 in. (2.616 m) to 14 ft 6 in. (4.420 m)	7 ft 0 in. (2.134 m) to 14 ft 6 in. (4.420 m)	17 ft 11 in. (5.461 m)	14 ft 0 in. (4.267 m)	15 ft 0 in. (4.572 m)
3	6 ft 11 in. (2.108 m) to 16 ft 1 in. (4.902 m)	8 ft 6 in. (2.591 m) to 13 ft 0 in. (3.962 m)	8 ft 1 in. (2.464 m) to 14 ft 6 in. (4.420 m)	NA	17 ft 0 in. (5.182 m)	14 ft 0 in. (4.267 m)	15 ft 0 in. (4.572 m)
4	7 ft 0 in. (2.134 m) to 12 ft 7 in. (3.835 m)	8 ft 6 in. (2.591 m) to 13 ft 0 in. (3.962 m)	7 ft 11 in. (2.413 m) to 14 ft 7 in. (4.445 m)	NA	9 ft 11 in. (3.023 m)	8 ft 0 in. (2.438 m) to 14 ft 0 in. (4.267 m)	NA

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TABLE XII. ROLL-ON/ROLL-OFF SHIPS' CHARACTERISTICS. - Continued.

Constraint	USNS Comet	USNS Meteor	GTS Adm. Wm. M. Callaghan	Defiance class1/	Maine class	Ponce/ Lurline classes	Great Land class
5	NA	NA	11 ft 8 in. (3.556 m) to 14 ft 2 in. (4.318 m)	NA	NA	NA	NA
6	NA	NA	11 ft 7 in. (3.531 m) to 14 ft 3 in. (4.343 m)	NA	NA	NA	NA
Tank top	NA	NA	NA	NA	NA	5 ft 9 in. (1.753 m)	13 ft 7 in. (4.140 m) to 15 ft 0 in. (4.572 m)
Stern area below deck	NA	NA	12 ft 4 in. (3.759 m) to 14 ft 3 in. (4.343 m)	NA	NA	NA	NA
Stern ramp (h x w)	12 ft 10 in. (3.912 m) by 19 ft 0 in. (5.791 m)	18 ft 0 in. (5.486 m) by 18 ft 0 in. (5.486 m)	18 ft 0 in. (5.486 m) by 17 ft 5 in. (5.309 m)	14 ft 6 in. (4.420 m) by 15 ft 0 in. (4.572 m)	16 ft 0 in. (4.877 m) by 38 ft 6 in. (11.735 m)	NA	16 ft 0 in. (4.877 m) by 16 ft 0 in. (4.877 m)
Side port ramp (h x w)	12 ft 0 in. (3.658 m) by 15 ft 0 in. (4.572 m)	15 ft 0 in. (4.572 m) by 13 ft 0 in. (3.962 m)	13 ft 0 in. (3.962 m) by 15 ft 0 in. (4.572 m)	NA	15 ft 0 in. (4.572 m) by 14 ft 0 in. (4.267 m) (port)	15 ft 3 in. (4.648 m) by 24 ft 0 in. (7.315 m) (fwd)	15 ft 3 in. (4.648 m) by 22 ft 8 in. (6.909 m)
						15 ft 3 in. (4.648 m) by 24 ft 0 in. (7.315 m) (mid)	15 ft 3 in. (4.648 m) by 24 ft 0 in. (7.315 m)
					10 ft 0 in. (3.048 m) by 24 ft 0 in. (7.315 m) (stbd)	15 ft 3 in. (4.648 m) by 22 ft 8 in. (6.909 m) (aft)	
No. of containers	NA	NA	NA	3/	NA	NA	NA
Max lift on/lift off (LTON)	60 (60 960 kg)	70 (71 120 kg)	120 (121 920 kg)	30 (30 480 kg)	4/	NA	NA

[1] RORO/containership.

[2] RORO parking on 2d deck and hold number 7 only.

[3] 484-8' by 8' by 20' (2.438 m by 2.438 m by 4.096 m) or 242-8' by 8' by 40' (2.438 m by 2.438 m by 12.192 m) containers below deck.
294-8' by 8' by 20' (2.438 m by 2.438 m by 4.096 m) or 147-8' by 8' by 40' (2.438 m by 2.438 m by 12.192 m) containers on deck.

[4] Twin deck crane, 15 LTON (15 240 kg) capacity each (30 LTON, or 30 480 kg, in tandem), aft of forward hatch.

[5] Hold numbers indicated for the Defiance, Maine, Ponce/Lurline and Great Land classes of ships are actually deck numbers starting with No. 1 below the weather deck and finishing at the tank top.

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5.3.8 Inland waterways. As a transportation system, inland waterways are quickly gaining in importance in some of the major economic areas of the world, since they generally provide the lowest cost transport. In some world areas, inland waterways are the primary mode for movement of a country's commerce. Although the controlling cargo dimensional limitations are normally the capacities and dimensions of the system equipment, vertical bridge clearances at various water stages may be a limiting factor; also limits are established by the channel width, depth, and lock sizes, and/or the capacities of typical waterway barges and boats. There is a wide variation among the waterways of the world. When transport by a specific waterway is required, or when it is necessary or desirable to move items larger than those that are rail transportable, a special analysis should be made (paragraph 4.1.1).

5.3.9 Logistics-over-the-shore (LOTS) operations. Maximum dimensional data on the cargo capacity of US Army lighters for a single item of cargo are given in table XIII. Additionally, lighter cargo space width, above the restrictions, is provided. Figures 32 to 35 present the interior dimensions at deck level, with the location of the restrictions, for various lighters and landing craft (LCU 1466).

5.3.10 Tactical delivery mode constraints.

5.3.10.1 Ship-to-ship vertical replenishment (VERTREP). For normal VERTREP operations, maximum single lift should be held to approximately 2 short tons (STON) (1 814 kg). However, any unit that cannot be manhandled requires special study of passageway and strike-down routes. Unless special elevators or hoists to preplanned stowage spaces are available, each unit (less container, if necessary) should be capable of passing through an opening 26 inches (0.660 m) by 45 inches (1.143 m), with rounded corners of an 8-inch (0.203-m) radius, and of being lowered through a hatch 30 inches (0.762 m) square, with corners rounded on a 7-1/2-inch (0.191-m) radius.

5.3.10.2 Ship-to-ship connected replenishment. Restrictions shall be the same as for vertical replenishment except that on smaller combatants, where house-fall rigging is required, the limit is 1 LTON (1 016 kg).

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TABLE XIII. US ARMY LIGHTERAGE FLEET

	Length	Restricted width (deck level)	Cargo width (above restrictions)	Ramp clearance width	Cargo Capacity
LARC V[a]	15 ft 11-1/4 in. (4.857 m)	8 ft 8-1/4 in.[b] (2.648 m)	9 ft 9 in. (2.972 m)		10,000 lb (4 536 kg)
LARC XV[a]	24 ft 0 in. (7.315 m)	9 ft 11-1/4 in.[c] (3.029 m)	13 ft 4 in. (4.064 m) 9 ft 0 in.[f] (2.743 m)	9 ft 0 in. (2.743 m)	30,000 lb (13 608 kg)
LARC LX[a]	42 ft 3 in. (12.878 m) 42 ft 6 in.[e] (12.954 m)	12 ft 11 in.[d] (3.937 m)	13 ft 8 in. (4.166 m) 14 ft 6 in.[f] (4.420 m)	13 ft 8 in. (4.166 m)	120,000 lb (54 432 kg)
LCM-8	42 ft 9 in. (13.030 m)	14 ft 6 in. (4.420 m)	14 ft 6 in. (4.420 m) 14 ft 6 in.[f] (4.420 m)	14 ft 6 in. (4.420 m)	120,000 lb (54 432 kg)
LCU-1466	52 ft 0 in. (15.850 m) 75 ft 6 in. (23.012 m)	29 ft 6 in.[g] (8.992 m) 14 ft 4 in.[h] (4.369 m)	29 ft 6 in. (8.992 m) 14 ft 4 in.[f] (4.369 m)	14 ft 4 in. (4.369 m)	336,000 lb (152 410 kg)
LCU-1646	110 ft 0 in. (33.528 m)	17 ft 0 in.[i] (5.182 m)	28 ft 0 in. (8.534 m)	Bow: 14 ft 0 in. (4.267 m) Stern: 18 ft 0 in. (5.486 m)	358,400 lb (162 568 kg)
BDL MARK I[l]	225 ft 0 in. (68.580 m)			22 ft 0 in. (6.706 m)	1,344,000 lb (609 524 kg)
Barge BK Design 7001	81 ft 0 in. (24.689 m)		22 ft 0 in. (6.706 m)		291,200 lb (132 086 kg)
Barge BC Design 7005	110 ft 0 in. (33.528 m)		32 ft 0 in. (9.754 m)		1,276,500 lb (579 147 kg)
Barge BC Design 231-A	120 ft 0 in. (36.576 m)		33 ft 0 in. (10.059 m)		1,310,400 lb (594 388 kg)
Barge BC[j] Design 231-B	120 ft 0 in. (36.576 m)		33 ft 0 in. (10.059 m)		1,294,720 lb[k] (587 276 kg)
LACV-30	Cargo deck: 51 ft 6 in. (15.697 m)		32 ft 6 in (9.906 m)		60,000 lb (27 216 kg)

[a] Wheeled vehicle.
[b] Inside handrail stanchions.
[c] Inside forward bulkhead/side gussets.
[d] Inside tiedown rings.
[e] When wood battens on aft end are removed.
[f] Ramp width.
[g] From aft end of cargo deck to aft end of forecastle, 29 feet 6 inches (8.992 m).
[h] From aft end of forecastle to ramp, width tapers down to 14 feet 4 inches (4.369 m).
[i] Width in vicinity of pilot house.
[j] Primarily used for liquid transport (tanker barge). A survey of usable deck area will be required prior to shipment of dry cargo, due to piping, hatches, and deckhouse.
[k] Deck cargo capacity.
[l] BDL MARK I: Bow ramp:
capacity: 120,000 pound (54 432 kg)
overhead clearance: 13 feet 6 inches (4.115 m).

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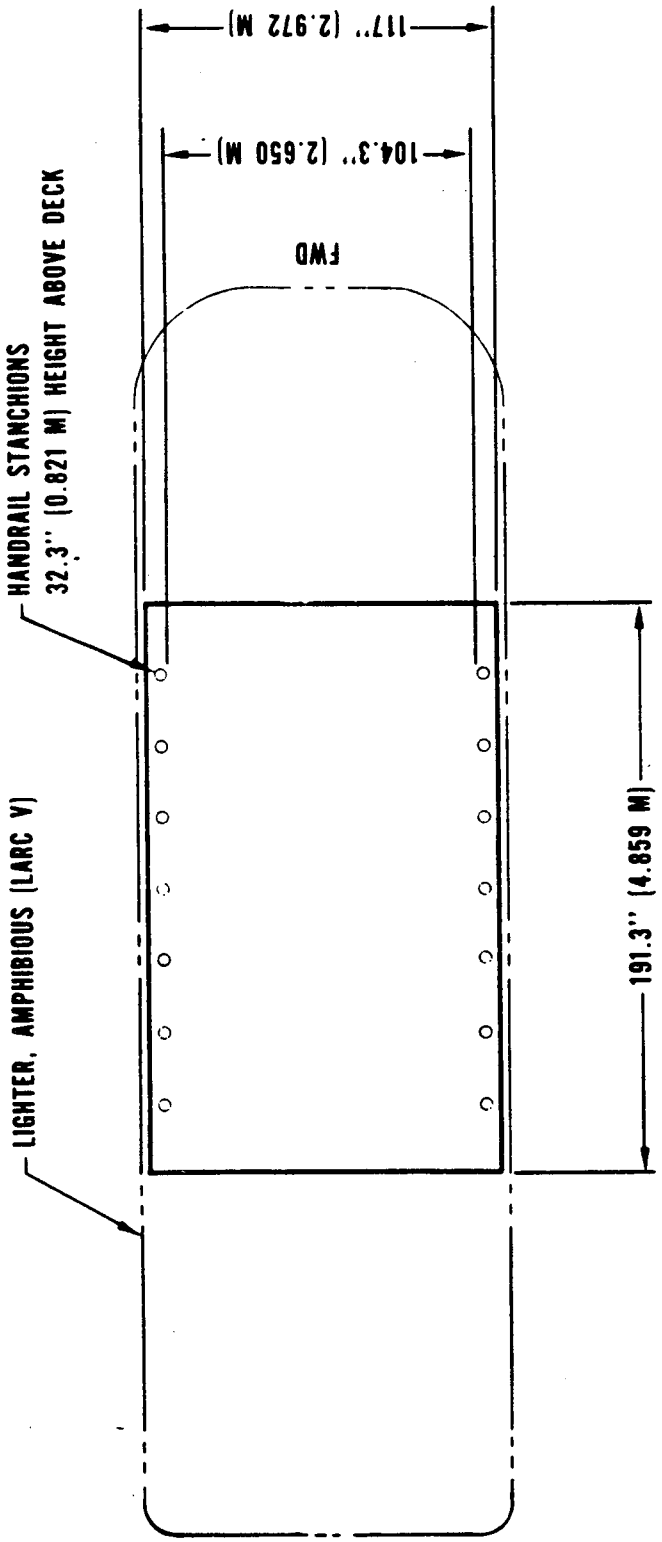


FIGURE 32. Lighter, amphibious (LARC V).

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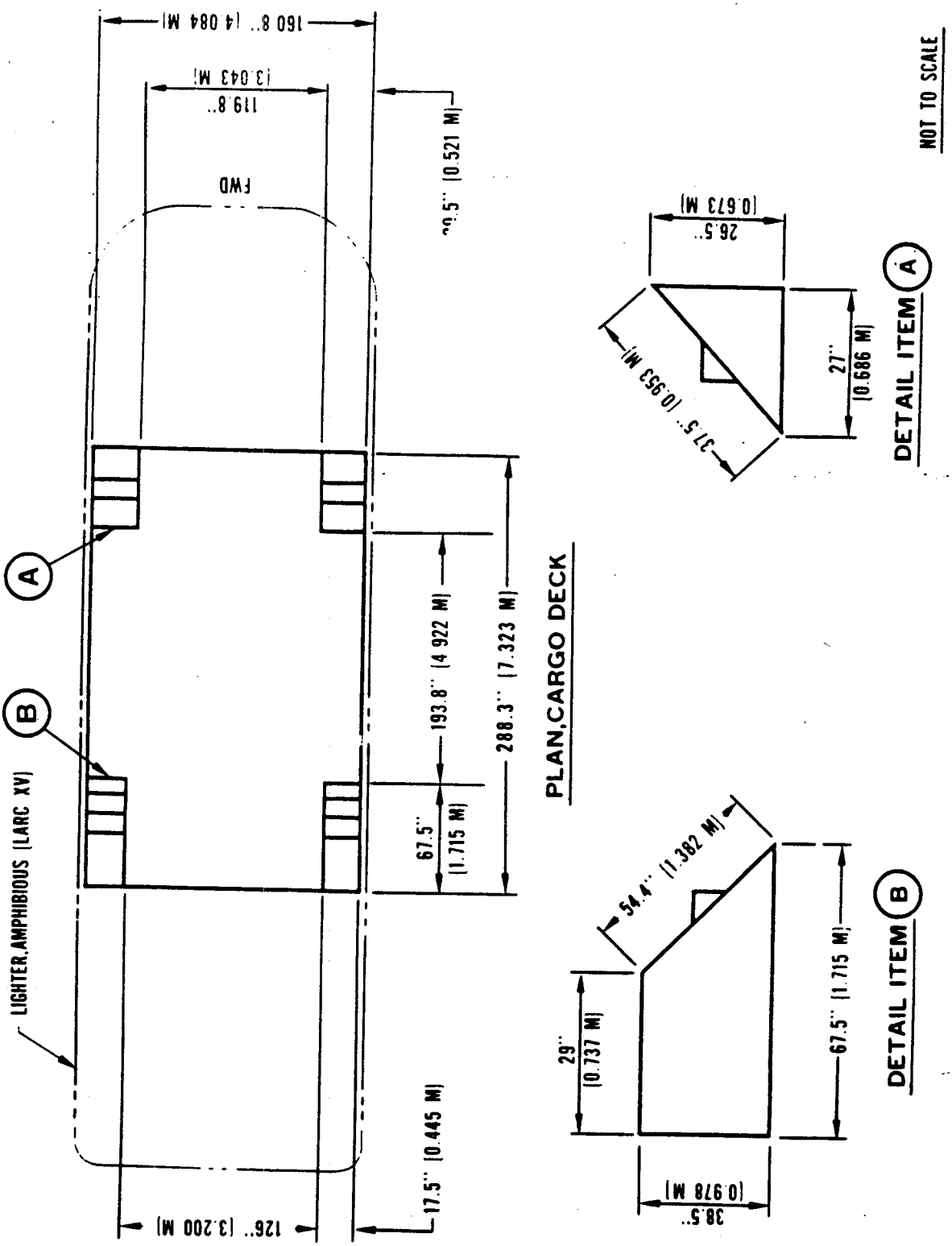


FIGURE 33. Lighter, amphibious (LARC XV).

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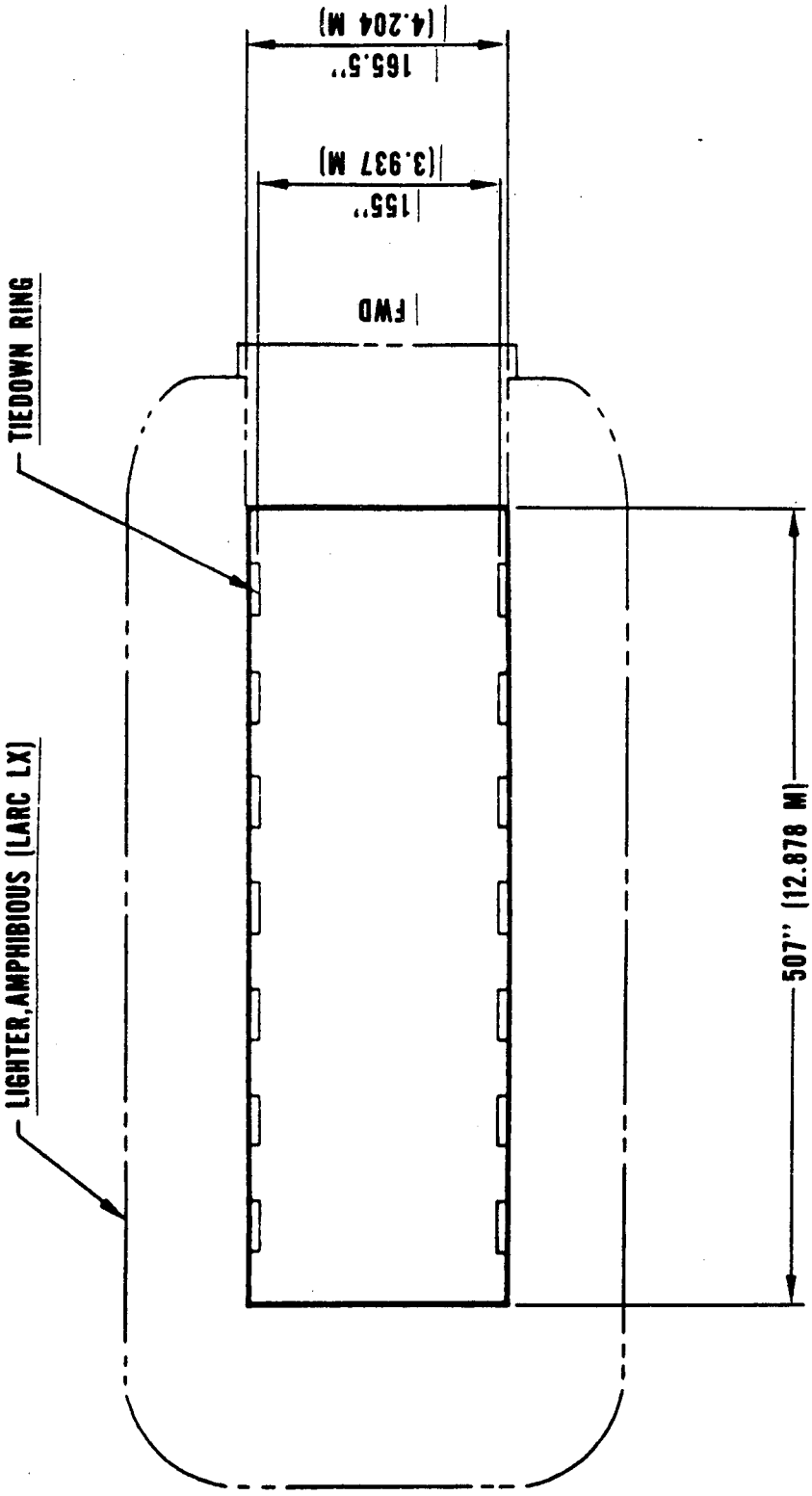


FIGURE 11. Lighter, amphibious (LARC LX).

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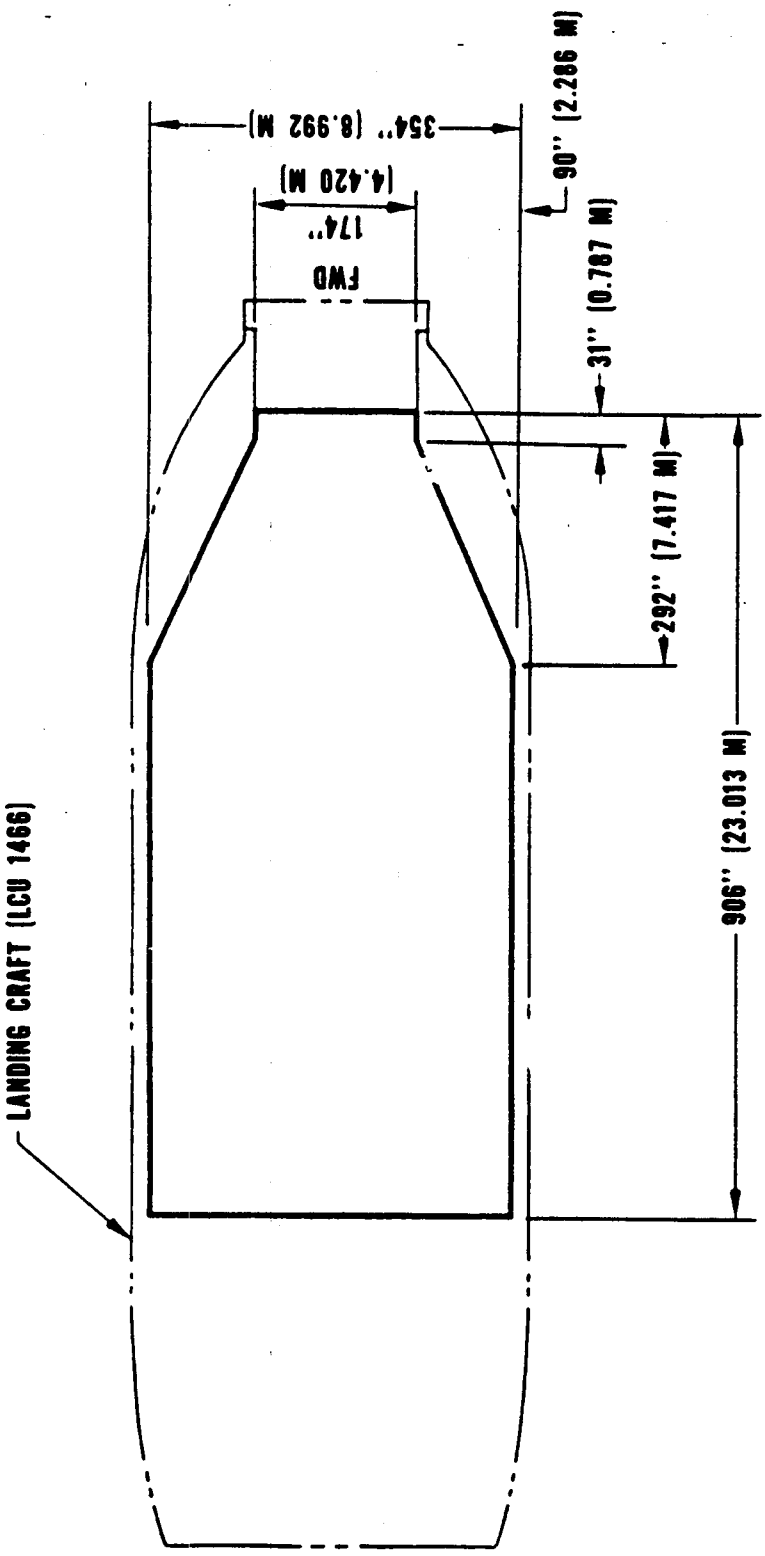


FIGURE 35. Landing craft utility (LCU 1466).

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5.3.10.3 Carrier on-board delivery (COD) requirements. Carrier on-board delivery requirements and dimensional constraints for C2-A aircraft are contained in supplemental NATOPS cargo-loading manual NAVAIR 01-85WBB-1-1; for C1-A aircraft, cargo-loading manual NAVAIR 01-85SAB-509. The term "carrier on-board delivery" means delivery is made by a fixed-wing cargo aircraft, landing on a carrier.

5.3.10.4 Shipboard elevators. Navy weapons that must be handled aboard surface ships shall be compatible with the weapons-handling elevators installed in such ships. Elevator size for aircraft carriers shall be taken as 193 inches (4.902 m) by 84 inches (2.134 m). Elevator size for ammunition-capable support ships shall be taken as 240 inches (6.096 m) by 78 inches (1.981 m). It shall be understood that Navy air-launched missiles shall be compatible with both sets of elevators, leaving a net platform size of 193 inches (4.902 m) by 78 inches (1.981 m).

5.3.10.5 Handling allowance. To allow efficient and safe handling at sea, a handling allowance should be subtracted from the dimensions contained in paragraph 5.3.12.4. A minimum allowance of 6 inches (0.152 m) all around is required, with 10 inches (0.254 m) preferred.

5.4 Air transportation.

5.4.1 General. While this handbook provides general guidance for the air transport and airdrop of items, the requirements of the latest edition of MIL-A-8421 are integral to, and shall be met in, the design and restraint of materiel that will be loaded and air transported aboard US Air Force cargo aircraft. The data contained here are based upon information available at the time of publication and are subject to change. Prior to design initiation and/or inclusion in any contractual action, the data should be verified by the US Air Force Systems Command, Aeronautical Systems Division (ASD/ENECA), Wright-Patterson Air Force Base, Ohio 45433, for US Air Force aircraft; or by the Commander, HQ, Aviation Readiness and Development Command, ATTN: DRDAV-EOT, 4300 Goodfellow Blvd, St. Louis, Missouri 63120, for US Army aircraft. This section contains current characteristics of US Air Force aircraft, airdrop systems, US Army aircraft, and US Navy C-2A carrier on-board delivery (COD) aircraft. In addition, it establishes data for provisions used for tiedown to platform or aircraft, for suspension from an external helicopter lift or an airdrop retardation system, or for extraction from the aircraft. A procedure is established for determining and evaluating the capability of the airdrop item to withstand the severe forces resulting from ground impact. Further detailed information on US Air Force aircraft and weight limitations for

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airdrop may be obtained from Air Force Systems Command (AFSC) Design Handbook DH 1-11 or from the applicable Air Force technical order (TO1C-XXX-9).

5.4.2 Materiel development. An item shall be made light and compact rather than merely capable of being air lifted. Weight and cubage thus saved can be utilized for personnel, bulk supplies, or other equipment. To utilize available air transport facilities most effectively and economically, a close relationship should exist between the military characteristics of items of equipment required to be moved by air and the numbers and types of aircraft that can be provided to move them. The development of materiel for air transport should be related to those aircraft that will be in service at the time of the introduction of the materiel and for a reasonable time thereafter. It is not feasible to design all items requiring air transport against the limitations of dimensions and weight imposed by the cargo capability of the most advanced type of aircraft available. To do so would eliminate the use of many aircraft, which, though less advanced, are still operational and have an air transport capability. For example, to use the C-5 aircraft as the criterion for design would eliminate the use of the C-130 and C-141 aircraft.

5.4.2.1 Materiel considerations. Weight and bulk reduction should not be permitted to impair essential operational characteristics or to increase overall costs beyond reasonable limits. However, the design of lighter and more compact materiel will increase transportability and reduce the costs associated with logistical movement of the item throughout the entire life cycle. Agencies developing or reviewing requirements should insure that lightness and compactness receive consideration consistent with other essential characteristics, including provisions for external lift. It is important that the procedures for reducing the weight and bulk of materiel not depend primarily upon the use of lightweight materials, but upon improved design and the use of lightweight materials.

5.4.2.2 Materiel sectionalization. Sectionalization must not impair the capability of an item to perform its fundamental mission. Items should be capable of being reassembled and placed in operation within the time frame stated in paragraphs 5.4.7.1 and 5.4.7.2, or specified in the operational requirements. If an item is required in the assault phase, the entire item, together with tools, equipment, and the crew required for assembly, should be capable of being transported in one aircraft. All mobile materiel requiring sectionalization should be designed to permit the basic element to be moved on its own wheeled or tracked system. Materiel should be clearly marked as a guide to disassembly and assembly procedures. Shipping weight and the location of the vertical and longitudinal

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centers of gravity should be marked on each component and end-item. Components that require lifting should be equipped with appropriate attaching points or lift provisions. Disassembly and assembly should be within the capability of using units, with air-transportable organizational tools and equipment moved with the sectionalized item.

5.4.3 Aircraft designation. Based upon requirements in the acquisition documents, the program office will designate the aircraft desired to transport the equipment or materiel. Action will be taken to insure compliance with MIL-A-8421.

5.4.4 Aircraft performance variables. Aircraft performance is subject to many variables such as altitude, temperature, humidity, operating range or radius, runway characteristics, availability of fuel at destination, and the nature of the operation. Air Force policy is to use maximum gross weights, specified in applicable TO1C-XXX-9, for all normal flight operations. Operation of aircraft at gross weights in excess of those specified in TO1C-XXX-9 increases operational hazards and will be authorized only by major Air Force Command.

5.4.5 Aircraft characteristics that affect transportability. Development of materiel with the required degree of air transportability requires careful consideration of the characteristics of the aircraft in which it is to be transported. The most important of these characteristics are:

a. Maximum allowable aircraft payload.

b. Size, location, and configuration of cargo-loading entries/openings, to include cargo jettison capabilities.

c. Size and configuration of cargo compartment, to include limiting features that may prevent full utilization of available space.

d. Structural limitations of the ramp and cargo compartment, including tiedown receptacles, and roller system.

e. Cargo compartment environment (temperature, pressure, vibrations, and G-load accelerations).

Cargo capability data and related information on US Air Force, US Navy C-2A COD aircraft, and US Army aircraft are shown in paragraphs 5.4.9, 5.4.10, and 5.4.11, respectively. Similar data for commercial cargo aircraft may be obtained from the Commander, Military Traffic Management

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Command Transportation Engineering Agency, PO Box 6276, Newport News, Virginia 23606. Items of equipment that are critical from the standpoint of weight, center of gravity, and dimension shall be treated separately. Data on new aircraft available for airlift of defense forces will be published, as available, by changes to this handbook.

5.4.6 Airdrop capabilities. Materiel developed to be air droppable shall meet the limitations imposed by the characteristics and capabilities of the current or developmental airdrop systems and the aircraft to be employed with the systems. Developing agencies may obtain engineering and design assistance from the US Army Natick Research and Development Command, ATTN: STSNL-UBS, Natick, Massachusetts 01760, including: (1) analysis of proposed designs to determine airdrop acceptability; (2) recommendations for component and systems designs and energy dissipater configurations to provide optimum airdrop capability; and (3) laboratory facilities for development testing of proposed materiel in controlled airdrop environments including static drop, roller test, and tiedown provision restraint test facilities. Information pertaining to testing of airdrop equipment and systems may be obtained from the US Army Test and Evaluation Command (TECOM), ATTN: DRSTE-CT-A, Aberdeen Proving Ground, Maryland. Testing facilities include controlled impact drops from static drop facilities and instrumented drop zones for actual airdrop. Engineering and testing assistance, along with operational information, should be obtained as early as possible in the design stage of development. Airdrops of development items normally should be confined to the final design submitted for developmental test. For agencies within the Department of the Army, developmental airdrop equipment may be used only with the approval of the Commander, US Army Materiel Development and Readiness Command (DARCOM).

5.4.7 Phases of airborne operations. The planning, preparation, and execution of an airborne operation generally fall into four interrelated operational phases: the mounting phase, the air movement phase, the assault phase, and the subsequent operations phase. Details are listed in FM 57-1. The following should be considered in designing materiel to be airdropped during these phases.

5.4.7.1 Assault phase. Items to be used in the initial support of forces deployed in an airborne assault, including supply by air, should be operational immediately except for selected engineer construction equipment, which should be capable of employment within 1 hour.

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5.4.7.2 Subsequent operations phase. The large, heavy materiel air-landed in logistic support of previously deployed forces, and materiel included in support of followup forces, should be capable of employment within 6 hours.

5.4.8 Requirements for airborne operation.

5.4.8.1 Required operational capability. The required operational capability (ROC) should include, as one of the fundamental characteristics of the new materiel, a statement of the required airdrop capability. This should include, where appropriate, availability of the type of aircraft required to move these items of equipment during a particular phase. Obviously, it is not feasible to design all items required during the assault phase of an operation against the maximum dimensions and weight permitted by the cargo capability of the most advanced type of aircraft available. Conversely, requiring all items of equipment to conform to the lower weight and dimensional capabilities of the aircraft in the current reserve and active service inventories may result in failure to exploit advances in technology. Priority should be given to developing materiel for transport in the type of cargo aircraft that meets operational requirements most effectively. Consideration should be given to minor modifications in the design of materiel to facilitate maximum utilization of all cargo aircraft available. Modifications should not affect the intended utilization of materiel being developed.

5.4.8.2 Required physical characteristics. The statement of required physical characteristics, contained in the ROC for air droppable materiel, should be consistent with the physical and performance capabilities of US Air Force tactical and strategic airlift aircraft currently available, or expected to be available, in large quantities, during the period that the equipment will be fielded.

5.4.8.3 System considerations. Development of materiel or equipment for air transport must meet the critical limitations of the aircraft, airdrop systems, and materials handling equipment that will be in use, or are planned to be in use, during the time period in which the item is to be transported. Verification and data pertaining to these limitations will be furnished by the sources identified in paragraph 5.4.1.

5.4.8.4 Materiel considerations. The agency originating the characteristics for new materiel/equipment for the military services must give consideration to the specifications of equipment expected to be available in the same time frame as the new aircraft. This will insure that characteristics such as lift capability and cargo compartment configuration of

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supporting aircraft will adequately accommodate equipment organic to supported units. Developers of materiel shall address the packaging and handling criteria contained in AFR 71-4/TM 38-250/NAVSUP Pub 505/ MCOP 4030.19/DSAM 4145.3 when designing hazardous items of materiel for air movement.

5.4.9 Characteristics of US Air Force aircraft. MIL-A-8421 and MIL-HDBK 318 establish the general design and performance requirements for military equipment to be air transportable on Air Force aircraft. It provides general design criteria and test requirements that this equipment shall meet. Specific characteristics of Air Force cargo aircraft are provided in Air Force Systems Command (AFSC) Design Handbook DH 1-11, and in the individual loading manual (TO1C-XXX-9) for each aircraft. The specific characteristics of each Air Force aircraft establish limiting parameters for the design of equipment to be air transportable in that aircraft. Air transportable equipment or materiel shall meet the following general requirements:

5.4.9.1 Air transportability analysis/test loading. In general, any item of materiel or equipment to be transported by C-130, C-141, or C-5 aircraft that exceeds any one of the conditions listed in paragraph 4.1.1e will require transportability analysis or test loading, as prescribed in appendix B, AR 70-44/OPNAVINST 4600.22B/AFR 80-18/MCO 4610.14C/ DSAR 4500.25.

5.4.9.2 Dimensional limitations. Overall dimensions of the equipment or materiel shall meet or be within the dimensional limits presented in MIL-A-8421 and shall conform to projection limitations provided in AFSC DH 1-11 and in the loading manual, TO1C-XXX-9, for each aircraft.

5.4.9.3 Aircraft floor loading. Axle, wheel, tire-footprint, and general-floor loads must conform to the fuselage zone and compartment limitations of the aircraft. Detailed floor loads are published in AFSC Design Handbook DH 1-11 and in the TO1C-XXX-9 for a particular aircraft. ASD/ENECA and MTMCTEA maintain files of current technical orders, and assistance should be requested from either agency.

5.4.9.4 Wheeled/tracked vehicles. Angles of approach and departure for wheeled and tracked vehicles shall conform to the limitations stated in MIL-A-8421. The front and rear-end overhang and the underside clearance above the ground of wheeled and tracked vehicles shall meet the overhang and ramp-cresting limitations presented in AFSC Design Handbook DH 1-11 and in the loading manual, TO1C-XXX-9 for each aircraft.

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5.4.9.5 463L system description. Where possible, materiel should be designed so that it can be handled by the 463L air cargo handling system (see MIL-A-8421). This system consists of separate, but interdependent, equipment families. Included are terminals, onboard aircraft cargo-handling systems, ground-handling equipment, pallets, and nets. Descriptive information and operating precautions for each item of 463L-system equipment is provided in Air Force TO36M-1-141. Pallets are used as a unitization module and are loaded into the aircraft on roller conveyors.

The C-130, C-141, and C-5 aircraft are all equipped with 463L roller conveyors and restraint rails with integral pallet locks. Three pallet sizes are listed in MIL-P-27443; however, only the HCU-6/E, 108-inch (2.743-m) by 88-inch (2.235-m) pallet with restraint nets is used on the C-130, C-141, and C-5 aircraft. Most materiel palletized on system 463L-type HCU-6/E pallets can also be transported on Civil Reserve Air Fleet (CRAF) cargo aircraft. Pallet couplers are available to join as many as five pallets together when cargo is too large to be placed on one 108-inch (2.743-m) by 88-inch (2.235-m) pallet. Because of restraint problems, pallet trains seldom exceed three pallets. The HCU-6/E pallet description is as follows:

a. HCU-6/E, MIL-P-27443, Type 1.

b. Overall dimensions: 108 inches (2.743 m) by 88 inches (2.235 m) by 2-1/4 inches (0.057 m).

c. Rated capacity: 10,000 pounds (4 536 kg).

d. Weight (empty pallet): not to exceed 300 pounds (136 kg)

e. Usable floor dimensions: 104 inches (2.642 m) by 84 inches (2.134 m).

f. Net assembly: two HCU-7/E side nets and one HCU-15/C top net conforming to MIL-N-27444.

5.4.9.6 Pallet and integral base design. Where equipment is to be transported on its own base or on a pallet designed as an integral part of that equipment, or, if a special pallet is designed for movement in a US Air Force aircraft, that pallet or base shall meet the requirements of MIL-A-8421.

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5.4.9.7 Tiedown provisions. Tiedown locations, sufficient in number and strength to provide the restraint required by MIL-A-8421, shall be available on any item of equipment or materiel designed to be air transportable. Tiedown provisions shall comply with the requirements of paragraph 4.7.

5.4.9.8 Restraint criteria. Materiel, except nuclear materiel, shall be capable of being restrained to withstand, without loss of serviceability, the load factors specified in MIL-A-8421 (see paragraph 4.8).

5.4.9.9 Accompanying loads. If equipment is capable of carrying additional equipment, or of having additional equipment attached, and the order or contract requires this accompanying load to be air transported with the equipment, the entire unit load will be required to meet the requirements of MIL-A-8421. The restraint of the accompanying load to the basic equipment, including the attachment points and backup structure on the basic item, shall be capable of withstanding the forces/load factors specified in MIL-A-8421.

5.4.9.10 Markings. As prescribed in MIL-A-8421, equipment should be marked to provide the information necessary to facilitate loading in the aircraft. Unless otherwise specified, the identification should be stenciled or marked by other means, as defined in the contract, in an appropriate location or on an equipment data plate. Tiedown attachment points and fitting capacities may be etched or stamped on the attachment point or fitting.

5.4.10 Characteristics of US Army aircraft. Materiel developed for air transport by US Army aircraft shall meet the requirements imposed by the characteristics of the aircraft, as well as those imposed by US Air Force aircraft. FM 101-20 contains data for all Army aircraft, including cargo loads for normal operating conditions. Additional information required for determining design criteria or for development of materiel may be requested from the appropriate developmental agency, MTMCTEA, or Office of the Deputy Chief of Staff for Research, Development, and Acquisition (DCSRDA), Department of the Army. Data for the cargo-carrying US Army aircraft are shown in Table XIV. Data for developmental aircraft will be published at a later date.

5.4.11 Characteristics of US Navy carrier on-board delivery (COD) C-2A aircraft. For detailed characteristics of US Navy aircraft C-2A refer to NAVAIR 01-85WBB-1-1.

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TABLE XIV. LOADING DIMENSION AND WEIGHT FOR US ARMY AIRCRAFT.

Aircraft	Cargo door		Cargo compartment			External hook or sling capacity[a]	Basic mission payload[b]
	width	height	length usable	width	height		
U-21[c]	4 ft 5-1/2 in. (1.359 m)	4 ft 3-1/2 in. (1.308 m)	12 ft 8 in. (3.861 m)	4 ft 6 in. (1.372 m)	4 ft 9 in. (1.448 m)	NA	3,000 lb (1 360 kg)
UH-1B/C/M	4 ft 0 in. (1.219 m)	4 ft 0 in. (1.219 m)	4 ft 0 in. (1.219 m)	6 ft 8 in. (2.032 m)	4 ft 8 in. (1.422 m)	4,000 lb (1 814 kg)	UH-1B 2,939 lb (1 333 kg) UH-1C 3,320 lb (1 506 kg) UH-1M 3,820 lb (1 733 kg)
UH-1D/H[d]	6 ft 2 in (1.880 m)	4 ft 1 in. (1.245 m)	7 ft 8-1/2 in. (2.350 m)	8 ft 0 in. (2.438 m)	4 ft 4 in. (1.321 m)	4,000 lb (1 814 kg)	3,344 lb (1 517 kg)
CH-47B/C	7 ft 6 in. (2.286 m)	6 ft 6 in. (1.981 m)	30 ft 6 in. (9.296 m)	7 ft 6 in. (2.286 m)	6 ft 6 in. (1.981 m)	20,000 lb (9 072 kg)	CH-47B 15,000 lb[e] (6 804 kg) CH-47C 18,200 lb[e] (8 255 kg)
CH-54A/B	NA	NA	NA	NA	NA	CH-54A 20,000 lb (9 072 kg) CH-54B 25,000 lb (11 340 kg)	CH-54A 13,773 lb (6 247 kg) CH-54B 16,173 lb (7 336 kg)
CH-54A/B universal military pod	8 ft 10 in. (2.692 m)	6 ft 6 in. (1.981 m)	27 ft 2-1/2 in. (8.293 m)	8 ft 10 in. (2.692 m)	6 ft 6 in. (1.981 m)	NA	16,980 lb (7 701 kg) (maximum)

a/ Maximum load for cargo hook. For external load missions each load must be analyzed, therefore, the maximum capacity may not be a valid limitation for external load missions.

b/ Basic mission payload is based on standard day conditions as defined by the International Civil Aeronautics Organization (an atmosphere of dry air of 29.92 inches (0.760 m) of mercury (H_g) pressure, 59°F (15°C) temperature, and zero wind) for a 50-mile (80.45-km) radius mission.

c/ Fixed-wing aircraft; all others shown are rotary-wing aircraft.

d/ Transmission housing in center of aft end of cargo compartment 35 inches (0.889 m) long and 28 inches (0.711 m) wide. For additional data see Army aircraft operator's manuals (TM 55- series).

e/ Respective payloads are for a 100-mile (160.9-km) radius mission for a nontactical configured helicopter at maximum gross weight (40,000 pounds (18 141 kg) for CH-47B and 46,000 pounds (20 862 kg) for CH-47C). The payloads are derived from data shown in figures 14-22 and 14-92, TM 55-1520-227-10, except that allowance is made for 30-minute fuel reserve, 2,000 feet (609.6 m) altitude and 95°F (35°C).

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5.4.12 External lift by Army rotary-wing aircraft. Materiel and combat developers may obtain engineering and design assistance, for materiel to be air transported as externally suspended cargo by rotary-wing aircraft, from the Applied Technology Laboratory, US Army Research and Technology Laboratories (AVRADCOM), Fort Eustis, Virginia 23604. The following criteria shall be adhered to when developing lift-point structural requirements for materiel that is to be externally transported by helicopter (AVRADCOM technical report No. 72-36).

5.4.12.1 Number of lift points. Four lift points are to be provided, where possible. In cases where four lift points are not feasible, a proposed modification incorporating less than four lift points requires prior approval, as stated in paragraph 5.4.12.9.

5.4.12.2 Angle of sling leg. Maximum resultant forces that are to be considered shall be calculated for a sling leg inclined no more than 45 degrees (0.79 rad) true angle with the vertical, or a maximum included angle of 90 degrees (1.57 rad). This is not applicable to items with only one lift point.

5.4.12.3 Basis for materiel classification. All types of materiel do not behave in the same manner when suspended from a helicopter in flight. However, materiel with similar flight characteristics may be classified using the ratio of the weight of the item to its maximum projected frontal area. When an item is suspended from a single point suspension system, the projected frontal area will be the maximum area projected on a vertical plane perpendicular to the line of flight, as the item is rotated about a vertical axis through the suspension point. For a rectangular solid, the projected frontal area will be at a maximum when the diagonal of the rectilinear solid is perpendicular to the line of flight. When an item is suspended from a multipoint system, the materiel is not permitted to rotate about a vertical center line; therefore, the maximum projected area will be the area of the front projection of the item.

5.4.12.4 Classes of materiel.

a. Type A materiel has a weight to maximum-projected-frontal-area ratio of 60 pounds per square foot (293 kg/m^2) or greater. Typical examples of Type A loads are trucks, trailers, trailer-mounted equipment, engineer equipment, self-propelled guns, howitzers, and forklift trucks.

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b. Type B materiel has a weight to maximum-projected-frontal-area ratio of less than 60 pounds per square foot (293 kg/m^2). These normally are large, light items. Typical examples of Type B loads are shop van boxes, empty cargo containers, communications huts, missile cases, and shelters.

5.4.12.5 Strength of lift points. Each multipurpose or slinging eye, including the connection to the structural frame, shall withstand without permanent deformation, a working load equal to the maximum static resultant (tension) force times a lift-point load factor. The maximum static resultant force is obtained by vectorially adding the maximum horizontal component and the maximum vertical component of the force components, calculated for the individual lift points under deadweight loading conditions, even if they do not occur at the same lift point. Ultimate strength should equal 1.5 times the maximum static resultant force, times a lift-point load factor. For items that can be shipped in a loaded or unloaded condition, the required lift-point strength shall be determined for the loaded condition.

5.4.12.6 Lift-point load factor. The following load factors shall be used in determining the strength of the lift points:

a. Type A materiel - 3.2 (materiel less than 20,000 pounds (9 072 kg)
- 2.3 (materiel 20,000 pounds or greater) (9 072 kg)

b. Type B materiel - Requires special consideration. Contact AVRADCOM (paragraph 5.4.12) for design data.

5.4.12.7 Dimensional requirements of lift points. To be compatible with the sling systems used to suspend the materiel, the required lift-point dimensional requirements are given in MIL-STD-209. If a shackle is attached to each lift point, all shackles shall conform to MIL-STD-209 and meet the lift-point ultimate strength requirements of paragraph 5.4.12.5.

5.4.12.8 Location of lift points. To achieve stability in flight and to be compatible with the slings used to suspend the materiel, the following procedures governing the lift points shall be used:

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a. For locations in the vertical plane, three- and four-point lift configurations shall be located above the center of gravity of the materiel, in both the empty and the fully loaded conditions. If this requirement cannot be met, the lift points shall be so located that the center of gravity (CG) falls within a triangle whose apex angle is 120 degrees (2.09 rad) and whose base leg is formed by the lift points (figure 36). For

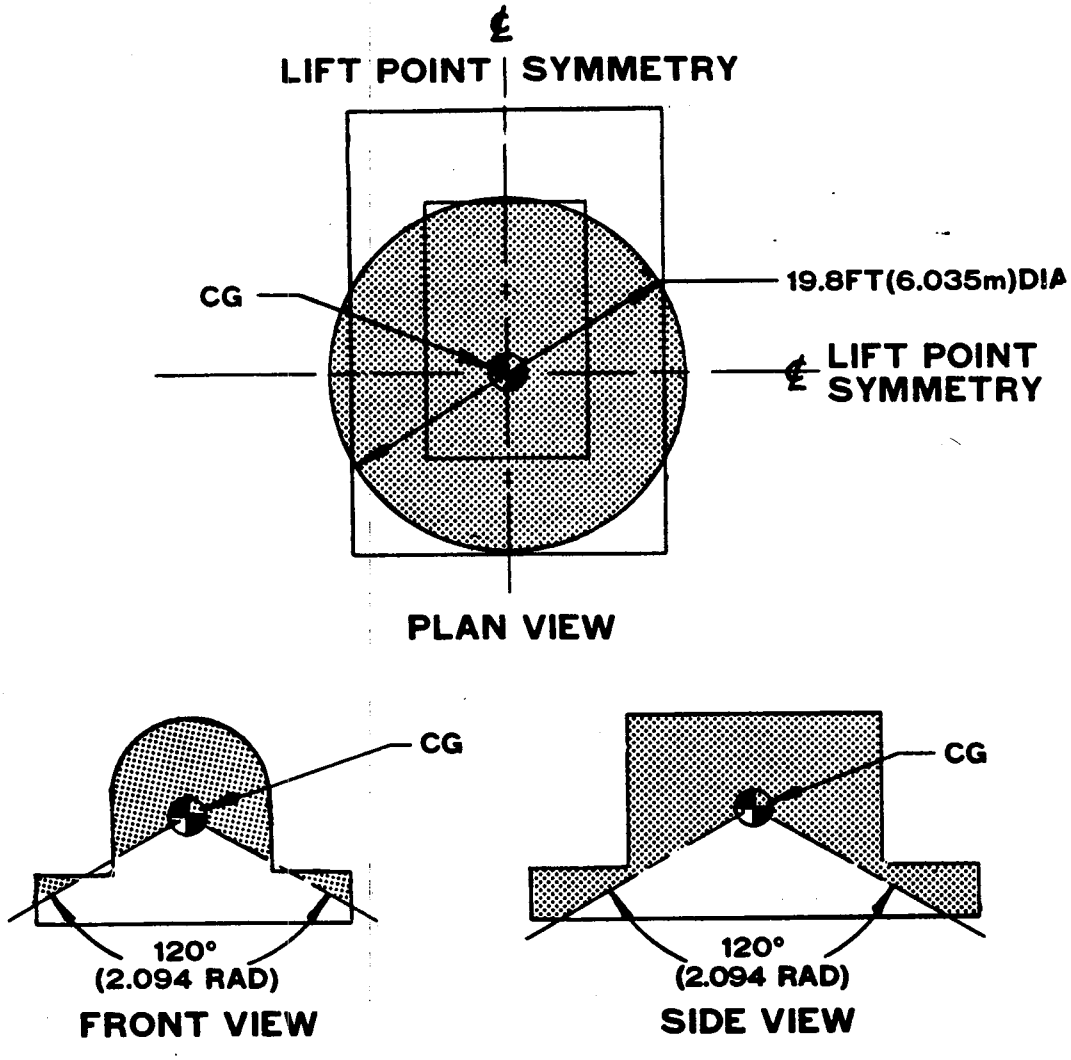


FIGURE 36. Lift-point center of gravity relationship, four- and three-point lift.

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single- and two-point lift configurations, the lift point(s) shall be located above the center of gravity of the materiel, high enough so that at least 60 percent of the maximum projected vertical area lies below the lift point(s).

b. For location in the horizontal plane, all lift points shall be located as far apart as possible inside a circle whose diameter is 19.8 feet (6.035 m), with the materiel center of gravity location as its center. All lift points shall be located symmetrically about a longitudinal axis and a lateral axis, both axes passing through the materiel center of gravity (figure 36). If exact symmetry cannot be achieved, limited asymmetry is permissible provided the ratio of the largest vertical force to the smallest vertical force does not exceed 1.2. In the case of helicopters with multipoint suspension, the materiel lift points shall be located so that the tension member fleet angle, which is the angle between the vertical and the line of action, does not exceed 20 degrees (0.35 rad) when the materiel is lifted clear of the ground. For those cases in which other conditions preclude use of these criteria (for example, containers), lift points should be designed for the most severe conditions; however, the design requires prior approval, as stated in paragraph 5.4.12.9.

c. To obtain the required sling-leg/materiel clearance, lift points shall be located so that, when the materiel is freely suspended, there is a minimum clearance of 8 inches (0.203 m) between the center line of the sling leg and the materiel. If the required clearance cannot be obtained, relocate the lift points, or remove or relocate the offending part on the materiel, or investigate the use of spreader bars or cushioning, and submit proposals, as stated in paragraph 5.4.12.9.

5.4.12.9 Deviations. Lift-point proposals, modifications, or special considerations that deviate from the above requirements shall be directed to the appropriate service transportability agent for approval (paragraph 4.1).

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5.4.12.10 Sling assembly criteria. The following criteria apply for the design of sling systems: resultant reactive forces due to sling leg inclinations to the vertical; aerodynamic forces imposed by a helicopter capable of producing a vertical acceleration of 2.5G at the point of attachment to the helicopter hook; a factor for load asymmetry based on a 10-percent horizontal plan center of gravity shift; and a system safety factor. Not included are factors for actinic and temperature degradation that are functions of the specific materials selected for the sling. The ultimate strength value of the sling legs is obtained by multiplying the weight of the load by appropriate factors shown in table XV and then dividing by the number of sling legs in the system.

TABLE XV. SLING SYSTEM FACTORS

Maximum included angle, degrees	Dynamic plus leg inclination factor	Load asymmetry factor	System safety factor
120 (2.094 rad)	5.8	1.45	1.5
100 (1.745 rad)	4.5	1.17	1.5
90 (1.571 rad)	4.1	1.12	1.5
65 (1.134 rad)	3.4	1.04	1.5

5.4.12.11 Tiedown provisions. Tiedown provisions should be in accordance with paragraph 4.7.

5.4.12.12 Restraint criteria. The internal cargo-restraint factors presented in table XVI are for the Army U-21 fixed-wing aircraft and for all rotary-wing aircraft in which personnel are not protected by aircraft structures and/or other barriers.

TABLE XVI. CARGO RESTRAINT FACTORS

Restraint	U-21 fixed-wing aircraft	Rotary-wing aircraft
Forward	9.0	4.0*
Aft	1.5	2.0
Vertical down	6.6	4.0
Vertical up	3.0	2.0
Lateral	1.5	1.5
*Where personnel are protected, a forward factor of 2.0 will be used.		

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5.4.13 Airdrop system. Combat and support materiel that is airdropped to combat forces by parachute and/or assault landings shall be capable of immediate effective employment and meet the requirements specified in MIL-STD-669. An airdropped item, from the time it is loaded in the aircraft until it is recovered on the ground, will have been subjected to a force-loading environment due to the following considerations:

a. Restraint in the aircraft for flight safety.

b. Deployment of the extraction system.

c. Deployment of the retardation system.

d. Deceleration at ground impact.

Although lightweight, small materiel (2,200 pounds (998 kg) gross rigged weight or less) may be airdropped in containers. This handbook addresses only materiel that may be airdropped on airdrop platforms. This includes gross-rigged weights from 2,500 pounds (1 134 kg) to 35,000 pounds (15 876 kg). Materiel developed shall be designed to withstand the forces encountered in an airdrop environment. All airdropped items shall receive prior approval by ASD/ENECA.

5.4.14 Platform airdrop systems. Current and developmental airdrop systems, for use with Air Force aircraft, employ a platform on which airdrop materiel is rigged, using the current energy-dissipater system to provide maximum protection against damage on ground impact. The airdrop materiel is restrained within the aircraft by strapping the materiel to the platform, using tiedown straps and the materiel tiedown provisions. At present, two different methods of extraction and suspension are used. For certain bulk cargo and light materiel (up to 8,000 pounds (3 629 kg)), the extraction and suspension provisions are mounted on the platform; for the majority of materiel, the extraction and suspension provisions shall be an integral part of the materiel to be airdropped. Platform-rigged airdrop loads experience a tumbling action during deployment of the parachute system, creating forces that require consideration in the design of suspension provisions. Force versus time histories show that extraction forces reach a magnitude of plus 1.25G in 4 seconds after initiation of extraction; suspension sling forces rarely exceed 1G levels. The loading environments for items of military materiel, when airdropped as platform loads, are of less magnitude than those experienced during helicopter external lift, except for ground impact loads.

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5.4.15 Standard heavy airdrop.

5.4.15.1 Maximum gross-rigged weight.

a. The following approximate allowable airdrop gross-rigged weight limitations (including platform, parachutes, webbing, or hardware) are shown for indicated aircraft:

C-130	35,000 pounds (15 876 kg)
C-141	35,000 pounds (15 876 kg)

Currently, due to roller load limitations, only a small number of loads have been approved to exceed 25,000 pounds (11 340 kg).

b. To estimate the gross-rigged weight of materiel to be airdropped from aircraft, use the following equation: gross-rigged weight equals approximately 1,200 pounds (544 kg) plus 1.16 times the weight of the unrigged item.

5.4.15.2 Dimensional constraints.

a. Materiel developed to be air droppable shall be designed so as not to exceed the dimensional limitations of the airdrop aircraft and airdrop platforms. The unit-rigged load shall meet the limitations specified in MIL-A-8421. Refer to applicable TO1C-XXX-9 for additional limitations.

b. For C-130 and C-141 aircraft, the width limitation is 108 inches (2.743 m). Base width of the materiel shall not be wider than the usable width of the airdrop platform which is 101-1/4 inches (2.571 m).

c. The overall height of the materiel, including energy dissipater, parachutes, or other retardation systems, when rigged on the airdrop platform, shall not exceed the dimensional limitation envelope curves shown in figures 37 and 38.

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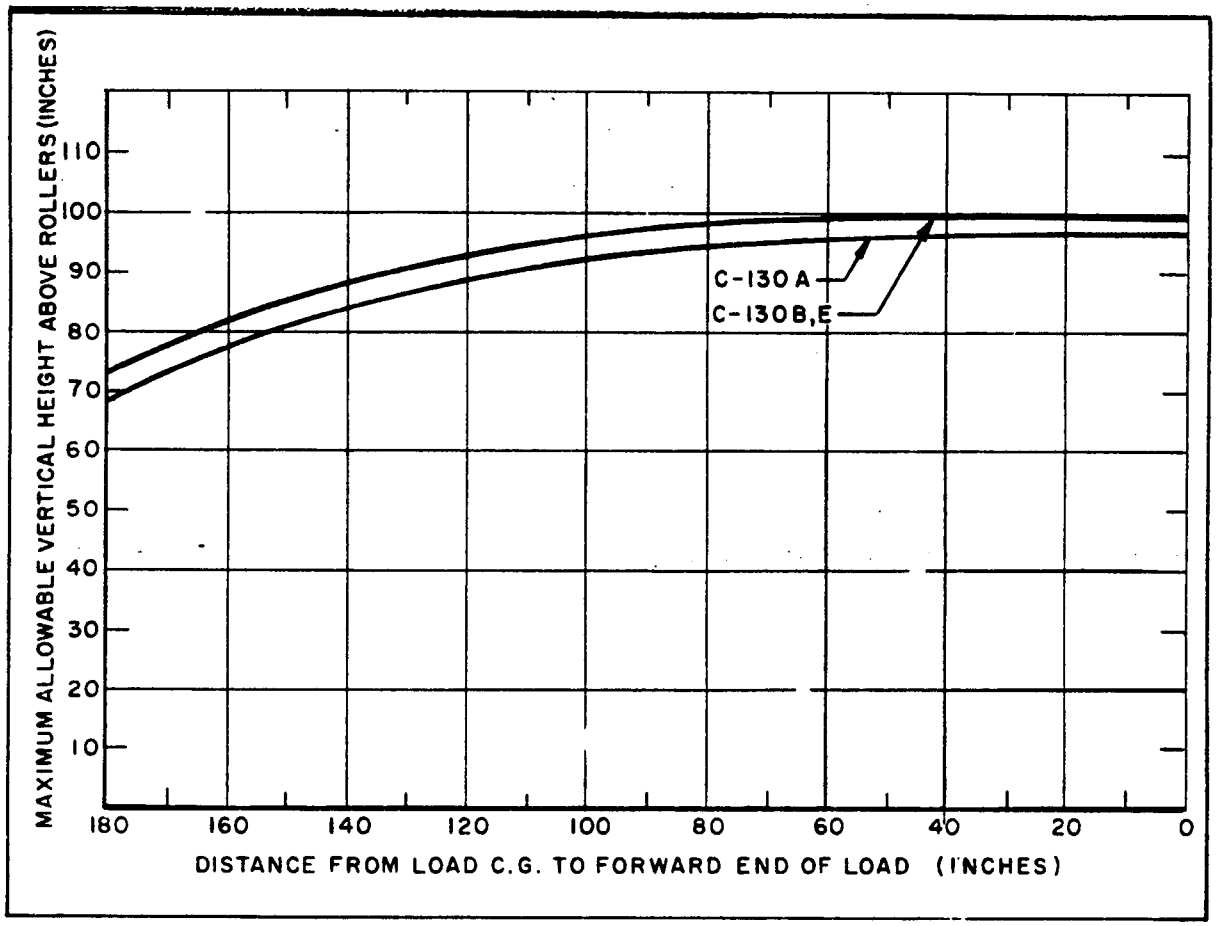


FIGURE 37. Dimension limitation envelope for extracted loads for the C-130 aircraft.

d. The longest airdrop platform that can be used with current airdrop aircraft is 288 inches (7.315 m) for the C-130 and C-141. Maximum forward and rear overhang of the load, from the edge of the platform, for the C-130 and C-141 aircraft, is 36 inches (0.914 m) and shall not extend below an imaginary plane extending up from the lower edge of the platform at an angle of 30° (0.524 rad) with the horizontal.

e. All standard platform loads are rigged in accordance with joint Air Force/Army publication FM 10-500/TO 18C7-1-5.

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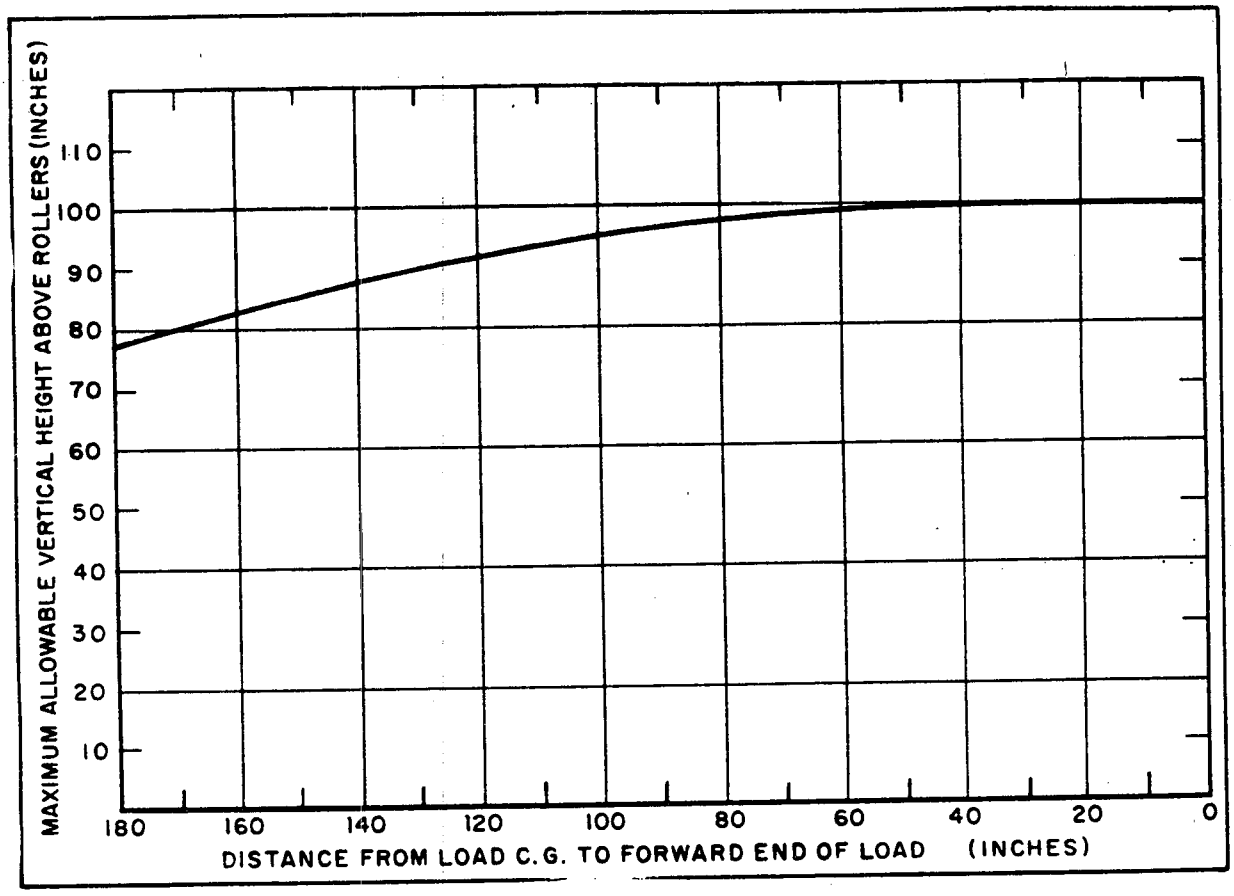


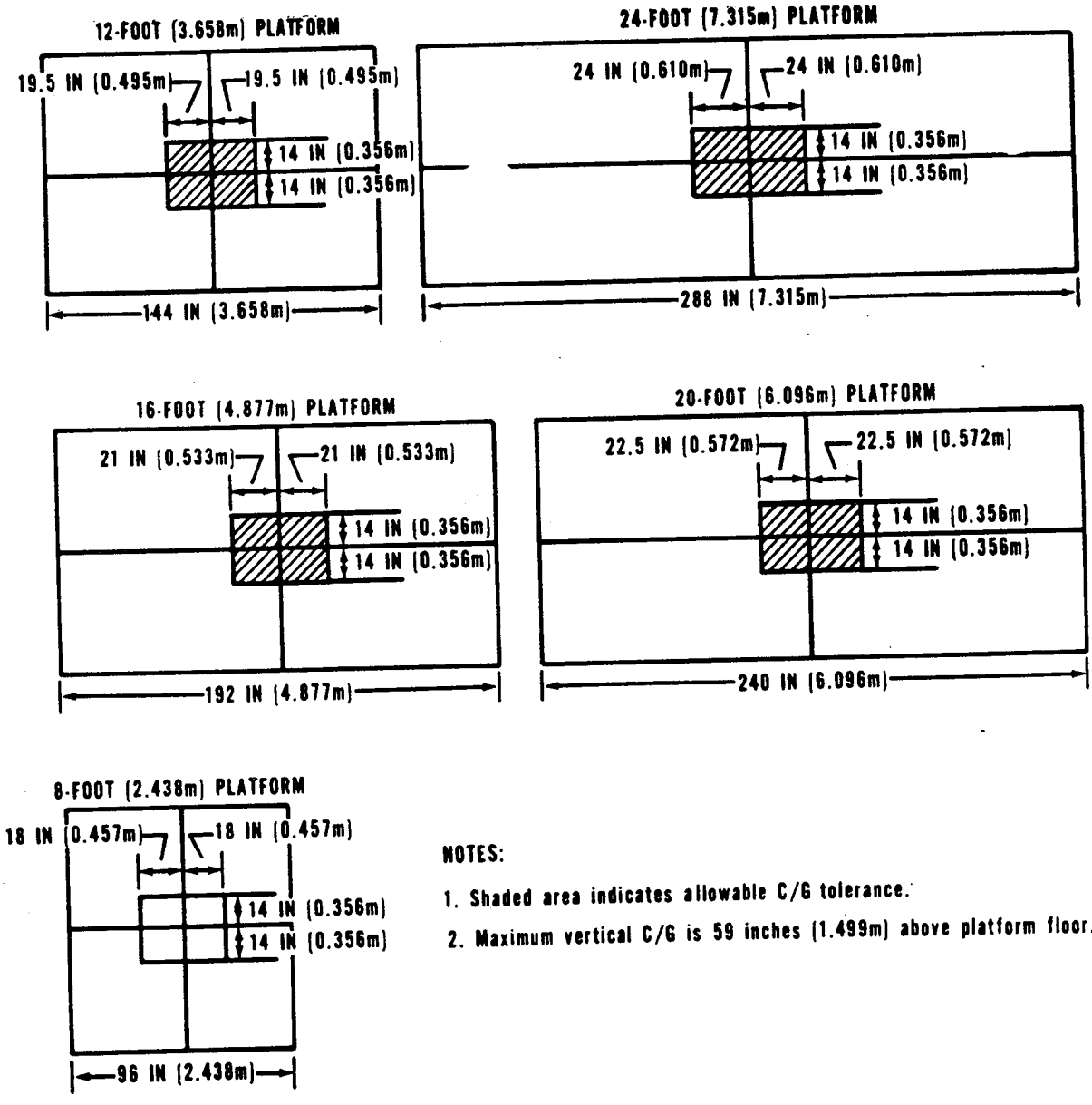
FIGURE 38. Dimension limitation envelope for extracted loads for the C-141 aircraft.

f. The center of gravity of the completely rigged load must be maintained as nearly as possible at the center of the platform. The center of gravity limits are shown in figure 39.

5.4.15.3 Energy-dissipater materiel. Materiel to be airdropped shall be designed to withstand the dynamic compressive stresses at impact and the stresses due to the relative motion of the parts of the items. There are a number of materials that have been, or are being, used to absorb the final kinetic energy of an airdropped load at ground impact. Paper honeycomb core material exhibits excellent characteristics as a shock absorber and is used extensively. This material is utilized in blocks, with the parallel cells aligned in the vertical direction between the load and the platform. Since its energy-absorbing characteristics are utilized in a

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direction normal to its expansion, the honeycomb core material shall be expanded prior to installation on the platform. The current paper honeycomb energy-dissipater system conforms to MIL-H-9884, and consists of class 3, double-faced, 2-inch (0.076-m) thick panels.



NOTES:

1. Shaded area indicates allowable C/G tolerance.
2. Maximum vertical C/G is 59 inches (1.499m) above platform floor.

FIGURE 39. Center of gravity limits for airdrop platforms.

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5.4.15.4 Theory of cushioning. The theory of cushioning is described by the relation between the forces applied to a mass and the acceleration of the mass. For additional information refer to MIL-HDBK-304. These factors are governed by Newton's Second Law of Motion:

$$F_r = ma$$

where

F_r = the resultant force applied to the mass, lb (newton)

$m = \frac{W}{g}$ = mass, slug (kg)

W = weight of mass, lb (kg)

a = acceleration of center of gravity of mass, ft/sec^2 (m/s^2)

g = acceleration due to gravity = 32.2 ft/sec^2 (9.815 m/s^2)

This equation can be written:

$$F_r = WG$$

where

$G = \frac{a}{g}$ = level of acceleration compared to gravitational acceleration (dimensionless quantity)

For example, in figure 40, if F is larger than W, the mass is instantaneously accelerated upward, due to the combined forces of F and W and

$$F - W = \frac{Wa}{g} = WG$$

or

$$F = SA = W(G + 1)$$

where

S = the instantaneous stress in the cushion material, lb/ft^2 (Pa)

A = effective area of cushion, ft^2 (m^2)

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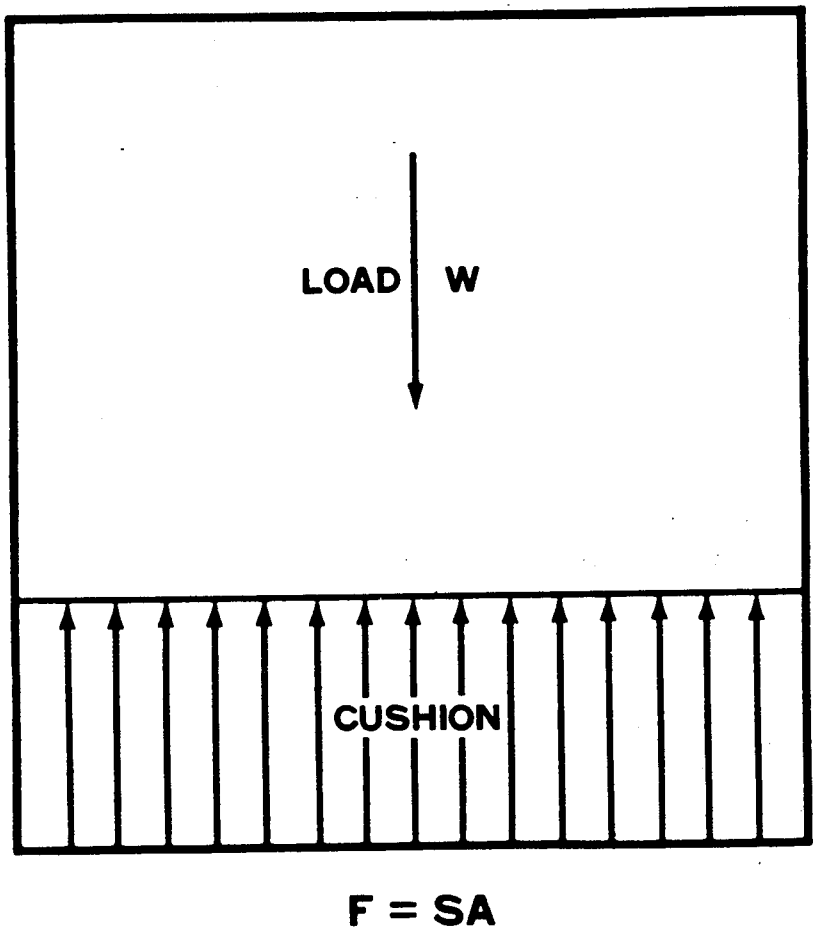


FIGURE 40. Mass-impacting cushion.

5.4.15.5 Energy-dissipater system for a standard load. For standard, low-velocity airdrops, a terminal velocity of 28.5 feet per second (8.69 m/s) is specified. A design value of 18.5 has been assigned for G. The G value is based on a desired maximum dissipater stack height of 12 inches (0.305 m). The deceleration force of G + 1 or 19.5 times the airdrop weight is met by using 3.25 square feet (0.30 m) of paper honeycomb crushing area for each 1,000 pounds (454 kg) of item airdrop weight. The 12-inch (0.305-m) thickness of paper honeycomb is composed of four layers of 3-inch (0.076-m) thick panels with a dynamic crushing stress of 6,000 pounds per square foot (287 280 Pa).

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5.4.15.6 High-velocity airdrop versus low-velocity airdrop. High-velocity airdrop items experience an impact velocity of 70 to 90 feet per second (21.34 to 27.43 m/s) compared with the 28.5 feet per second (8.69 m/s) specified for low-velocity airdrop items. The criteria in paragraph 5.4.15.5 of 18.5-G deceleration and 1-foot (0.305-m) thickness should not be utilized for high-velocity airdrop. The design value of 18.5 assigned for G in a low-velocity airdrop requires an impractical height of cushioning material when the impact velocity is increased to that experienced in a high-velocity airdrop. The specified cushion height of 1 foot (0.305 m) used in a low-velocity airdrop is also impractical for most items in a high-velocity airdrop. To accommodate a practical energy-dissipater system, high-velocity airdrop should be limited to items that will withstand a higher deceleration force than that specified for low-velocity airdrop Furthermore, high-velocity airdrop requires the use of a container. A value of 60 for G will yield a dissipater thickness under 2.5 feet (0.762 m) and is indicated as a reasonable initial test configuration.

5.4.15.7 Load spreader. The layers of paper honeycomb shall be glued to each other and to the skid or platform. A wooden load spreader shall be used when the underside of the item in contact with the energy-dissipater stack does not present a flat contact surface. The load spreader shall be glued to the top of the stack and shall be of sufficient size and thickness to insure full crushing of the stack over its entire area.

5.4.15.8 Load distribution. Consideration shall be given to the distribution of the dissipater area on the under surface of the item and to the total area of the dissipater material used. For items such as vehicles, the required bearing area of the dissipater material is less than the under surface of the item. In these instances, the item shall withstand the dynamic compressive stresses at impact and the stresses due to the relative motion of the parts of the item. The distribution of the dissipater will affect the aircraft roller loadings, and this must be considered to prevent overloading the rollers.

5.4.15.9 Airdrop impact tests. Initial tests shall be made with deceleration force levels less than the G + 1 level specified in paragraph 5.4.15.5 for a standard load, while retaining a total 12-inch (0.305-m) thickness of honeycomb stack. The developing agency shall specify the force levels to be used in this test. The agency shall also provide the impact velocities and the total areas of the energy-dissipater stack for levels selected, and shall specify whether the impact test will be performed at contractor or Government facility. The US Army Natick Research and Development Command (STSNL-UBS), Natick, Massachusetts 01760, will

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provide engineering and design assistance to materiel and combat developers. The requirement of 28.5 feet per second (8.69 m/s) ground-impact velocity, specified in paragraph 5.4.15.5 for a standard load, will be attained when the item is free-dropped from a height of 12.7 feet (3.886 m), measured from the lowest point on the bottom of the skid of platform upon which the item is positioned to the impact surface. The item shall be rigged for tests, using a skid or standard platform with the requisite number and sizes of energy-dissipater stacks, and load spreaders, if they are required. The dissipater configuration shall be designed to minimize rebound by placing the energy-dissipater material under rigid frame members of the item. The test shall be conducted using a level concrete impact surface. The platform skid should be approximately parallel to the impact surface prior to the drop, and shall strike the surface at an angle of not more than 2.5 degrees (0.044 rad) in any direction. After impact, the test item shall meet the performance requirements of the applicable end-item specification. The developing agency shall specify the instrumentation to be used with the simulated airdrop impact test. Instrumentation may include, but is not restricted to, high frame-rate (2,000 to 4,000 feet per second (610 to 1219 m/s)) motion-picture cameras, accelerometers, and strain gauges.

5.4.15.10 Airdrop materiel restraint. The unit-rigged load shall meet the restraint requirements of the applicable TO1C-XXX-9. Provisions and supporting structures shall be designed to secure materiel to airdrop platforms to comply with the strength requirement, number, location, and dimensions of MIL-STD-814.

5.4.15.11 Equipment suspension points. The equipment shall be designed to withstand rotational loading and to provide four suspension points to attach to the retardation system. Suspension provisions on airdrop items with a gross rigged weight of 25,000 pounds (11 340 kg) or less are required to have a hole of no less than 1-1/16-inch (0.026 m) diameter, with the centerline at least 2 inches (0.051 m) clear of the airdrop item. Drill the hole through material no thicker than 2 inches (0.051 m). Insure that the centerline of the hole is at a right angle to the longitudinal centerline of the airdrop item, or, to minimize bending loads, orient suspension provisions to match sling and clevis geometry. Comply with the dimensions shown on figure 41 for suspension provisions on airdrop items having a gross rigged weight under 25,000 pounds (11 340 kg). Comply with the dimensions shown on figure 42 for suspension provisions on airdrop items having a gross rigged weight over 25,000 pounds (11 340 kg). The limit load of each suspension provision shall be 1.5 times the suspended gross rigged weight of the item. The yield strength shall be 1.5 times the limit load. These forces are applicable in any direction within the shaded area

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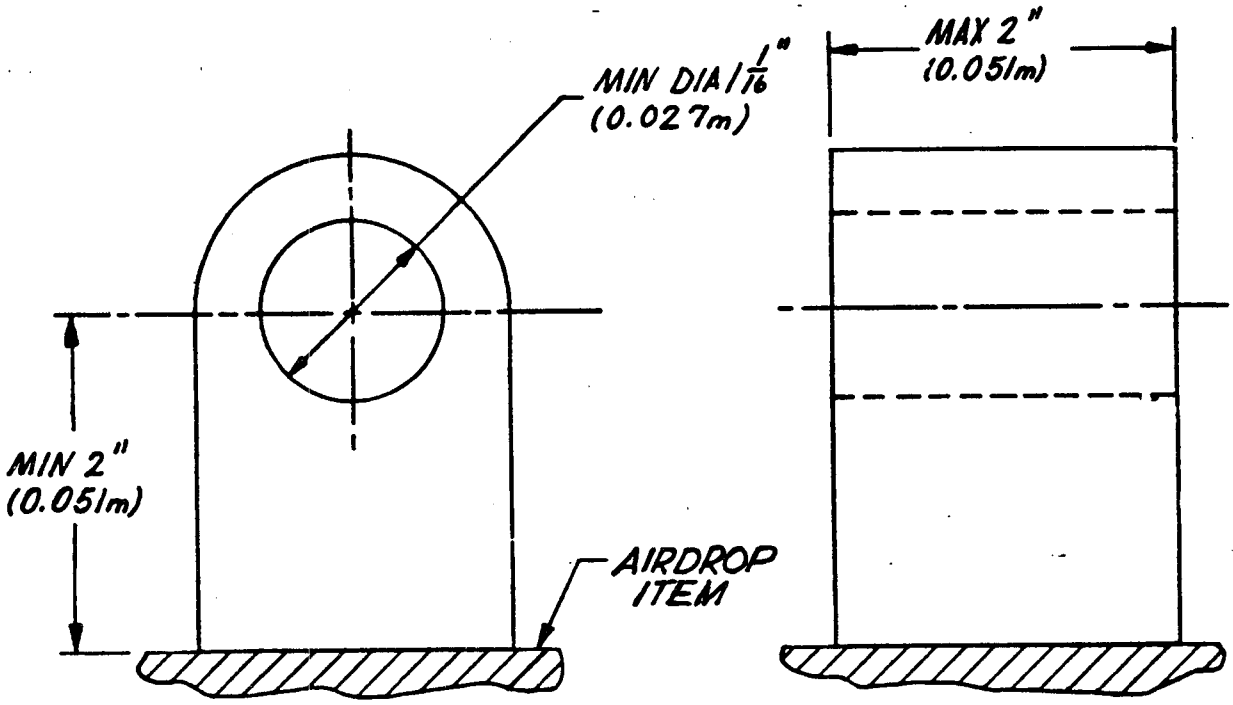


FIGURE 41. Suspension provision, dimensional requirements for gross rigged weights under 25,000 pounds (11 340 kg).

shown in figure 43. The required ultimate strength of each suspension provision shall be a minimum of 1.65 times the limit load. Locate the four provisions for each airdrop item as far apart as practicable, and, wherever possible, in a rectangular pattern about the vertical axis through the CG of the suspended load. Locate suspension provisions above the horizontal plane passing through the CG of the suspended load at such a height that interference between suspension slings and the item is minimized. The use of spreader bars is not authorized.

5.4.16 Equipment extraction points.

5.4.16.1 General. In the extraction of a load, various forces are applied to the cargo, requiring special design considerations. These forces are applied through two primary means, either through the airdrop platform or to the cargo itself.

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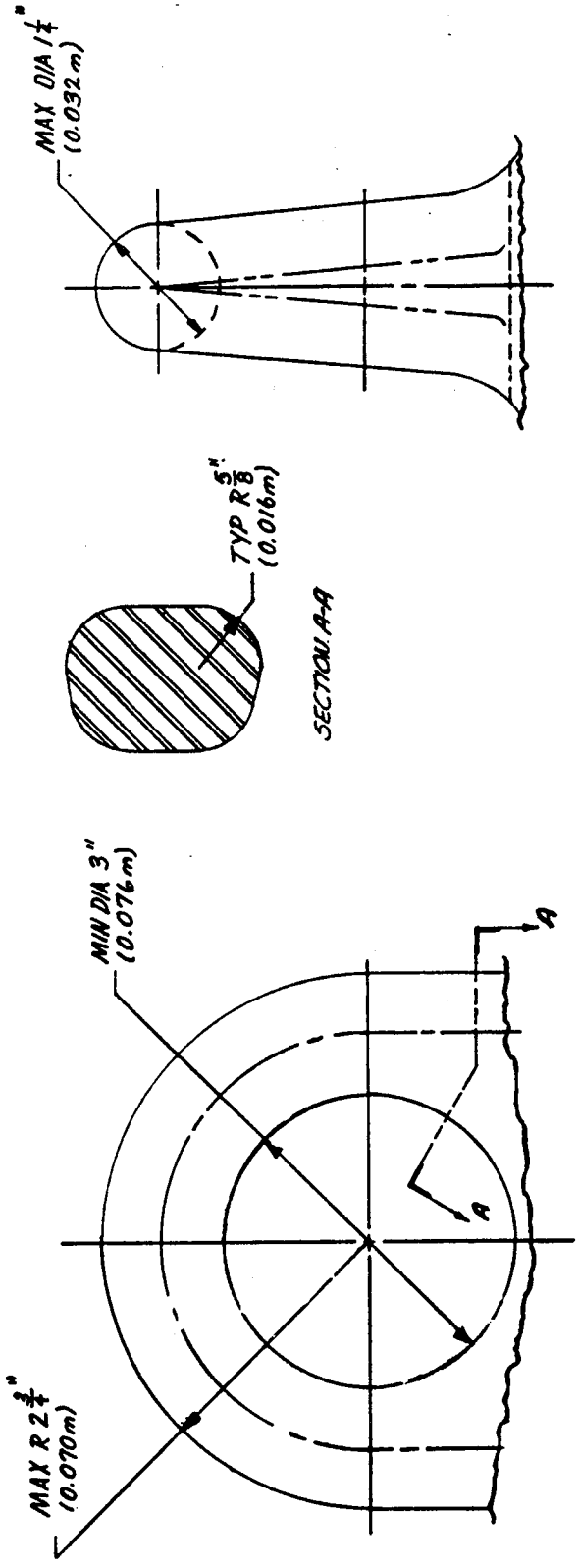


FIGURE 42. Suspension provision, dimensional requirements for gross rigged weights over 25,000 pounds (11 340 kg).

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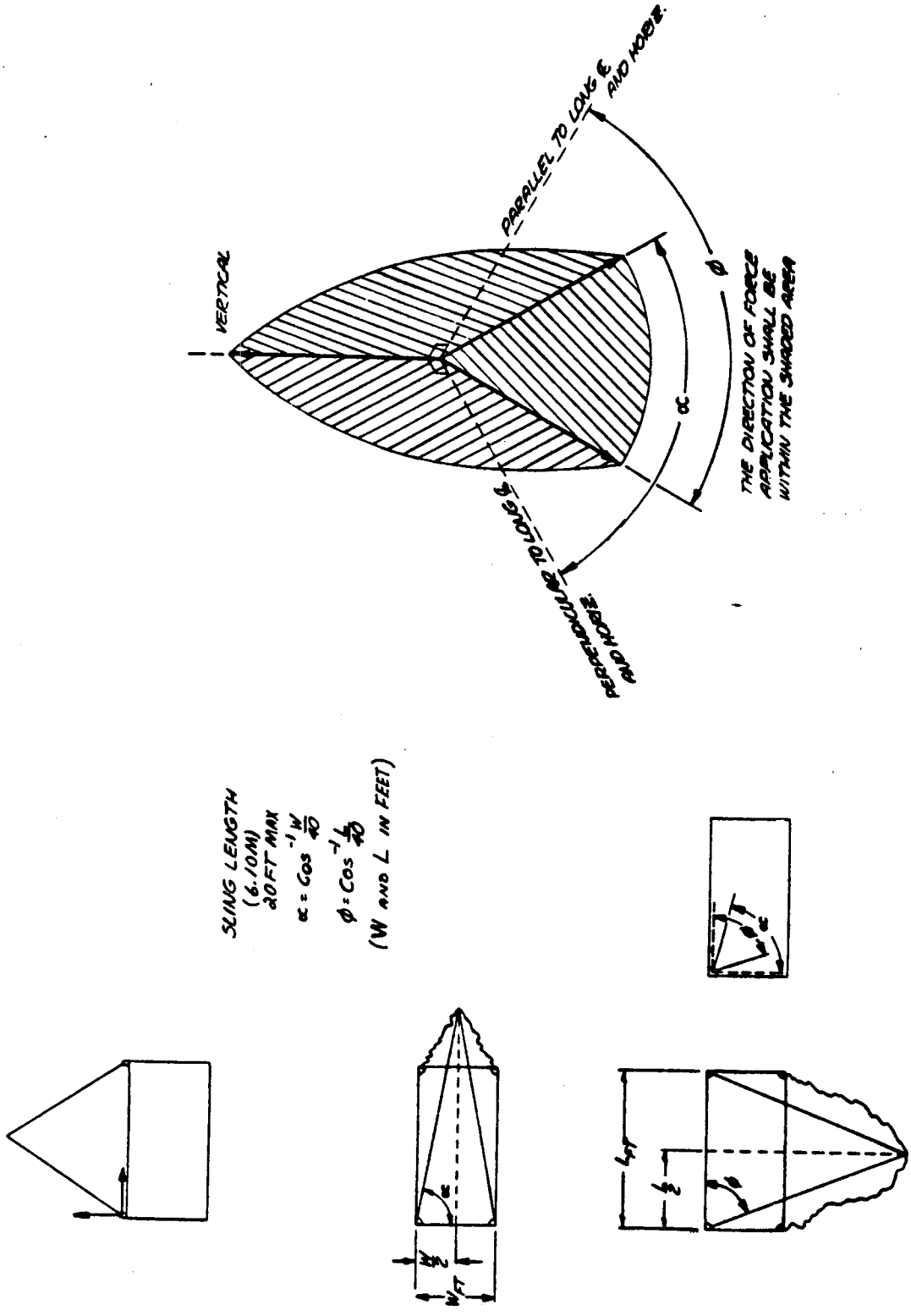


FIGURE 43. Suspension provision, direction of loading for gross rigged weights over 25,000 pounds (11 340 kg).

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5.4.16.2 Pintle extraction. In this method, a vehicle is secured to an airdrop platform. The vehicle has a pintle that is attached to the extraction parachutes for removal of the load from the aircraft. This extraction provision shall be located on the longitudinal centerline and below the CG of the item. The opening in the provision for attachment of the extraction system should be a minimum of 1-1/16 inches (0.026 m) in diameter and the centerline at least 1.375 inches (0.034 m) clear of the airdrop item. Typical dimensional requirements for extraction provisions on airdrop items under 25,000 pounds (11 340 kg) are shown in figure 44 and for those over 25,000 pounds (11 340 kg), figure 45. The limit load capacity of the extraction provision shall be 1.5 times the gross rigged weight. The yield strength shall be a minimum of 1.5 times the limit load. The greatest load capacity of the extraction provision used on an airdrop item that has a gross rigged weight less than 25,000 pounds (11 340 kg) shall be a minimum of 1.65 times the limit load and shall be a minimum of 1.75 times the limit load for gross rigged weight greater than 25,000 pounds (11 340 kg). These forces are considered as applied to the extraction provision shown in figure 46. In addition, the attachment point shall be rated at a yield strength of 3.0 times the equipment weight.

5.4.16.3 Platform extraction. The need for an extraction provision on a vehicle may be eliminated by use of three currently available aerial delivery platforms. The force is equally divided from the extraction line by a metal link to each side rail of the platform. The Type II modular and Type A/E29H-1 platforms, using the platform extraction force transfer coupling (PEFTC), may be used for loads under 14,000 pounds (6 350 kg). However, for loads in excess of this weight, the joint service Type V platform, using the extraction force transfer coupling (EFTC), may be utilized.

5.4.17 Low altitude parachute extraction system.

5.4.17.1 Maximum gross-rigged weight. Only the C-130 aircraft has been approved to perform the low altitude parachute extraction system (LAPES) mission. The maximum load (gross rigged weight) that may be extracted over the cargo ramp is 36,700 pounds (16 644 kg).

5.4.17.2 LAPES configuration criteria.

a. Platform. Only platforms specified in standard technical orders may be used in an airdrop test unless prior approval has been granted by ASD/ENECA.

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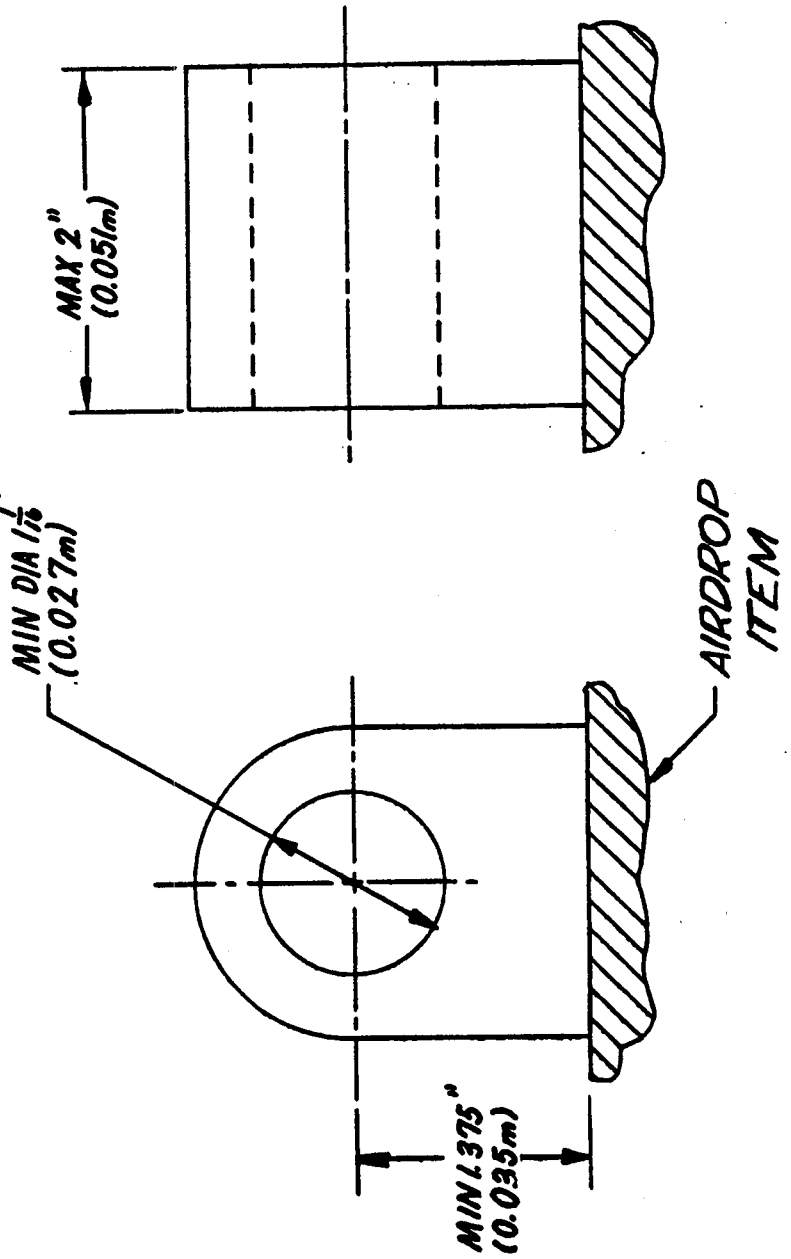


FIGURE 44. Extraction provision, dimensional requirements for gross rigged weights under 25,000 pounds (11 340 kg).

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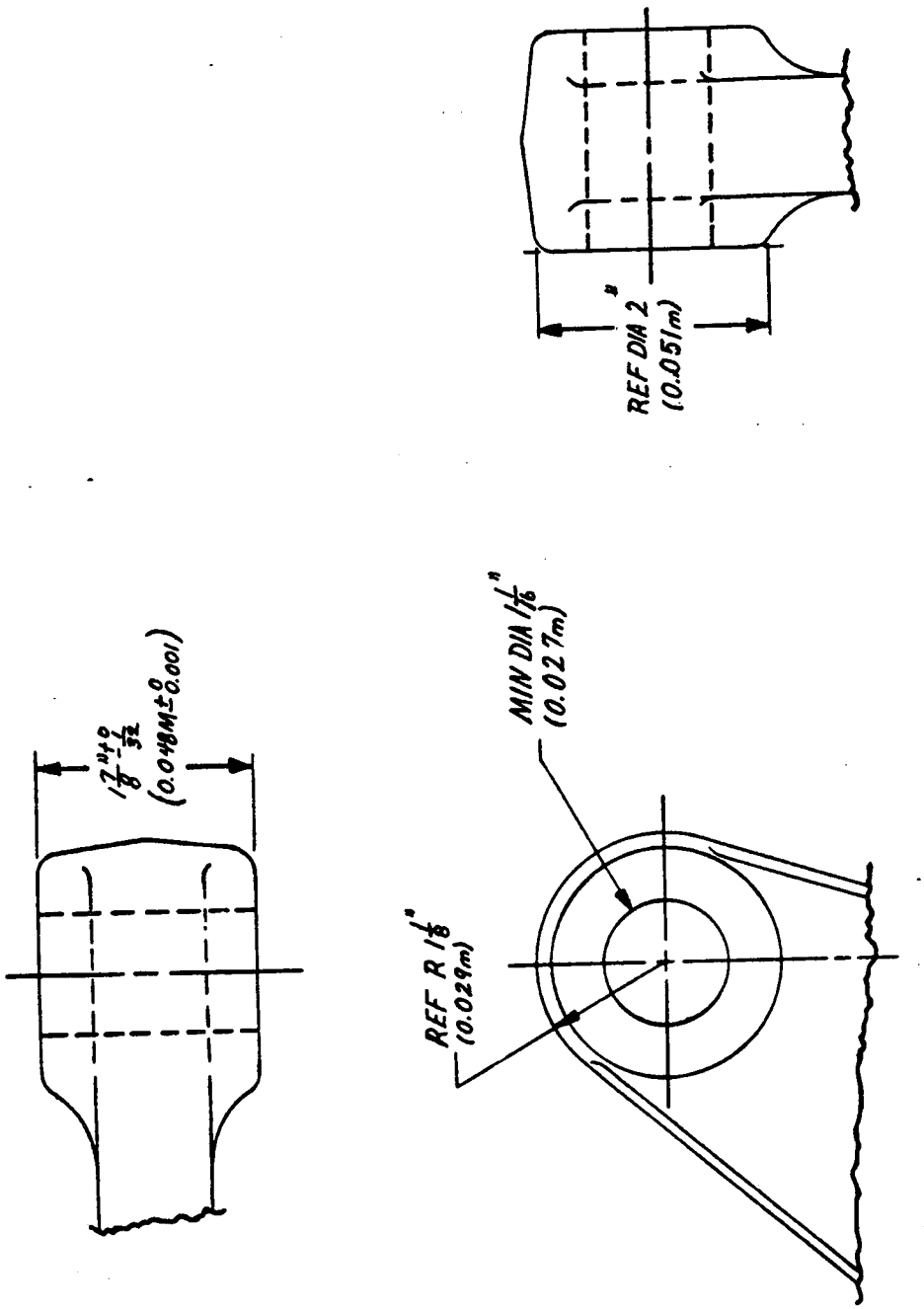


FIGURE 45. Extraction provision, dimensional requirements for gross rigged weights over 25,000 pounds (11 340 kg).

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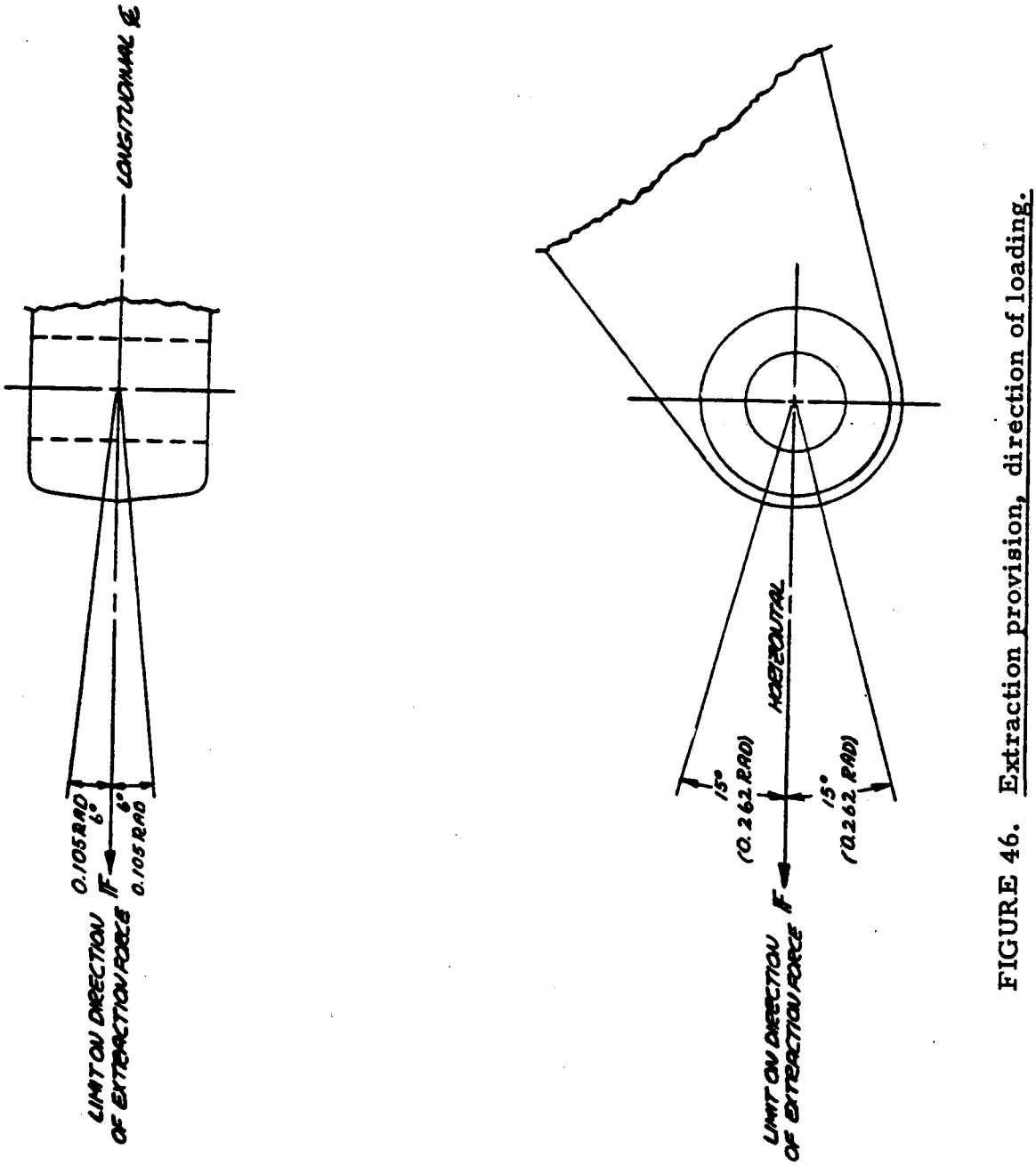


FIGURE 46. Extraction provision, direction of loading.

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b. Energy-dissipating material. When energy-dissipating material is placed between the equipment and the platforms, it shall be arranged to provide maximum load stability so as to preclude sway of the vehicle under turbulent flight conditions and during extraction. In addition, the energy-dissipating material is to be positioned in such a manner as to preclude overloading the aircraft floor/rollers. Refer to paragraph 5.4.15.3 for a description of energy-dissipating material and paragraph 5.4.15.4 for the theory of cushioning.

c. Load preparation. The load will be prepared as prescribed in the equipment maintenance handbook and appropriate TO 13C-series technical orders/FM 10-500 manuals.

NOTE: When equipment contains dangerous material, the packing and packaging of the dangerous material shall conform to AFR 71-4. When the dangerous material contains ammunition or explosives not qualified for LAPES, a safety certification must be obtained from MAC/IGFX, Scott AFB, Illinois, before requesting airdrop approval from ASD/ENECA.

d. Restraint. The load shall be restrained to the platform to withstand the following ultimate load factors:

	Restraint Required	
Type of Restraint	Platform Extraction	Item Extraction
Forward	12.0 G(1)	8.0 G
Aft	6.0 G	(2)
Lateral	6.0 G	6.0 G
Up	6.0 G	6.0 G

(1) G is based on the restrained load gross weight.
(2) Restraint required for item extraction using LAPES is equal to two times towed extraction parachute force.

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e. Load tiedown provisions. Tiedown provisions shall conform to MIL-STD-814. Physical dimensions and material composition, or static test results, shall be submitted to ASD/ENECA for approval.

f. Accompanying loads restraint. Accompanying loads will be restrained to the primary load in accordance with restraint criteria stated above. The structure of the primary load to which the accompanying load is restrained shall be carefully evaluated in regard to its ability to transmit (without failure) the forces introduced by the accompanying load. Accompanying loads placed on the primary load will in no case exceed the off-highway load capacity of the vehicle nor cause an overloading of the aircraft floor area/ roller conveyors.

g. Tiedowns. The use of tiedowns will conform to the rules outlined in the applicable TO 13C-series technical orders/FM 10-500 manuals. Nonstandard tiedowns require approval by ASD/ENECA. Requests for approval of nonstandard tiedown components submitted to ASD will be accompanied by design analysis and static test results.

h. Load dimensional limitation.

(1) No portion of the loaded equipment shall extend outboard of the vertical restraint flange of the modular platform (101-1/4 inches (2.571 m)).

(2) The allowable longitudinal height of the load (measured from the bottom of the platform) at any platform station shall be determined by figure 37.

(3) The projected lateral view of the rigged load shall allow a safety aisle, as described in MACR 55-130 Aircraft Operational Manual.

(4) Equipment that overhangs the ends of the platform shall not extend below an imaginary plane extending up from the lower edge of the platform at an

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angle of 30° (0.524 rad) with the horizontal. In no case will the overhang measure greater than 3 feet. The linear platform-loading distribution must be supplied by the requesting agency prior to load approval by ASD engineering. Aircraft linear floor and roller loading limits must not be exceeded.

(5) The center of gravity of the completely rigged load must be maintained as nearly as possible at the center of the platform. The center of gravity limits are shown in figure 39.

i. Platform loading limits. Depending upon the strength of the platform, the maximum load that can be placed on any platform is limited by the strength of the aircraft floor, as outlined in the TO 1C-130-9 loading handbook.

j. Equipment extraction points. In the extraction of a load, forces are applied to the cargo requiring special design considerations. These forces are applied by two primary means: through the airdrop platform or to the cargo itself.

(1) Load extraction. In this method, a vehicle is secured to an airdrop platform. The vehicle has provisions to attach the extraction parachutes for removal of the load from the aircraft. The opening in each of the provisions for attachment of the extraction system should be a minimum of 1-1/16 inches (0.026 m) in diameter, and the centerline at least 1.375 inches (0.034 m) clear of the airdrop item. The limit load capacity of the extraction provision(s) shall be 3.0 times the gross rigged weight. The yield strength shall be a minimum of 1.5 times the limit load. The ultimate strength shall be a minimum of 1.65 times the limit load.

(2) Platform extraction. The need for an extraction provision on a vehicle may be eliminated through the use of extraction slings and platform side plates which allow the extraction force to be applied directly to the platform.

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5.4.18 Nonstandard military airdrop materiel requirements. This handbook does not include airdrop provision requirements for many special-purpose items of military materiel being developed. The unique configuration or other characteristics of an item may require special consideration. The US Army Natick Research and Development Command (STSNL-UBS) and ASD/ENECA are available for assistance on problems relating to airdrop provision requirements for these items of military materiel.

5.5 Intermodal cargo containers.

5.5.1 General. A cargo container is transport equipment designed and constructed to facilitate the international and intermodal exchange of goods. It is permanent, strong enough for repeated use, and fitted with devices that permit its ready handling and transfer from one mode of transport to another. To take full advantage of the intermodal benefits of containerization, materiel should be transportable, where practicable, in ISO or ANSI containers (ANSI MH5.1.1).

5.5.2 Classification and dimensions of cargo containers.

5.5.2.1 ISO Series-1 containers. One series of cargo containers is approved for intercontinental traffic in compliance with ISO recommendations. Designated Series 1, these containers are 8 feet (2.438 m) wide and 8 feet (2.438 m) or 8 feet 6 inches (2.603 m) high.

5.5.2.2 ISO Series-2 containers. This series of containers has been deleted from ISO recommendations, but they are still in use in some of the European railway systems. They are smaller than Series-1 containers and have a uniform exterior height of 6 feet 10-1/2 inches (2.096 m) and widths of:

Series Designated	Width
2A	7 ft 6-1/2 in. (2.299 m)
2B	6 ft 10-1/2 in. (2.096 m)
2C	7 ft 6-1/2 in. (2.299 m)

CAUTION: Because the inventory of Series-2 containers may be limited, check their availability prior to designing items requiring their use, or before specifying their use for the movement of existing items.

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5.5.2.3 ANSI containers. The ANSI standard, ANSI-5.1.1, identifies 40-, 30-, 20-, and 10-foot (12.192-m, 9.144-m, 6.096-m, and 3.048-m) containers that have the same length and width as ISO 1A through 1D containers.

5.5.2.4 Length and capacities of containers. Capacities and dimensional characteristics of ISO and ANSI containers are shown in table XVII.

5.5.3 Gross operating weights. Gross operating weights of ANSI containers are as follows:

a. Marine and rail. Same as design gross weight, except the US Coast Guard (USCG) restrains the gross weight of some commercial containers used for ammunition and explosives.

b. Highway. Subject to federal, state, and local regulatory requirements, but not to exceed design gross weight.

c. Air. To assure that the container and its payload do not exceed the structural limitations of the transporting aircraft, container gross operating weights shall not exceed those shown in table XVIII for the specific type of aircraft. Fixed-wing data are for commercial aircraft. Refer to paragraph 4.1.1e for US Air Force aircraft.

5.5.4 Door openings. Standards require that each cargo container be provided with a door opening at one end. The door opening will be the full rectangular opening, measured from the top of the floor to the underside of the door header and from the inside faces of the door posts or any other interfering door member. Minimum door clearance height should be the nominal outside height minus 12 inches (3.048 m). Minimum door clearance width should be the nominal outside width minus 6 inches (1.524 m). These dimensions establish the maximum theoretical height and width envelope for a single item of cargo capable of unrestricted movement in standard commercial cargo containers.

5.5.5 Interior dimensions. Minimum inside length shall be the minimum outside length permissible by tolerance minus 7-1/2 inches (1.905 m). Tolerances are given in ANSI MH 5.1-1.

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TABLE XVII. STANDARD ANSI AND ISO CARGO CONTAINERS

Nominal length in feet (meters)	Nominal height/ISO classification[1]					Door opening in inches (meters)		Interior dimensions in inches (meters)			Design gross weight, lbs (kg)
	⩽8'0" (2.438 m)	8'0" (2.438 m)	8'6" (2.591 m)	9'0" (2.743 m)	9'6" (2.896 m)	width[2]	height[6]	length[3]	width[4]	height[7]	
40 (12.192)	1AX	1A	1AA	ANSI	ANSI[5]	90 (2.286)		472.5 (12.002)	92 (2.337)		67,200 (30 480)
35 (10.668)	NA	ANSI	ANSI	ANSI	ANSI[5]	90 (2.286)		412.5 (10.478)	92 (2.337)		61,600 (27 940)
30 (9.144)	1BX	1B	1BB	ANSI	ANSI[5]	90 (2.286)		352.5 (8.954)	92 (2.337)		56,000 (25 400)
24 (7.315)	NA	ANSI	ANSI	ANSI	ANSI[5]	90 (2.286)		280.5 (7.125)	92 (2.337)		52,500 (23 810)
20 (6.096)	1CX	1C	1CC	ANSI	ANSI[5]	90 (2.286)		232.5 (5.906)	92 (2.337)		44,800 (20 320)
10 (3.048)	1DX	1D	NA	NA	NA	90 (2.286)		112.5 (2.858)	92 (2.337)		22,400 (10 160)

Legend:

ISO = International Organization for Standardization
ANSI = American National Standards Institute
NA = Not applicable

Notes:

1 Container sizes identified as "ANSI" are approved only by ANSI.
2 Nominal width minus 6.0 inches (0.152 m).
3 Nominal length minus 7.5 inches (0.191 m).
4 Nominal width minus 4.0 inches (0.102 m).
5 Equipped with "tunnel sections" around kingpin.
6 Nominal height minus 12.0 inches (0.305 m).
7 Nominal height minus 9.5 inches (0.241 m).

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TABLE XVIII. GROSS OPERATING WEIGHTS

Container nominal length (ft)	Rotary wing (lb)	Fixed wing (lb)
40 (12.192 m)	45,000 (20 412 kg)	45,000 (20 412 kg)
30 (9.144 m)	45,000 (20 412 kg)	35,000 (15 876 kg)
20 (6.098 m)	44,800 (20 321 kg)	25,000 (11 340 kg)
10 (3.048 m)	22,400 (10 161 kg)	12,500 (5 670 kg)

5.6 Special-purpose containers.

5.6.1 General. The quantity of special-purpose containers for DOD requirements is limited. Prior to specifying a particular type, a determination of the inventory available should be made.

5.6.2 Platform containers. These containers have the same exterior dimensions as other standard cargo containers and basically consist of a platform with corner posts. One of their advantages is that loading is not restricted by the size of the door opening. They are available in lengths of 40, 35, 24, and 20 feet (12.192 m, 10.668 m, 7.315 m, and 6.096 m). Payload capacity varies between 40,000 pounds (18 144 kg) and 50,000 pounds (22 680 kg). Platform containers are particularly suited for carrying equipment and vehicles that will not fit into an 8-foot (2.438-m) wide cargo container, but will fit into container cells.

5.6.3 Bulk liquid containers. These containers are designed for transporting bulk liquids and powders of many varieties, both hazardous and nonhazardous. They carry liquid cargo or cargo capable of being liquefied. Their lengths vary from 6 feet 10 inches (2.083 m) to 40 feet (12.192 m). Capacities range from 600 gallons (2.28 m^3) to 5,000 gallons (19.00 m^3).

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5.6.4 Open-top containers. These containers meet ANSI/ISO standards for size, corner fittings, load factors, and handling and coupling devices. Open-top containers are used primarily for heavy, bulky cargo that can be loaded better through the top than through a door. A tarpaulin is used when a cover is required.

5.6.5 Half-height containers. These containers meet ANSI/ISO standards for corner fittings, load factors, and handling and coupling devices and are available in several configurations, such as platform and open-top with collapsible corner posts or sides when empty. Such containers are used for transporting dense loads.

5.6.6 Collapsible small containers. Collapsible containers are not constructed to ANSI/ISO standards. They are used primarily for consolidation operations and are adaptable for stowing two-high into standard ISO cargo containers. These small containers are available in two types.

5.6.6.1 Type 1 containers. Type 1 containers have a capacity of 46 cubic feet (1.303 m^3) with a load of 2,205 pounds (1 000 kg) and a tare weight of 176 pounds (36 kg). Exterior dimensions are 3 feet 8 inches (1.117 m) by 3 feet 9 inches (1.117 m) by 3 feet 5 inches (1.041 m). Interior dimensions are 3 feet 5 inches (1.041 m) by 3 feet 5 inches (1.041 m) by 3 feet 1 inch (0.904 m).

5.6.6.2 Type 2 containers. Type 2 containers have a capacity of 93 cubic feet (2.654 m) with a load of 4,409 pounds (2 000 kg) and a tare weight of 353 pounds (160 kg). Exterior dimensions are 7 feet 4 inches (2.235 m) by 3 feet 8 inches (1.117 m) by 3 feet 4 inches (1.041 m). Interior dimensions are 7 feet 1 inch (2.160 m) by 3 feet 5 inches (1.041 m) by 3 feet 1 inch (0.940 m).

5.6.7 Air mode containers.

5.6.7.1 Commercial airlift. Jumbo commercial aircraft, such as the Boeing 747F freighter, can carry air mode van-type containers up to 40 feet (12.192 m) long. These containers are generally designed to the configuration of a particular aircraft. For specific container dimensional or weight constraints, contact the International Air Transport Association (IATA), 1000 Connecticut Avenue, NW, Washington, DC 20036, or refer to the Air Cargo Guide. However, because of the limited numbers of containers and aircraft of this capacity, it is not recommended that shipment normally be planned by this mode.

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5.6.7.2 Military airlift. Container airlift is primarily intertheater, and even this is inhibited by circumstances. The 8-foot (2.438-m) by 8-foot (2.438-m) by 20-foot (6.098-m) military-owned demountable container (MILVAN) or ANSI/ISO container is compatible with C-130, C-141, and C-5A only when the correct loading procedure is used. The tare weight of the 8-foot (2.438-m) by 8-foot (2.438-m) by 20-foot (6.098-m) container varies from 3,200 (1 451 kg) to 4,700 pounds (2 132 kg), imposing a relatively high tare weight penalty against the payload of an aircraft. ANSI/ISO standard containers do not use cargo space within the aircraft efficiently. While the commercial air industry does use several IATA standard containers, neither their configuration nor their strength meets military requirements. Currently, the 463L pallet system is incorporated into Air Force cargo transport aircraft. This system involves the use of 108-inch (2.743-m) by 88-inch (2.235-m) pallets to which the cargo is restrained. Because the 463L pallet was designed as a part of a materials handling system for airlift, it is not compatible with Army materials handling equipment and some vehicles, and is not sufficiently rugged for field use. Because of the incompatibility between the MILVAN and the HCU-6/E pallet the HCU-6/E pallet should not be used as an adapter platform to airlift MILVAN. This problem does not exist with the 20-foot (6.098-m) metric platform (A/E-29H-1) or with the 20-foot (6.098-m) type II airdrop platform; therefore, these should be specified as the adapter platforms for airlift of MILVANs. Air transport of the MILVAN container should not be routine but rather on an emergency or high-priority basis. The capability for intertheater airlift of containerized cargo will be further increased when the freighter version of commercial jumbo-type aircraft, such as the 747F, is introduced in sufficient numbers into the civil reserve air fleet (CRAF).

5.6.8 Container express (CONEX). This container is a reusable metal shipping box utilized by both the US Army and US Air Force. There are two types of CONEX containers. Cargo capacity of each type is 9,000 pounds (4 082 kg), and the interior dimensional constraints are shown in table XIX. To facilitate handling and loading small items, fiberboard

TABLE XIX. CONEX CONTAINERS

Type	Width	Height	Length
Type 1 (half-size)	5 ft 11-3/4 in. (1.822 m)	6 ft 0-3/8 in. (1.838 m)	3 ft 10-5/8 in. (1.184 m)
Type 2 (standard)	5 ft 11-3/4 in. (1.822 m)	6 ft 0-3/8 in. (1.838 m)	8 ft 1-5/8 in. (2.479 m)

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insert containers, with a maximum capacity of 1,000 pounds (454 kg) when fastened to a wooden pallet, are provided. Six insert containers may be stowed in the standard CONEX container (type 2) and four in the half-size CONEX container (type 1).

5.6.9 Military-owned demountable container (MILVAN). This is a US Army 8-foot (2.438-m) by 8-foot (2.438-m) by 20-foot (6.098-m) cargo container constructed to ANSI/ISO standards. The basic container, with plywood interior paneling, is a dry-cargo container that has a tare weight of 4,700 pounds (2 132 kg). With installation of an internal, mechanical load-bracing system, MILVAN containers are used to transport ammunition; however, they are also utilized for other cargo. Using the mechanical bracing system, the MILVAN has a tare weight of 5,640 pounds (2 558 kg). The maximum gross weight for either container is 44,800 pounds (20 321 kg). MILVANs have door openings 7 feet 6 inches (2.286 m) wide and 7 feet 0 inches (2.134 m) high and an interior length of 19 feet 3 inches (5.867 m).

5.7 Induced environments.

5.7.1 Handling and transport shock and vibration. This section explains the mechanical shock and vibration environments that packaged and unpackaged equipment and materiel may encounter during rough handling and transportation by highway, rail, ocean, and air. The transportation shock and vibration environment is complex, and the severity of an individual impact is unpredictable. Therefore, it is impossible to describe such environments accurately in simple terms; they must be described statistically, and the derived data must be presented in a form directly related to their end use. Moreover, in practice, there is often no precise distinction between shock and vibration environments since any item may experience shocks that can excite vibration responses. Examples of this could be a truck negotiating a potholed highway or a ship in heavy seas. The transit shock and vibration conditions given represent input conditions at the floor or cargo platform of the transit vehicle. They do not take into account the damping or energy-absorbing characteristics of the equipment or materiel and its packaging. When significant motion is anticipated between the cargo and floor or between the cargo platform and the transit vehicle, a mechanical analysis should be made to establish reasonable criteria. Detailed specifications for and information on shock and vibration are given in MIL-STD-810, MIL-P-116, MIL-STD-648, FED-STD-101, MIL-S-901, and MIL-STD-1670.

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5.7.2 Test facilities. Details of the facilities normally used to evaluate the effects of handling and transportation shock and vibration conditions are contained in standards covering environmental test methods.

5.7.3 Physical handling and delivery shocks. Shocks caused by handling procedures can be more severe than those experienced during transit; however, the probability of severe handling shock is lower where mechanical materials handling equipment is used. The severity of shock from physical handling during maintenance and installation is difficult to predict but, for normal design purposes, impact resulting from dropping or toppling onto critical faces, edges, and corners may be taken as a guide.

5.7.3.1 Drop height. Generally, the lighter the item the more likely it is to be dropped from greater heights. Drop heights that can be encountered in unloading highway and rail vehicles should be used as a normal criterion.

5.7.3.2 Shock pulse dependency factors. Designers should note that the shock pulse experienced by an item of materiel/equipment due to dropping, while extremely variable, does depend on several well-established factors. The more important of these factors are:

a. Height of drop.

b. Nature and condition of surface upon which the item is dropped.

c. Attitude of item on striking surface.

d. Package material and its construction, where applicable.

e. Elasticity and natural frequency of item dropped.

f. Elasticity and strength of striking surface.

5.7.4 Dockside handling.

5.7.4.1 Shock environment--light cargo. Packaged materiel/equipment (except hazardous materiel) weighing less than 1,000 pounds (454 kg) generally will be handled in a hoisting net with other packages and items. When lifted, the packages and items become crushed together and, on release, may fall and roll over several times after impact.

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5.7.4.2 Shock environment--heavy cargo. Items of equipment weighing more than 1,000 pounds (454 kg) are more likely to be crane-handled, separately, and thereby are subject to misjudgment by the crane operator, causing shocks on dropping or by impact against a wall. The maximum severity of this environment is unpredictable; however, guidance for an equivalent free-fall drop height may be found in one of the documents cited in paragraph 5.7.1.

5.7.5 Aerial delivery.

5.7.5.1 Shock environment. Shocks up to 20g may be sustained by equipment and packages when the parachute opens.

5.7.5.2 Ground impact velocity. The velocity at impact with the ground varies with the parachute configuration, but 24 feet per second (7.3 m/s) is considered to be the minimum, and 30 feet per second (9.1 m/s) the maximum unless the parachute is damaged.

5.7.5.3 Crushing and rolling. Conditions of crushing and rolling, similar to those stated in paragraph 5.7.4.1, also can occur during aerial delivery.

5.7.6 Ammunition handling.

5.7.6.1 Drop height--packaged ammunition. Ammunition, which is likely to be subjected to ship and aircraft loading and unloading, should be designed to withstand 40-foot (12.192-m) drop heights without exploding, and to remain safe during disposal, in accordance with individual procedures. For Navy ammunition this explosive safety test must be performed on the bare round.

5.7.6.2 Drop height--unpackaged ammunition. Unpackaged tank and artillery ammunition, which is likely to be loaded or unloaded with combat vehicles, should be designed to withstand 10-foot (3.048-m) drop heights without exploding. If the ammunition is intended for truckage or gun loading, it should be designed to withstand 5-foot (1.524-m) drop heights without visible damage and should be safe to fire. If visibly damaged, the ammunition should be safe to dispose of in accordance with individual procedures. These requirements do not apply to Navy gun ammunition, since that ammunition must meet the 40-foot (12.192-m) explosive safety test on the bare round.

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5.7.7 Transportation shock.

5.7.7.1 General. To describe the total transportation environment, trials are required to evaluate both the effects of repeated shock and the damage effects from high-intensity shock that occur infrequently during transportation. Some criteria apply in general; but in specific instances, as for items of large mass, items subject to infrequent transportation, and items requiring special handling, the severity and duration will need to be modified.

5.7.7.2 Repeated shock procedure. To evaluate the effects of repeated shock, the following design-evaluation methods shall be used.

a. For loose cargo, a method can be used based on the motional simulation, created by an approved facility, of a vehicle platform operating at a nominal speed and double amplitude in either a synchronous or its equivalent nonsynchronous mode. Design criteria should be based on a duration ranging from 1 to 3 hours (3 600s to 10 800s).

b. For restrained cargo, the application of a repeated half-sine wave shock with a specified peak amplitude and pulse duration may be applied. Design criteria normally will be based on a minimum number of such shocks.

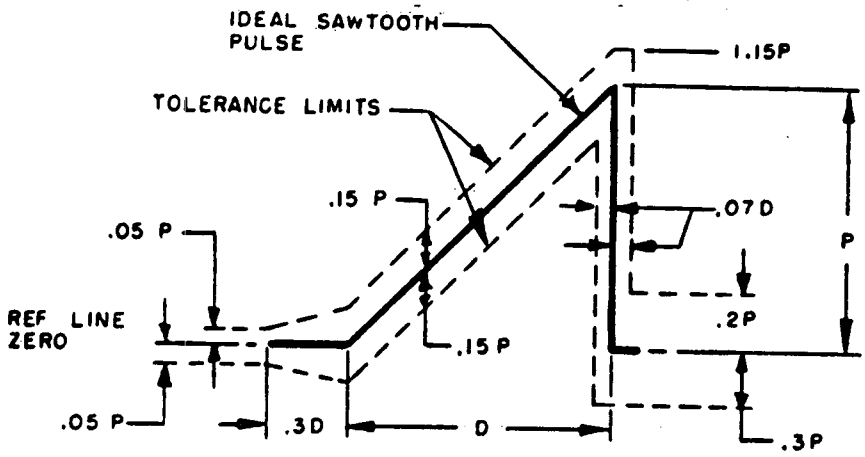
5.7.7.3 High-intensity shock procedure. High-intensity shock is described as an ideal terminal peak pulse having a high peak acceleration with a short pulse duration. The pulse may have a square or saw-toothed shape. A sample saw-toothed pulse is shown in figure 47. Design criteria should be based on the application of this shock pulse in each direction of the three principal axes of the item that may be subject to this type of impact.

5.7.7.4 Transit situations. The above design criteria should provide a high level of confidence of survival of equipment or materiel in the following transit situations:

a. Wheeled highway vehicles backing into loading docks, negotiating railway crossings, and traversing pot-holes.

b. Rail vehicles during car switching and shunting operations, starting, stopping, and slack takeup in coupling systems.

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PROCEDURE	TEST	PEAK VALUE (P) g's		NOMINAL DURATION (D) ms	
		FLIGHT VEHICLE EQUIPMENT a 1/	GROUND EQUIPMENT b	FLIGHT VEHICLE EQUIPMENT c 1/	GROUND EQUIPMENT d
I	BASIC DESIGN	20	40 2/	11	11
III	CRASH SAFETY	40	75	11	6
IV	HIGH INTENSITY	100	100	6	11

1/ SHOCK PARAMETERS a AND c: RECOMMENDED FOR EQUIPMENT NOT SHOCK MOUNTED AND WEIGHING LESS THAN 300 POUNDS.

2/ EQUIPMENT MOUNTED ONLY IN TRUCKS AND SEMITRAILERS MAY USE A 20g PEAK VALUE.

NOTE: THE OSCILLOGRAM SHALL INCLUDE A TIME ABOUT 3D LONG WITH A PULSE LOCATED APPROXIMATELY IN THE CENTER. THE PEAK ACCELERATION MAGNITUDE OF THE SAWTOOTH PULSE IS P AND ITS DURATION IS D. THE MEASURED ACCELERATION PULSE SHALL BE CONTAINED BETWEEN THE BROKEN LINE BOUNDARIES AND THE MEASURED VELOCITY CHANGE (WHICH MAY BE OBTAINED BY INTEGRATION OF THE ACCELERATION PULSE) SHALL BE WITHIN THE LIMITS OF $V_i \pm 0.1\ V_i$, WHERE V_i IS THE VELOCITY-CHANGE ASSOCIATED WITH THE IDEAL PULSE WHICH EQUALS 0.5 DP. THE INTEGRATION TO DETERMINE VELOCITY CHANGE SHALL EXTEND FROM 0.4D BEFORE THE PULSE TO 0.1D AFTER THE PULSE.

FIGURE 47. High-intensity shock pulse envelope.

c. Ships docking or slamming when in motion in heavy seas.

d. Aircraft during landing and braking.

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5.8 Natural environments.

5.8.1 Climatic conditions. Climatic conditions vary widely from point to point over the surface of the earth. Furthermore, at any particular location, daily as well as yearly, variations in these conditions occur and, in many instances, the variations are extreme. It is beyond the scope of this handbook to present detailed information on climatic conditions around the world.

5.8.1.1 Worldwide climatic extremes. Worldwide climatic extremes are presented in figure 48. A summary of the general range of temperature, solar radiation, and relative humidity for those extremes is shown in table XX and figure 49. The eight climatic categories are differentiated on the basis of temperature and/or humidity extremes and are useful for consideration of their effect on military materiel (AR 70-38). Operational conditions, as stated in table XX, are the climatic conditions in the open to which military materiel might be subjected during operations or standby for operations. Storage and transit conditions may differ from operational temperature and humidity because of the induced effects of heat gains or losses of air in confined spaces. Knowledge of worldwide climatic extremes is useful to materiel developers and testers as background information; however, since the extreme values occur so infrequently, they normally are not used for design and test criteria. Additional environmental criteria and guidelines are presented in MIL-STD-1670 and MIL-STD-210. Other climatic environments that should be considered in the design of military materiel are altitude (pressure and temperature); blowing sand, dust, and snow; ozone; and a combination of environmental factors (MIL-STD-810 and MIL-STD-1670). For standby and transit conditions, the environmental conditions between origin and destination should be considered in the preparation for shipment.

5.9 Transportability testing.

5.9.1 General. Materiel must be able to survive transportation in a military environment without reduction in functional performance. The environment imposes numerous constraints involving impacts, vibrations, interferences, and repetitive motions. These constraints require attention to blocking, bracing, slinging, tiedown, and containerization in connection with the stowage, orientation, suspension, or transport of cargo. Adequacy of design depends upon compatibility with the transportation media and performance after handling and transport. The media cover a wide range of air and surface vehicles and materials handling

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equipment. A single item in transit may be subjected to a wide variation of environment and orientation, and to forces, ranging from negligible to potentially destructive.

5.9.2 Tests. TECOM test operations procedure (TOP) 1-2-500 provides the Army guidance for preparing test plans to evaluate the transportability characteristics of military materiel whether towed, self-propelled, or moved by carrier over highway, off-road terrain, rail, ocean, or by air. The tests listed in the TOP, sections II and III, are used as pertinent to the particular test item and test type. A development test II plan, for example, should include the subtests that would satisfy the criteria of the ROC, development plan (DP), or other governing documents. A development test III plan should include the subtests that would satisfy the contractual provisions of the applicable military specifications and suitability criteria, as established by test directive MIL-STD-810. It should be consulted for further guidance.

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TABLE XX. SUMMARY OF TEMPERATURE, SOLAR RADIATION, AND RELATIVE HUMIDITY DIURNAL EXTREMES

Climatic category	Operational conditions			Storage and transit conditions	
	Ambient air temperature °F	Solar radiation BTU/ft²/hr	Ambient relative humidity %	Induced air temperature °F	Induced relative humidity %
1 Wet-Warm	Nearly constant 75	Negligible	95 to 100	Nearly constant 80	95 to 100
2 Wet-Hot	78 to 95	0 to 360	74 to 100	90 to 160	10 to 85
3 Humid-Hot Coastal Desert	85 to 100	0 to 360	63 to 90	90 to 160	10 to 85
4 Hot-Dry	90 to 125	0 to 360	5 to 20	90 to 160	2 to 50
5 Inter-mediate Hot-Dry	70 to 110	0 to 360	20 to 85	70 to 145	5 to 50
6 Inter-mediate Cold	-5 to -25	Negligible	Tending toward saturation	-10 to -30	Tending toward saturation
7 Cold	-35 to -50	Negligible	Tending toward saturation	-35 to -50	Tending toward saturation
8 Extreme Cold	-60 to -70	Negligible	Tending toward saturation	-60 to -70	Tending toward saturation

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6. NOTES

6.1 International standardization agreements. Certain provisions of this document are the subject of international standardization agreements (ASTM E 380-76; STANAG 2832; STANAG 3548; and AIR STD 44/21). When a change notice, revision, or cancellation of this handbook is proposed that will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels, including departmental standardization offices, if required.

Custodians:
Army - MT
Navy - SA
Air Force - 11

Review activities:
Army - ME, AL, AT, AV
Navy
Air Force
Other - CA, GS

International interest (see 6.1)

User:
Army - GL, AR
Corps of Engineers - CE

Preparing activity:
Army - MT
Project No. PACK 0590

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APPENDIX

CONVERSION TABLES

1. Common Metric Abbreviations

m = meter
dm = decimeter
cm = centimeter
mm = millimeter
Hz = hertz
gal = gallon
mhp = metric horsepower
kg = kilogram
km = kilometer
t = metric ton
Pa = pascal
rad = radian
s = second
km/hr = kilometers per hour

2. Length

1 mi = 1,609.35 m
1 yd = 0.9144 m
1 ft = 0.3048 m
1 in. = 0.0254 m
1 m = 10dm = 100 cm = 1,000 mm
1 km = 0.6214 mi
1 m = 1.0936 yd
1 m = 3.2808 ft
1 m = 39.3700 in.

3. Area

1 sq yd = 0.8361 sq m
1 sq ft = 0.0929 sq m
1 sq in. = 0.00065 sq m
1 sq m = 1.196 sq yd
1 sq m = 10.764 sq ft
1 sq m = 1,550 sq in.

4. Volume

1 cu yd = 0.76455 cu m
1 cu ft = 0.02831 cu m
1 cu in. = 0.000016 cu m
1 litre = 1.00×10^{-3} cu m
1 cu m = 1.31 cu yd
1 cu m = 35.30 cu ft
1 cu m = 61 023 cu in.
1 gal = 0.0038 cu m

5. Mass

1 STON = 907.185 kg
1 LTON = 1 016 kg
1 lb = 0.45359 kg
1 kg = 2.2046 lb
1 t = 1 000 kg
1 t = 2,204.62 lb

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6. Acceleration

1 foot/second^2 = 0.3048 m/s^2
freefall, standard = 9.807 m/s^2

7. Velocity (includes speed)

1 foot/second = 0.3048 m/s
1 knot (international) = 0.5144 m/s
1 mile/hour = 0.4470 m/s
1 rev/min = 0.1047 rad/s
1 kilometer/hour = 0.278 m/s

8. Mass/Area

1 pound-mass/foot^2 = 4.882 kg/m^2
1 pound-mass/inch^2 = 703.1 kg/m^2

9. Pressure/Stress

1 pound-force/inch^2 = 6 895 Pa
1 pound-force/foot^2 = 47.88 Pa

10. Angular Measure

1 degree = 0.01745 rad

11. Force

1 pound-force = 4.448 newton

12. Power

1 horsepower = 1.014 metric horsepower

13. The following simplified conversion factors are accurate to within 2 percent for quick computations:

a. Inches to centimeters - Multiply in. by 10 and divide by 4.

b. Yards to meters - Multiply yd by 9 and divide by 10.

c. Miles to kilometers - Multiply mi by 8 and divide by 5.

d. Pounds to kilograms - Multiply lb by 5 and divide by 11.

14. A measurement ton equals 40 cubic feet of volume.

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